

Stochastic Calculus and Applications in Finance

March 4, 2025

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1 Basics of measure-theoretic probability

1.1 Probability spaces and random variables

In this section we gather some basic topics from measure theory that will be used throughout these notes. We will introduce the concepts of σ -algebras, measures, integrals and we will see how they can be used in order to have a formalisation of probability. Let us consider a non empty set Ω which we will think of as the set of all possible outcomes. For example, if we want to model a dice, we will choose $\Omega = \{1, 2, 3, 4, 5, 6\}$. The event that the outcome is even is $\{2, 4, 6\}$ which is clearly a subset of Ω . Events, in general, will be subsets of the sample space Ω . If “ A happens” is an event, then “ A does not happen” should also be an event. If we have events “ A_i happens”, for $i \in \mathbb{N}$, then “at least one of the A_i happens” should also be an event. This suggests that the collection of all events should be modelled by a σ -algebra.

Definition 1.1.1. A σ -algebra $\mathcal{F}$ on Ω is a collection of subsets of Ω such that

- (i) $\Omega \in \mathcal{F}$
- (ii) If $A \in \mathcal{F}$ then $A^c := \Omega \setminus A \in \mathcal{F}$
- (iii) If $A_n \in \mathcal{F}$ for $n \in \mathbb{N}$, then $\bigcup_{n=1}^{\infty} A_n \in \mathcal{F}$.

The pair $(\Omega, \mathcal{F})$ is called a measurable space.

Exercise 1.1.2. Show that $\emptyset \in \mathcal{F}$. Show that if $A_n \in \mathcal{F}$ for $n \in \mathbb{N}$, then $\bigcap_{n=1}^{\infty} A_n \in \mathcal{F}$

Solution: Let $(\Omega, \mathcal{F})$ be a measurable space.

- Since $\mathcal{F}$ is a σ -algebra, $\Omega \in \mathcal{F}$. Thus using that σ -algebras are closed under complementation: $\Omega^c \in \mathcal{F}$, i.e. $\emptyset \in \mathcal{F}$.
- Let $A_1, A_2, A_3, \dots \in \mathcal{F}$. Now, making use of De Morgan’s identity:

$$\bigcap_n A_n = \bigcap_n (A_n^c)^c = \left(\bigcup_n A_n^c \right)^c$$

which is in $\mathcal{F}$ because σ -algebras are closed under complementation and countable union.

□

Exercise 1.1.3. Consider the set of real numbers $\mathbb{R}$.

- (1) Find the smallest σ -algebra on $\mathbb{R}$.
- (2) Find the smallest σ -algebra on $\mathbb{R}$ that contains the interval $(0, 1)$.

Solution:

1. The smallest σ -algebra on $\mathbb{R}$ is $\{\emptyset, \mathbb{R}\}$.
(In fact on any nonempty set Ω , the smallest σ -algebra is $\{\emptyset, \Omega\}$, and the largest σ -algebra is the power set $\mathcal{P}(\Omega)$.)
2. Let $\mathcal{F}$ be the smallest σ -algebra on $\mathbb{R}$ that contains the interval $(0, 1)$. Then

- $\emptyset, \mathbb{R} \in \mathcal{F}$ by the def. of σ -algebras and by Exercise 2.1.2.
- By assumption, $(0, 1) \in \mathcal{F}$, so since σ -algebras are closed under complementation: $(-\infty, 0] \cup [1, \infty) \in \mathcal{F}$.

Since $\{\emptyset, \mathbb{R}, (0, 1), (-\infty, 0] \cup [1, \infty)\}$ is a σ -algebra, it follows that

$$\mathcal{F} = \{\emptyset, \mathbb{R}, (0, 1), (-\infty, 0] \cup [1, \infty)\}.$$

□

Definition 1.1.4. Let Ω be a non empty set and let $\mathcal{A}$ be a collection of subsets of Ω . We denote by $\sigma(\mathcal{A})$ the intersection of all σ -algebras on Ω that contain $\mathcal{A}$, that is

$$\sigma(\mathcal{A}) := \bigcap_{\mathcal{B} \in \mathbf{S}_{\mathcal{A}}} \mathcal{B},$$

where $\mathbf{S}_{\mathcal{A}} = \{\mathcal{B} : \mathcal{B} \text{ is a } \sigma\text{-algebra on } \Omega \text{ and } \mathcal{A} \subset \mathcal{B}\}$.

It turns out that $\sigma(\mathcal{A})$ is a σ -algebra. We call it “the σ -algebra generated by $\mathcal{A}$ ”. In fact, it is the smallest σ -algebra that contains $\mathcal{A}$.

Exercise 1.1.5. Show that $\mathbf{S}_{\mathcal{A}}$ is non empty. Show that intersection of σ -algebras is a σ -algebra. Consequently, $\sigma(\mathcal{A})$ is indeed a σ -algebra.

Solution: Let Ω be nonempty set and let $\mathcal{A}$ be a collection of subsets of Ω . Let $\mathbf{S}_{\mathcal{A}} := \{\mathcal{B} : \mathcal{B} \text{ is a } \sigma\text{-algebra and } \mathcal{A} \subset \mathcal{B}\}$. Denote the power set of Ω by $\mathcal{P}(\Omega)$.

Then $\mathcal{P}(\Omega) \in \mathbf{S}_{\mathcal{A}}$, because $\mathcal{P}(\Omega)$ is a σ -algebra, and $\mathcal{A} \subset \mathcal{P}(\Omega)$. Hence $\mathbf{S}_{\mathcal{A}}$ is non empty.

We will now show that intersection of σ -algebras is a σ -algebra. Let H be a family of σ -algebras. Then

- (i) By the first σ -algebra axiom: $\Omega \in \mathcal{F}$ for all $\mathcal{F} \in H$. Therefore

$$\Omega \in \bigcap_{\mathcal{F} \in H} \mathcal{F}$$

- (ii) Let $A \in \bigcap_{\mathcal{F} \in H} \mathcal{F}$. Then

$$A \in \mathcal{F} \quad \forall \mathcal{F} \in H.$$

So since σ -algebras are closed under complementation:

$$A^c \in \mathcal{F} \quad \forall \mathcal{F} \in H.$$

Hence

$$A^c \in \bigcap_{\mathcal{F} \in H} \mathcal{F}.$$

So $\bigcap_{\mathcal{F} \in H} \mathcal{F}$ is closed under complementation.

(iii) Let $A_1, A_2, A_3, \dots \in \bigcap_{\mathcal{F} \in H} \mathcal{F}$. Then

$$A_1, A_2, A_3, \dots \in \mathcal{F} \quad \forall \mathcal{F} \in H.$$

So since σ -algebras are closed under countable union:

$$\bigcup_n A_n \in \mathcal{F} \quad \forall \mathcal{F} \in H.$$

Hence

$$\bigcup_n A_n \in \bigcap_{\mathcal{F} \in H} \mathcal{F}.$$

So $\bigcap_{\mathcal{F} \in H} \mathcal{F}$ is closed under countable union.

By (i),(ii),(iii): $\bigcap_{\mathcal{F} \in H} \mathcal{F}$ is a σ -algebra.

We have shown that intersection of σ -algebras is indeed a σ -algebra. In particular this implies that

$$\sigma(\mathcal{A}) = \bigcap_{\mathcal{B} \in \mathcal{S}_{\mathcal{A}}} \mathcal{B}$$

is a σ -algebra.

□

Definition 1.1.6. The Borel σ -algebra on $\mathbb{R}^d$, denoted by $\mathcal{B}(\mathbb{R}^d)$ is the σ -algebra generated by the collection of open sets of $\mathbb{R}^d$.

Exercise 1.1.7. Show that $\mathcal{B}(\mathbb{R})$ contains all sets of the form (a, b) , $(a, b]$, $[a, b)$, $[a, b]$, $(-\infty, a]$, $(-\infty, a)$, $(a, +\infty)$, $[a, +\infty)$, for $a, b \in \mathbb{R}$. Moreover, show that the σ -algebra $\mathcal{B}(\mathbb{R})$ is generated by any of these classes.

Solution:

- (i) Recall that $\mathcal{B}(\mathbb{R})$ is defined to be the σ -algebra generated by the collection of all open sets of $\mathbb{R}$. Hence $\mathcal{B}(\mathbb{R})$ contains all open sets of $\mathbb{R}$. Thus $(a, b) \in \mathcal{B}(\mathbb{R})$.
- (ii) $(a, b] = \bigcap_{n=1}^{\infty} (a, b + \frac{1}{n}) \in \mathcal{B}(\mathbb{R})$, since open sets are Borel, and σ -algebras are closed under countable intersection.
- (iii) $[a, b) = \bigcap_{n=1}^{\infty} (a - \frac{1}{n}, b) \in \mathcal{B}(\mathbb{R})$ by the same reasoning.
- (iv) $[a, b] = \bigcap_{n=1}^{\infty} (a - \frac{1}{n}, b + \frac{1}{n}) \in \mathcal{B}(\mathbb{R})$ by the same reasoning.
- (v) $(-\infty, a] = \bigcup_{n=1}^{\infty} [-n, a] \in \mathcal{B}(\mathbb{R})$ since (by (iv)) closed intervals are Borel and σ -algebras are closed under countable union.
- (vi) $(-\infty, a) = \bigcup_{n=1}^{\infty} (-n, a) \in \mathcal{B}(\mathbb{R})$ since open sets are Borel and σ -algebras are closed under countable union.
- (vii) $(a, \infty) = \bigcup_{n=1}^{\infty} (a, n) \in \mathcal{B}(\mathbb{R})$ by the same reasoning as in (vi).

(viii) $[a, \infty) = \bigcup_{n=1}^{\infty} [a, n] \in \mathcal{B}(\mathbb{R})$ by the same reasoning as in (v).

Hence $\mathcal{B}(\mathbb{R})$ contains all sets of the form (a, b) , $(a, b]$, $[a, b)$, $[a, b]$, $(-\infty, a]$, $(-\infty, a)$, (a, ∞) , $[a, \infty)$. We will now show that any of these classes generates $\mathcal{B}(\mathbb{R})$.

(I) Since $\mathcal{B}(\mathbb{R})$ contains all sets of the form (a, b) , and $\sigma(\{(a, b) : a < b\})$ is the smallest σ -algebra containing all sets of the form (a, b) , we have

$$\sigma(\{(a, b) : a < b\}) \subset \mathcal{B}(\mathbb{R}).$$

Since every open set in $\mathbb{R}$ is a countable union of (disjoint) open intervals, and σ -algebras are closed under countable union, it follows that $\sigma(\{(a, b) : a < b\})$ must contain all open sets in $\mathbb{R}$. Since $\mathcal{B}(\mathbb{R})$ is the smallest σ -algebra containing all open sets of $\mathbb{R}$, we have

$$\mathcal{B}(\mathbb{R}) \subset \sigma(\{(a, b) : a < b\})$$

Therefore we conclude that

$$\sigma(\{(a, b) : a < b\}) = \mathcal{B}(\mathbb{R})$$

(II) Similarly, as $\mathcal{B}(\mathbb{R})$ contains all sets of the form $(a, b]$, and $\sigma(\{(a, b] : a < b\})$ is the smallest σ -algebra containing all sets of the form $(a, b]$, we have

$$\sigma(\{(a, b] : a < b\}) \subset \mathcal{B}(\mathbb{R})$$

Since

$$(a, b) = \bigcup_{n=1}^{\infty} \left(a, b - \frac{b-a}{2n} \right],$$

every open interval is a countable union of intervals of the form $(a, b]$. So since σ -algebras are closed under countable union, all open intervals must be contained in $\sigma(\{(a, b] : a < b\})$. So since $\sigma(\{(a, b] : a < b\}) = \mathcal{B}(\mathbb{R})$ is the smallest σ -algebra containing all open intervals, we have

$$\mathcal{B}(\mathbb{R}) \subset \sigma(\{(a, b] : a < b\}).$$

Hence we conclude that

$$\mathcal{B}(\mathbb{R}) = \sigma(\{(a, b] : a < b\})$$

(III) Similarly:

$$\sigma(\{[a, b) : a < b\}) \subset \mathcal{B}(\mathbb{R})$$

and since

$$(a, b) = \bigcup_{n=1}^{\infty} \left[a + \frac{b-a}{2n}, b \right)$$

we also have that

$$\mathcal{B}(\mathbb{R}) \subset \sigma(\{[a, b) : a < b\}).$$

Therefore

$$\sigma(\{[a, b) : a < b\}) = \mathcal{B}(\mathbb{R}).$$

(IV) Similarly:

$$\sigma(\{[a, b] : a < b\}) \subset \mathcal{B}(\mathbb{R})$$

and since

$$(a, b) = \bigcup_{n=1}^{\infty} \left[a + \frac{b-a}{3n}, b - \frac{b-a}{3n} \right]$$

we also have

$$\mathcal{B}(\mathbb{R}) \subset \sigma(\{[a, b] : a < b\})$$

Therefore

$$\sigma(\{[a, b] : a < b\}) = \mathcal{B}(\mathbb{R})$$

(V) We have

$$\sigma(\{(-\infty, a] : a \in \mathbb{R}\}) \subset \mathcal{B}(\mathbb{R}).$$

Furthermore

$$(a, b] = (-\infty, b] \setminus (-\infty, a],$$

so since σ -algebras are closed under taking differences of sets, $\sigma(\{(-\infty, a] : a \in \mathbb{R}\})$ must contain all sets of the form $(a, b]$. So since $\sigma(\{[a, b] : a < b\}) = \mathcal{B}(\mathbb{R})$ is the smallest σ -algebra containing all sets of the form $(a, b]$, it follows that

$$\mathcal{B}(\mathbb{R}) \subset \sigma(\{(-\infty, a] : a \in \mathbb{R}\}).$$

Hence we conclude that

$$\sigma(\{(-\infty, a] : a \in \mathbb{R}\}) = \mathcal{B}(\mathbb{R})$$

(VI) Similarly,

$$\sigma(\{(-\infty, a) : a \in \mathbb{R}\}) \subset \mathcal{B}(\mathbb{R}),$$

furthermore

$$[a, b) = (-\infty, b) \setminus (-\infty, a)$$

and thus

$$\sigma(\{[a, b) : a < b\}) \subset \sigma(\{(-\infty, a) : a \in \mathbb{R}\}),$$

i.e.

$$\mathcal{B}(\mathbb{R}) \subset \sigma(\{(-\infty, a) : a \in \mathbb{R}\}).$$

So we conclude that

$$\mathcal{B}(\mathbb{R}) = \sigma(\{(-\infty, a) : a \in \mathbb{R}\})$$

(VII) Similarly,

$$\sigma(\{(a, \infty) : a \in \mathbb{R}\}) \subset \mathcal{B}(\mathbb{R}),$$

furthermore

$$(a, b] = (a, \infty) \setminus (b, \infty)$$

and thus

$$\sigma(\{(a, b] : a < b\}) \subset \sigma(\{(a, \infty) : a \in \mathbb{R}\})$$

i.e.

$$\mathcal{B}(\mathbb{R}) \subset \sigma(\{(a, \infty) : a \in \mathbb{R}\}).$$

So we conclude that

$$\sigma(\{(a, \infty) : a \in \mathbb{R}\}) = \mathcal{B}(\mathbb{R}).$$

(VIII) Similarly,

$$\sigma(\{[a, \infty) : a \in \mathbb{R}\}) \subset \mathcal{B}(\mathbb{R}),$$

furthermore

$$[a, b) = [a, \infty) \setminus [b, \infty)$$

and thus

$$\sigma(\{[a, b) : a < b\}) \subset \sigma(\{[a, \infty) : a \in \mathbb{R}\})$$

i.e.

$$\mathcal{B}(\mathbb{R}) \subset \sigma(\{[a, \infty) : a \in \mathbb{R}\}).$$

So we conclude that

$$\sigma(\{[a, \infty) : a \in \mathbb{R}\}) = \mathcal{B}(\mathbb{R}).$$

This finishes the proof.

Note: In our proof we used the fact that σ -algebras are closed under taking differences of sets. This can be seen as follows: Let A, B be measurable sets. Then $A \setminus B = A \cap B^c$ is measurable, since σ -algebras are closed under complementation and (countable and thus also finite) intersection. $\square$

On $\mathbb{R}^d$ we will always consider the Borel σ -algebra $\mathcal{B}(\mathbb{R}^d)$.

Definition 1.1.8. Let $(\Omega, \mathcal{F})$ and $(\Omega', \mathcal{F}')$ be two measurable spaces. A function $f : \Omega \rightarrow \Omega'$ is called $\mathcal{F}/\mathcal{F}'$ -measurable if $f^{-1}(B) \in \mathcal{F}$ for all $B \in \mathcal{F}'$.

If $\Omega' = \mathbb{R}^d$, then as already mentioned we will always consider $\mathcal{F}' = \mathcal{B}(\mathbb{R}^d)$. Moreover, in this case, for a $\mathcal{F}/\mathcal{B}(\mathbb{R}^d)$ -measurable function we will simply say it is $\mathcal{F}$ -measurable.

Lemma 1.1.9. Let $(\Omega, \mathcal{F})$ and $(\Omega', \mathcal{F}')$ be two measurable spaces. Let $\mathcal{A}'$ be a collection of subsets of Ω' and assume that $\mathcal{F}' = \sigma(\mathcal{A}')$. Then, a function $f : \Omega \rightarrow \Omega'$ is $\mathcal{F}/\mathcal{F}'$ -measurable if and only if $f^{-1}(B) \in \mathcal{F}$ for all $B \in \mathcal{A}'$.

Exercise 1.1.10. Let $(\Omega_1, \mathcal{F}_1)$, $(\Omega_2, \mathcal{F}_2)$, and $(\Omega_3, \mathcal{F}_3)$ be measurable spaces. Let $f : \Omega_1 \rightarrow \Omega_2$ be $\mathcal{F}_1/\mathcal{F}_2$ -measurable and let $g : \Omega_2 \rightarrow \Omega_3$ be $\mathcal{F}_2/\mathcal{F}_3$ -measurable. Show that $g \circ f : \Omega_1 \rightarrow \Omega_3$ is $\mathcal{F}_1/\mathcal{F}_3$ -measurable.

Solution: Let $A \in \mathcal{F}_3$. Since g is $\mathcal{F}_2/\mathcal{F}_3$ -measurable:

$$g^{-1}(A) \in \mathcal{F}_2$$

So since f is $\mathcal{F}_1/\mathcal{F}_2$ -measurable:

$$f^{-1}(g^{-1}(A)) \in \mathcal{F}_1$$

As $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$, we thus have

$$(g \circ f)^{-1}(A) \in \mathcal{F}_1.$$

We have shown that $(g \circ f)^{-1}(A) \in \mathcal{F}_1$ for all $A \in \mathcal{F}_3$. Therefore $g \circ f$ is $\mathcal{F}_1/\mathcal{F}_3$ -measurable. $\square$

Exercise 1.1.11. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function. Show that it is $\mathcal{B}(\mathbb{R})$ -measurable.

Solution: We need to show that f is $\mathcal{B}(\mathbb{R})/\mathcal{B}(\mathbb{R})$ -measurable. Since $\mathcal{B}(\mathbb{R})$ is generated by the collection of open sets in $\mathbb{R}$, recalling Lemma 1.1.9 it suffices to show that

$$f^{-1}(A) \in \mathcal{B}(\mathbb{R}) \quad \forall A \subset \mathbb{R} \text{ such that } A \text{ is open.}$$

To this end, let $A \subset \mathbb{R}$ be open. Since f is continuous, the preimage $f^{-1}(A)$ is also open. So since open sets are Borel, $f^{-1}(A) \in \mathcal{B}(\mathbb{R})$, as required. $\square$

Proposition 1.1.12. *Let $(\Omega, \mathcal{F})$ be a measurable space and let $f, g : \Omega \rightarrow \mathbb{R}$ be $\mathcal{F}$ -measurable functions. Then $f + g$ and $f \cdot g$ are $\mathcal{F}$ -measurable.*

Exercise 1.1.13. Let $(\Omega, \mathcal{F})$ be a measurable space and let $f_n : \Omega \rightarrow \mathbb{R}$ be $\mathcal{F}$ -measurable functions for $n \in \mathbb{N}$.

- (1) Show that $\sup_n f_n, \inf f_n$ are measurable.
- (2) Show that $\limsup_{n \rightarrow \infty} f_n$ is a measurable function

Solution:

1. We need to show that $\sup_n f_n$ is $\mathcal{F}/\mathcal{B}(\mathbb{R})$ -measurable. Since the collection of sets of the form $(-\infty, a]$ generates $\mathcal{B}(\mathbb{R})$, it suffices to show that

$$\left(\sup_n f_n \right)^{-1}((-\infty, a]) \in \mathcal{F} \quad \forall a \in \mathbb{R}.$$

Now,

$$\begin{aligned} \left(\sup_n f_n \right)^{-1}((-\infty, a]) &= \{\omega \in \Omega : \sup_n f_n(\omega) \leq a\} \\ &= \{\omega \in \Omega : \forall n \in \mathbb{N}, f_n(\omega) \leq a\} \\ &= \bigcap_{n \in \mathbb{N}} \{\omega \in \Omega : f_n(\omega) \leq a\} \\ &= \bigcap_{n \in \mathbb{N}} f_n^{-1}((-\infty, a]). \end{aligned}$$

Note that $(-\infty, a]$ is Borel, so since f_n is assumed to be $\mathcal{F}/\mathcal{B}(\mathbb{R})$ -measurable it follows that $f_n^{-1}((-\infty, a]) \in \mathcal{F}$. So since σ -algebras are closed under countable intersection, we have that

$$\bigcap_{n \in \mathbb{N}} f_n^{-1}((-\infty, a]) \in \mathcal{F}.$$

So by our calculations above:

$$\left(\sup_n f_n \right)^{-1}((-\infty, a]) \in \mathcal{F}.$$

Therefore $\sup_n f_n$ is indeed measurable.

Similarly, since the collection of sets of the form $[a, \infty)$ generates $\mathcal{B}(\mathbb{R})$, it suffices to show that

$$\left(\inf_n f_n\right)^{-1}([a, \infty)) \in \mathcal{F} \quad \forall a \in \mathbb{R}.$$

Now,

$$\begin{aligned} \left(\inf_n f_n\right)^{-1}([a, \infty)) &= \{\omega \in \Omega : \inf_n f_n \in [a, \infty)\} \\ &= \{\omega \in \Omega : \forall n \in \mathbb{N}, f_n(\omega) \geq a\} \\ &= \bigcap_{n \in \mathbb{N}} \{\omega \in \Omega : f_n(\omega) \geq a\} \\ &= \bigcap_{n \in \mathbb{N}} f_n^{-1}([a, \infty)) \end{aligned}$$

which is in $\mathcal{F}$ since $[a, \infty)$ is Borel, f_n is $\mathcal{F}/\mathcal{B}(\mathbb{R})$ -measurable, and σ -algebras are closed under countable intersection. Hence $\inf_n f_n$ is also measurable.

2. If for each $\omega \in \Omega$ the limit $\lim_{n \rightarrow \infty} f_n(\omega)$ exists, then

$$f(\omega) = \lim_{n \rightarrow \infty} f_n(\omega) = \lim_{n \rightarrow \infty} \inf f_n(\omega) = \sup_{n \geq 0} \inf_{m \geq n} f_m(\omega)$$

which is measurable by part 1. of the exercise.

□

Definition 1.1.14. Let $(\Omega, \mathcal{F})$ be a measurable space. A non-negative function $\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$ is called a probability measure if

1. $\mathbb{P}(\emptyset) = 0$
2. If $A_n \in \mathcal{F}$, for $n \in \mathbb{N}$, and $A_n \cap A_m = \emptyset$ for $n \neq m$, then

$$\mathbb{P}(\cup_{n=1}^{\infty} A_n) = \sum_{n=1}^{\infty} \mathbb{P}(A_n).$$

3. $\mathbb{P}(\Omega) = 1$

The triple $(\Omega, \mathcal{F}, \mathbb{P})$ is called a probability space.

A non-negative function $\mu : \mathcal{F} \rightarrow [0, +\infty]$ satisfying 1 and 2 of the above definition will be called a measure. The triple $(\Omega, \mathcal{F}, \mu)$ is called a measure space. Obviously, a probability space is a particular case of a measure space.

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. A set $A \subset \Omega$ is called null, if there exists $B \in \mathcal{F}$ such that $\mathbb{P}(B) = 0$ and $A \subset B$. The probability space $(\Omega, \mathcal{F}, \mathbb{P})$ is called complete if $\mathcal{F}$ contains all the null sets.

Exercise 1.1.15. Show that if $A, B \in \mathcal{F}$ and $A \subset B$, then $\mathbb{P}(A) \leq \mathbb{P}(B)$.

Solution: We have that $B = A \cup (B \setminus A)$. Since A and $(B \setminus A)$ are disjoint, we get $\mathbb{P}(B) = \mathbb{P}(A) + \mathbb{P}(B \setminus A)$. Since $\mathbb{P}(B \setminus A) \geq 0$, we get $\mathbb{P}(B) \geq \mathbb{P}(A)$. $\square$

Proposition 1.1.16 (Continuity from above and below). *Let $A_n, B_n \in \mathcal{F}$, for $n \in \mathbb{N}$ such that $A_n \subset A_{n+1}$ and $B_{n+1} \subset B_n$ for all $n \in \mathbb{N}$. Then,*

$$i) \mathbb{P}(\cup_n A_n) = \lim_{n \rightarrow \infty} \mathbb{P}(A_n)$$

$$ii) \mathbb{P}(\cap_n B_n) = \lim_{n \rightarrow \infty} \mathbb{P}(B_n)$$

Exercise 1.1.17. Let A_1, A_2 be events such that $\mathbb{P}(A_1) = \mathbb{P}(A_2) = 1$. Show that $\mathbb{P}(A_1 \cap A_2) = 1$.

Solution: Let A_1, A_2 be events such that $\mathbb{P}(A_1) = \mathbb{P}(A_2) = 1$. Show that $\mathbb{P}(A_1 \cap A_2) = 1$.

Proof. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let A_1, A_2 be events such that $\mathbb{P}(A_1) = \mathbb{P}(A_2) = 1$. Now, since $A_1 \setminus A_2, A_2 \setminus A_1, A_1 \cap A_2$ are pairwise disjoint, by the second measure axiom we have

$$\begin{aligned} \mathbb{P}(A_1 \cup A_2) &= \mathbb{P}((A_1 \setminus A_2) \cup (A_2 \setminus A_1) \cup (A_1 \cap A_2)) \\ &= \mathbb{P}(A_1 \setminus A_2) + \mathbb{P}(A_2 \setminus A_1) + \mathbb{P}(A_1 \cap A_2) \end{aligned}$$

We will now show that the first two terms in this expression are zero:

(i) Since $A_1 \setminus A_2 \subset \Omega \setminus A_2 = A_2^c$, by Exercise 2.1.9 we have

$$\mathbb{P}(A_1 \setminus A_2) \leq \mathbb{P}(A_2^c)$$

But $1 = \mathbb{P}(\Omega) = \mathbb{P}(A_2 \cup A_2^c) = \mathbb{P}(A_2) + \mathbb{P}(A_2^c) = 1 + \mathbb{P}(A_2^c)$, therefore

$$\mathbb{P}(A_2^c) = 0.$$

Hence by nonnegativity of measures it follows that

$$\mathbb{P}(A_1 \setminus A_2) = 0.$$

(ii) Switching A_2 with A_1 in the above computation, we can also see that

$$\mathbb{P}(A_2 \setminus A_1) = 0.$$

By (i) and (ii), the expression for $\mathbb{P}(A_1 \cup A_2)$ simplifies to

$$\mathbb{P}(A_1 \cup A_2) = \mathbb{P}(A_1 \cap A_2)$$

Since $A_1 \subset A_1 \cup A_2$, by Exercise 1.1.15 we have $\mathbb{P}(A_1) \leq \mathbb{P}(A_1 \cup A_2)$, so since $\mathbb{P}(A_1) = 1$ and probabilities cannot be greater than 1, it follows that $\mathbb{P}(A_1 \cup A_2) = 1$. Therefore

$$\mathbb{P}(A_1 \cap A_1) = 1$$

as required.

$\square$

Exercise 1.1.18. Let A_n be events such that $\mathbb{P}(A_n) = 1$ for all $n \in \mathbb{N}$. Show that $\mathbb{P}(\cap_{n=1}^{\infty} A_n) = 1$.

Hint: Use Proposition 1.1.16

Solution: We will first show by induction that $\mathbb{P}(\bigcap_{n=1}^k A_n) = 1$ for all $k \in \mathbb{N}$.

Initial case: for $k = 1$ we have

$$\bigcap_{n=1}^k A_n = \bigcap_{n=1}^1 A_n = A_1,$$

which has probability 1 by assumption. Hence the statement holds for $k = 1$.

Now suppose that the statement holds for some $k = K \in \mathbb{N}$, i.e. that $\mathbb{P}(\bigcap_{n=1}^K A_n) = 1$. Then it also holds for $k = K + 1$, because

$$\mathbb{P}\left(\bigcap_{n=1}^{K+1} A_n\right) = \mathbb{P}\left(\left(\bigcap_{n=1}^K A_n\right) \cap A_{K+1}\right) = 1,$$

where the last equality holds because by the inductive hypothesis $\mathbb{P}(\bigcap_{n=1}^K A_n) = 1$, furthermore by assumption $\mathbb{P}(A_{K+1}) = 1$, and by Exercise 1.1.17 the intersection of two events of probability 1 has also probability 1.

By the principle of mathematical induction, we have shown that $\mathbb{P}(\bigcap_{n=1}^k A_n) = 1$ for all $k \in \mathbb{N}$.

Define

$$B_k := \bigcap_{n=1}^k A_n$$

Note the following:

- (i) $\bigcap_{n=1}^{\infty} A_n = \bigcap_{k=1}^{\infty} \bigcap_{n=1}^k A_n = \bigcap_{k=1}^{\infty} B_k$
- (ii) $B_{k+1} \subset B_k$, hence $(B_k)_{k \in \mathbb{N}}$ satisfies the assumptions of Proposition 1.1.16
- (iii) as we have shown inductively: $\mathbb{P}(B_k) = 1$ for all $k \in \mathbb{N}$.

Using (i), (ii) and (iii) (in this order), we get that

$$\begin{aligned} \mathbb{P}\left(\bigcap_{n=1}^{\infty} A_n\right) &= \mathbb{P}\left(\bigcap_{k=1}^{\infty} B_k\right) \\ &= \lim_{k \rightarrow \infty} \mathbb{P}(B_k) \\ &= \lim_{k \rightarrow \infty} 1 \\ &= 1 \end{aligned}$$

as required.

□

Lemma 1.1.19 (Borel-Cantelli). *Let $(A_k)_{k=1}^{\infty}$ be a sequence of events. If $\sum_{k=1}^{\infty} \mathbb{P}(A_k) < \infty$, then*

$$\mathbb{P}(A_k \text{ i.o.}) = 0,$$

where $\{A_k \text{ i.o.}\} = \bigcap_{n=1}^{\infty} \bigcup_{j=n}^{\infty} A_j$

Definition 1.1.20. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. A function $X : \Omega \rightarrow \mathbb{R}$ is called *random variable* if it is $\mathcal{F}$ -measurable.

Exercise 1.1.21. Let X be a random variable on $(\Omega, \mathcal{F}, \mathbb{P})$. Show that the collections of sets

$$\{X^{-1}(A) : A \in \mathcal{B}(\mathbb{R})\}$$

is a σ -algebra. In particular, it then follows that

$$\{X^{-1}(A) : A \in \mathcal{B}(\mathbb{R})\} = \sigma(\{X^{-1}(A) : A \in \mathcal{B}(\mathbb{R})\}).$$

The σ -algebra above is called the σ -algebra generated by X and is denoted by $\sigma(X)$.

Solution: Let us denote $\sigma(X) := \{X^{-1}(A) : A \in \mathcal{B}(\mathbb{R})\}$.

- Note, that $\mathbb{R} \in \mathcal{B}(\mathbb{R})$. Since X is real-valued: $X(\omega) \in \mathbb{R}$ for all $\omega \in \Omega$, therefore

$$X^{-1}(\mathbb{R}) = \{\omega \in \Omega : X(\omega) \in \mathbb{R}\} = \Omega$$

Hence $\Omega \in \sigma(X)$.

- Let $B \in \sigma(X)$. Then by the def. of $\sigma(X)$:

$$B = X^{-1}(A)$$

for some $A \in \mathcal{B}(\mathbb{R})$. Therefore

$$\begin{aligned} B^c &= (X^{-1}(A))^c \\ &= \{\omega \in \Omega : X(\omega) \in A\}^c \\ &= \{\omega \in \Omega : X(\omega) \notin A\} \\ &= \{\omega \in \Omega : X(\omega) \in A^c\} \\ &= X^{-1}(A^c) \end{aligned}$$

which is in $\sigma(X)$, because since $A \in \mathcal{B}(\mathbb{R})$, it follows that $A^c \in \mathcal{B}(\mathbb{R})$. Hence $B^c \in \sigma(X)$.

This shows that $\sigma(X)$ is closed under complementation.

- Let $B_1, B_2, B_3, \dots \in \sigma(X)$. Then for all $n \in \mathbb{N}$,

$$B_n = X^{-1}(A_n)$$

for some $A_n \in \mathcal{B}(\mathbb{R})$. Therefore

$$\begin{aligned} \bigcup_n B_n &= \bigcup_n X^{-1}(A_n) \\ &= \bigcup_n \{\omega \in \Omega : X(\omega) \in A_n\} \\ &= \{\omega \in \Omega : X(\omega) \in \bigcup_n A_n\} \\ &= X^{-1}\left(\bigcup_n A_n\right) \end{aligned}$$

which is in $\sigma(X)$, because $A_1, A_2, A_3, \dots \in \mathcal{B}(\mathbb{R})$, and thus $\bigcup_n A_n \in \mathcal{B}(\mathbb{R})$. Hence $\bigcup_n B_n \in \sigma(X)$.

This shows that $\sigma(X)$ is closed under countable union. This finishes the proof.

□

Remark 1.1.22. If we have two (or more) random variables X and Y then the σ -algebra generated by X and Y is

$$\sigma(X, Y) := \sigma(\{X^{-1}(A) : A \in \mathcal{B}(\mathbb{R})\} \cup \{Y^{-1}(B) : B \in \mathcal{B}(\mathbb{R})\}).$$

Notice that the collection $\{X^{-1}(A) : A \in \mathcal{B}(\mathbb{R})\} \cup \{Y^{-1}(B) : B \in \mathcal{B}(\mathbb{R})\}$ is not a σ -algebra in general(!).

In order to ease the notation, most of the times that we deal with a random variable X we will drop the argument ω . For example, the set $\{\omega \in \Omega : X(\omega) \leq 1\}$ will simply be denoted by $\{X \leq 1\}$ of the probability $\mathbb{P}(\{\omega \in \Omega : X(\omega) \leq 1\})$ will simply be denoted by $\mathbb{P}(X \leq 1)$.

Definition 1.1.23. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $X : \Omega \rightarrow \mathbb{R}$ be a random variable. The distribution of X is a probability measure $\mathbb{Q}_X$ on the $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$, which is given by

$$\mathbb{Q}_X(A) = \mathbb{P}(X \in A) \quad \text{for all } A \in \mathcal{B}(\mathbb{R}).$$

1.2 Expectation

Next, we move to the concept of expectation. We consider a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

Definition 1.2.1. A random variable X is called simple if there exist $c_1, \dots, c_n \in \mathbb{R}$ and $A_1, \dots, A_n \in \mathcal{F}$ such that

$$X = \sum_{i=1}^n c_i \mathbf{1}_{A_i}. \quad (1.1)$$

If X is a simple with representation (1.1), then we define the expectation of X by

$$\mathbb{E}[X] := \sum_{i=1}^n c_i \mathbb{P}(A_i)$$

It turns out that the expectation does not depend on the representation of X . Moreover, notice that the expectation is a linear functional, that is, if X, Y are simple random variables and $a, b \in \mathbb{R}$, then

$$\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y].$$

It is also easy to see that if X, Y are simple random variables with $X \leq Y$, then $\mathbb{E}[X] \leq \mathbb{E}[Y]$. Next, we define the integral for all non-negative random variables X . If X is a non-negative random variable, then there exists random variables $X_n, n \in \mathbb{N}$ such that for all $\omega \in \Omega$, $X_n(\omega) \uparrow X(\omega)$, as $n \uparrow \infty$. Indeed, we can set $X_n = h_n(X)$, where $h_n(x) = \min\{[x2^n]2^{-n}, 2^n\}$ where $[a]$ denotes the lower integer part of $a \in \mathbb{R}$. We then set

$$\mathbb{E}[X] = \lim_{n \rightarrow \infty} \mathbb{E}[X_n]$$

Notice that the limit always exists, since it is an increasing sequence, however, it can be infinite.

Finally, a random variable $X : \Omega \rightarrow \mathbb{R}$ is called integrable if

$$\mathbb{E}[X^+] < \infty, \quad \mathbb{E}[X^-] < \infty,$$

where $X^+ = \max\{X, 0\}$ and $X^- = \max\{-X, 0\}$. Equivalently, X is integrable if and only if

$$\mathbb{E}[|X|] < \infty.$$

Notice that the above expectations are well defined since X^+ , X^- , and $|X|$ are all non-negative random variables. We denote the set of all integrable random variables by $L_1(\Omega)$. For $X \in L_1(\Omega)$, we set

$$\mathbb{E}[X] = \mathbb{E}[X^+] - \mathbb{E}[X^-].$$

The expectation $\mathbb{E}[X]$ is also denoted by

$$\int_{\Omega} X \, d\mathbb{P} \quad \text{or} \quad \int_{\Omega} X(\omega) \, d\mathbb{P}(\omega).$$

It is really *the integral of X with respect to the measure $\mathbb{P}$* . The linearity and the monotonicity of the expectation is preserved on $L_1(\Omega)$.

Lemma 1.2.2. *Let $X, Y \in L_1(\Omega)$ and $a, b \in \mathbb{R}$. Then the following hold:*

1. $\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y]$
2. *If $X \leq Y$, then $\mathbb{E}[X] \leq \mathbb{E}[Y]$.*

Theorem 1.2.3. *Let X be a random variable. Then, for any non-negative $\mathcal{B}(\mathbb{R})$ -measurable $g : \mathbb{R} \rightarrow \mathbb{R}$ we have*

$$\mathbb{E}[g(X)] = \int_{\mathbb{R}} g(x) \mathbb{Q}_X(dx). \tag{1.2}$$

Actually the above theorem remains valid for functions g taking values with different signs as long as one of the sides of (1.2) is well defined. This follows by applying the above theorem to g^+ and g^- and using that $g = g^+ - g^-$.

The equality in (1.2) probably looks familiar. You were taught in previous years that if X has a density function f , then

$$\mathbb{E}[g(X)] = \int_{\mathbb{R}} g(x) f(x) dx.$$

Actually, this is not a definition (as you were taught) but a consequence of Theorem 1.2.3, since, the expression " X has a density f " means exactly that the distribution of X satisfies $\mathbb{Q}_X(dx) = f(x) dx$.

For $p \in [1, \infty)$ let us set

$$\|X\|_{L_p} = (\mathbb{E}[|X|^p])^{1/p}$$

and let us also set

$$\|X\|_{L_\infty} = \inf\{K \geq 0 : \mathbb{P}(|X| \leq K) = 1\}.$$

For $p \in [1, \infty]$, we denote by $L_p(\Omega)$ the linear space of all random variables $X : \Omega \rightarrow \mathbb{R}$ such that

$$\|X\|_{L_p} < \infty.$$

Lemma 1.2.4 (Markov's inequality). *Let X be a random variable. Then, for any $\varepsilon > 0$ we have*

$$\mathbb{P}(|X| \geq \varepsilon) \leq \frac{1}{\varepsilon} \mathbb{E}[|X|].$$

Lemma 1.2.5 (Hölders inequality). *For any random variables X and Y we have*

$$\|XY\|_{L_1} \leq \|X\|_{L_p} \|Y\|_{L_q},$$

where for all $p, q \in [1, \infty]$ with $p^{-1} + q^{-1} = 1$.

Lemma 1.2.6 (Minkowski's inequality or triangle inequality). *Let $p \in [1, \infty]$. For any random variables X and Y we have*

$$\|X + Y\|_{L_p} \leq \|X\|_{L_p} + \|Y\|_{L_p}.$$

Exercise 1.2.7. Show that if $p \geq q$, then $\|X\|_{L_q} \leq \|X\|_{L_p}$.

Solution: We have

$$\|X\|_{L_q}^q = \mathbb{E}[|X|^q] = \mathbb{E}[1 \cdot |X|^q] \leq (\mathbb{E}[1]) (\mathbb{E}[|X|^q]^{p/q})^{q/p} = (\mathbb{E}[|X|^p])^{q/p} = \|X\|_{L_p}^q,$$

where we have used Hölder's inequality. Hence, raising both sides of the inequality to the $1/q$ finishes the proof. $\square$

Theorem 1.2.8 (Completeness of $L_p(\Omega)$). *Let $p \in [1, \infty]$. Let $X_n \in L_p(\Omega)$, for $n \in \mathbb{N}$ such that $\lim_{n,m \rightarrow \infty} \|X_n - X_m\|_{L_p} = 0$. Then there exists $X \in L_p(\Omega)$, such that*

$$\lim_{n \rightarrow \infty} \|X_n - X\|_{L_p} = 0.$$

1.3 Convergence of random variables

As we have seen, random variables are functions defined on a set Ω . As such, we have different modes of convergence.

Definition 1.3.1 (convergence almost surely). Let $(X_n)_{n \in \mathbb{N}}$, be a sequence of random variables. We say that X_n converges to a random variable X *almost surely* if

$$\mathbb{P}\left(\left\{\omega \in \Omega : \lim_{n \rightarrow \infty} X_n(\omega) = X(\omega)\right\}\right) = 1.$$

We usually write $X_n \rightarrow X$ almost surely.

Definition 1.3.2 (convergence in probability). Let $(X_n)_{n \in \mathbb{N}}$, be a sequence of random variables. We say that X_n converges to a random variable X *in probability* if for any $\varepsilon > 0$

$$\lim_{n \rightarrow \infty} \mathbb{P}\left(\left\{\omega \in \Omega : |X_n(\omega) - X(\omega)| \geq \varepsilon\right\}\right) = 0.$$

We usually write $X_n \rightarrow X$ in probability.

Definition 1.3.3 (convergence in L_p). Let $(X_n)_{n \in \mathbb{N}}$ be a sequence of random variables that belong to $L_p(\Omega)$. We say that X_n converges to a random variable $X \in L_p(\Omega)$ in L_p if

$$\lim_{n \rightarrow \infty} \|X_n - X\|_{L_p} = 0$$

We usually write $X_n \rightarrow X$ in L_p .

Theorem 1.3.4. Let $(X_n)_{n \in \mathbb{N}}$, X be random variables. The following hold.

- (i) If $X_n \rightarrow X$ almost surely, then $X_n \rightarrow X$ in probability.
- (ii) If $X_n \rightarrow X$ in probability, then there exists a subsequence $(X_{n_k})_{k \in \mathbb{N}}$ such that $X_{n_k} \rightarrow X$ almost surely.
- (iii) If $X_n \rightarrow X$ in L_p , then $X_n \rightarrow X$ in probability.

Exercise 1.3.5. Show (iii) from Theorem 1.3.4.

Solution: Let $\varepsilon > 0$. By Markov's inequality, we have

$$\mathbb{P}(|X_n - X| > \varepsilon) = \mathbb{P}(|X_n - X|^p > \varepsilon^p) \leq \frac{1}{\varepsilon^p} \mathbb{E}[|X_n - X|^p] = \frac{1}{\varepsilon^p} \|X_n - X\|_{L_p}^p.$$

The right hand side of the above inequality tends to zero as $n \rightarrow \infty$, by assumption. Hence, so does the left hand side. This shows that $X_n \rightarrow X$ in probability. $\square$

1.4 Limit theorems for expectations

Theorem 1.4.1 (Monotone Convergence). Let $X_n, X : \Omega \rightarrow \mathbb{R}$, for $n \in \mathbb{N}$, be non-negative random variables such that almost surely $X_n \uparrow X$ as $n \uparrow \infty$. Then

$$\mathbb{E}[X] = \lim_{n \rightarrow \infty} \mathbb{E}[X_n]$$

Lemma 1.4.2 (Fatou's Lemma). Let $X_n : \Omega \rightarrow \mathbb{R}$, for $n \in \mathbb{N}$, be non-negative random variables. Then

$$\mathbb{E}[\liminf_{n \rightarrow \infty} X_n] \leq \liminf_{n \rightarrow \infty} \mathbb{E}[X_n]$$

Theorem 1.4.3 (Lebesgue's theorem on Dominated Convergence). Suppose that $X_n \rightarrow X$ almost surely and that there exists $Y \in L_1(\Omega)$ such that $|X_n| \leq Y$. Then $X \in L_1(\Omega)$ and

$$\mathbb{E}[X] = \lim_{n \rightarrow \infty} \mathbb{E}[X_n].$$

1.5 Independence and conditioning

We consider a probability space $(\Omega, \mathcal{F}, \mathbb{P})$

Definition 1.5.1. The events $A_1, \dots, A_n$ are called independent if

$$\mathbb{P}(\cap_{i=1}^n A_i) = \prod_{i=1}^n \mathbb{P}(A_i).$$

The sigma algebras $\mathcal{G}_1, \dots, \mathcal{G}_n$ are called independent if for any $A_1 \in \mathcal{G}_1, \dots, A_n \in \mathcal{G}_n$, the events $A_1, \dots, A_n$ are independent. The random variables $X_1, \dots, X_n$ are independent if the σ -algebras $\sigma(X_1), \dots, \sigma(X_n)$ are independent. Finally, the elements of a family of events, σ -algebras, random variables are independent the element of any finite sub-family are independent.

Lemma 1.5.2. Let $\mathcal{G}_1, \mathcal{G}_2 \subset \mathcal{F}$ be σ -algebras. Let $\mathcal{A}_1 \subset \mathcal{G}_1$ and $\mathcal{A}_2 \subset \mathcal{G}_2$ be π -systems (classes that are closed under finite intersection) such that $\sigma(\mathcal{A}_1) = \mathcal{G}_1$ and $\sigma(\mathcal{A}_2) = \mathcal{G}_2$. Then $\mathcal{G}_1$ and $\mathcal{G}_2$ are independent if and only if

$$\mathbb{P}(A_1 \cap A_2) = \mathbb{P}(A_1)\mathbb{P}(A_2) \quad \text{for all } A_1 \in \mathcal{A}_1, A_2 \in \mathcal{A}_2.$$

Theorem 1.5.3. If $X, Y \in L_1(\Omega)$ are independent random variables then $XY \in L_1(\Omega)$ and

$$\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$$

A proof of the next lemma can be found in [?, Lemma 4.15]

Lemma 1.5.4. Let $X_1, \dots, X_n$ be independent random variables, $m_1 + \dots + m_k = n$ and $f_1, \dots, f_k$ be measurable functions of $\mathbb{R}^{m_1}, \dots, \mathbb{R}^{m_k}$ respectively. Then the random variables $Y_1 = f_1(X_1, \dots, X_{m_1})$, $Y_2 = f_2(X_{m_1+1}, \dots, X_{m_1+m_2})$, $\dots$, $Y_k = f_k(X_{m_1+\dots+m_{k-1}+1}, \dots, X_n)$ are independent.

Definition 1.5.5. Let $X \in L_1(\Omega)$ and let $\mathcal{G} \subset \mathcal{F}$ be a sub- σ -algebra of $\mathcal{F}$. One can show that there exists a unique $Y \in L_1(\Omega)$ with the following two properties:

1. Y is $\mathcal{G}$ -measurable
2. $\mathbb{E}[\mathbf{1}_A X] = \mathbb{E}[\mathbf{1}_A Y]$ for all $A \in \mathcal{G}$

Then we will call Y the *conditional expectation of X given $\mathcal{G}$* and we will write $Y = \mathbb{E}[X|\mathcal{G}]$. The existence and the uniqueness of the conditional expectation can be found in [?, p 71 Theorem 2 and p72 Theorem 5].

Exercise 1.5.6. Suppose that $\Omega = \cup_{i=1}^n A_i$ where A_i are disjoint events with $\mathbb{P}(A_i) > 0$. Let $\mathcal{G} = \sigma(\{A_1, \dots, A_n\})$ show that $\mathbb{E}[X|\mathcal{G}]$ is a simple random variable given by

$$\mathbb{E}[X|\mathcal{G}] = \sum_{i=1}^n c_i \mathbf{1}_{A_i},$$

where the constants c_i are given by

$$c_i = \frac{1}{\mathbb{P}(A_i)} \mathbb{E}[\mathbf{1}_{A_i} X].$$

It follows from the above that if $\mathcal{G} = \{\emptyset, \Omega\}$ is the trivial σ -algebra, then $\mathbb{E}[X|\mathcal{G}] = \mathbb{E}[X]$. We list below some properties of conditional expectations. For the proofs see for example [?, pp 71-75] and/or [?, p88].

Solution: Let

$$Y = \sum_{i=1}^n \mathbf{1}_{A_i} \frac{1}{\mathbb{P}(A_i)} \mathbb{E}[\mathbf{1}_{A_i} X]$$

We want to show that almost surely on Ω , we have $\mathbb{E}(X|\mathcal{G}) = Y$, i.e. that

$$\mathbb{E}[\mathbf{1}_G X] = \mathbb{E}[\mathbf{1}_G Y] \quad \forall G \in \mathcal{G}.$$

Note that since $\mathcal{G} = \sigma(\{A_1, \dots, A_n\})$ and $A_1, \dots, A_n$ is a partition of Ω , we have

$$G = \bigcup_{j \in H} A_j$$

for some $H \subset \{1, \dots, n\}$.

The simplest case is $G = A_j$ for some $j \in \{1, \dots, n\}$. Then we have that

$$\begin{aligned} \mathbb{E}[\mathbf{1}_G Y] &= \mathbb{E}[\mathbf{1}_{A_j} Y] \\ &= \mathbb{E} \left[\mathbf{1}_{A_j} \sum_{i=1}^n \mathbf{1}_{A_i} \frac{1}{\mathbb{P}(A_i)} \mathbb{E}(\mathbf{1}_{A_i} X) \right] \\ &= \mathbb{E} \left[\sum_{i=1}^n \mathbf{1}_{A_i \cap A_j} \frac{1}{\mathbb{P}(A_i)} \mathbb{E}(\mathbf{1}_{A_i} X) \right] \\ &=: (\star) \end{aligned}$$

Note that since $A_1, \dots, A_n$ are disjoint, so

$$\begin{aligned} \mathbf{1}_{A_i \cap A_j} &= \begin{cases} \mathbf{1}_\emptyset & \text{if } i \neq j \\ \mathbf{1}_{A_i} & \text{if } i = j \end{cases} \\ &= \begin{cases} 0 & \text{if } i \neq j \\ \mathbf{1}_{A_i} & \text{if } i = j. \end{cases} \end{aligned}$$

Therefore in the sum in the expression $(\star)$ only the $i = j$ term is nonzero, and in particular,

$$(\star) = \mathbb{E} \left[\mathbf{1}_{A_j} \frac{1}{\mathbb{P}(A_j)} \mathbb{E}(\mathbf{1}_{A_j} X) \right]$$

factoring out the nonrandom terms:

$$\begin{aligned} &= \frac{1}{\mathbb{P}(A_j)} \mathbb{E}[\mathbf{1}_{A_j} X] \mathbb{E}[\mathbf{1}_{A_j}] \\ &= \frac{1}{\mathbb{P}(A_j)} \mathbb{E}[\mathbf{1}_{A_j} X] \mathbb{P}(A_j) \\ &= \mathbb{E}[\mathbf{1}_{A_j} X] \end{aligned}$$

since we are considering the $G = A_j$ case:

$$= \mathbb{E}[\mathbf{1}_G X]$$

Thus the statement $\mathbb{E}[\mathbf{1}_G X] = \mathbb{E}[\mathbf{1}_G Y]$ holds for the $G = A_j$ case.

Now let us consider the general case, i.e. that $G = \bigcup_{j \in H} A_j$ for some $H \subset \{1, \dots, n\}$. Then we have

$$\mathbb{E}[\mathbf{1}_G Y] = \mathbb{E} \left[\mathbf{1}_{\bigcup_{j \in H} A_j} Y \right]$$

as $A_1, \dots, A_n$ are disjoint:

$$\begin{aligned}
&= \mathbb{E} \left[\sum_{j \in H} \mathbf{1}_{A_j} Y \right] \\
&= \sum_{j \in H} \mathbb{E}[\mathbf{1}_{A_j} Y]
\end{aligned}$$

as we have proven that $\mathbb{E}[\mathbf{1}_G X] = \mathbb{E}[\mathbf{1}_G Y]$ holds for the $G = A_j$ case:

$$\begin{aligned}
&= \sum_{j \in H} \mathbb{E}[\mathbf{1}_{A_j} X] \\
&= \mathbb{E} \left[\sum_{j \in H} \mathbf{1}_{A_j} X \right]
\end{aligned}$$

as $A_1, \dots, A_n$ are disjoint:

$$\begin{aligned}
&= \mathbb{E} \left[\mathbf{1}_{\bigcup_{j \in H} A_j} X \right] \\
&= \mathbb{E}[\mathbf{1}_G X].
\end{aligned}$$

Therefore the statement $\mathbb{E}[\mathbf{1}_G X] = \mathbb{E}[\mathbf{1}_G Y]$ also holds for the general $G = \bigcup_{j \in H} A_j$ case. This finishes the proof. $\square$

Proposition 1.5.7. *Let $X, Y \in L_1(\Omega)$ and $\mathcal{G} \subset \mathcal{F}$ be a σ -algebra. The following hold*

1. *Linearity: If a, b are constants, then $\mathbb{E}[aX + bY|\mathcal{G}] = a\mathbb{E}[X|\mathcal{G}] + b\mathbb{E}[Y|\mathcal{G}]$.*
2. *Taking out what is known: If X is $\mathcal{G}$ -measurable and XY is integrable, then $\mathbb{E}[XY|\mathcal{G}] = X\mathbb{E}[Y|\mathcal{G}]$. In particular, if X is $\mathcal{G}$ -measurable, then $\mathbb{E}[X|\mathcal{G}] = X$.*
3. *Conditioning on independent information: If $\mathcal{W} \subset \mathcal{F}$ is a σ -algebra independent of $\sigma(\mathcal{G}, X)$, then $\mathbb{E}[X|\sigma(\mathcal{G}, \mathcal{W})] = \mathbb{E}[X|\mathcal{G}]$. In particular, if X is independent of $\mathcal{W}$, then $\mathbb{E}[X|\mathcal{W}] = \mathbb{E}[X]$.*
4. *Tower property: If $\mathcal{W} \subset \mathcal{G}$ is a σ -algebra, then we have $\mathbb{E}[X|\mathcal{W}] = \mathbb{E}[\mathbb{E}[X|\mathcal{G}]|\mathcal{W}]$.*
5. *Monotonicity: If $X \geq 0$ then $\mathbb{E}[X|\mathcal{G}] \geq 0$.*
6. *Conditional Jensen's inequality: If $\phi : \mathbb{R} \rightarrow \mathbb{R}$ is convex, such that $\phi(X) \in L_1(\Omega)$, then $\phi(\mathbb{E}[X|\mathcal{G}]) \leq \mathbb{E}[\phi(X)|\mathcal{G}]$.*

Exercise 1.5.8. Show that the conditional expectation is a continuous operator from $L_1(\Omega)$ to $L_1(\Omega)$. In other words, show that if $X_n, X \in L_1(\Omega)$ such that $\lim_{n \rightarrow \infty} \|X_n - X\|_{L_1} = 0$, then $\lim_{n \rightarrow \infty} \|\mathbb{E}[X_n|\mathcal{G}] - \mathbb{E}[X|\mathcal{G}]\|_{L_1} = 0$.

Solution: Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, and let $\mathcal{G} \subset \mathcal{F}$ be a sub- σ -algebra. Suppose that $X_n, X \in L_1(\Omega, \mathcal{F}, \mathbb{P})$ and that $\lim_{n \rightarrow \infty} \|X_n - X\|_{L_1} = 0$. Then

$$\lim_{n \rightarrow \infty} \|\mathbb{E}[X_n|\mathcal{G}] - \mathbb{E}[X|\mathcal{G}]\|_{L_1(\Omega)} =$$

as $\mathbb{E}[\cdot|\mathcal{G}]$ is a linear operator:

$$= \lim_{n \rightarrow \infty} \|\mathbb{E}[X_n - X | \mathcal{G}]\|_{L_1(\Omega)}$$

by the def. of the $L_1(\Omega)$ -norm:

$$= \lim_{n \rightarrow \infty} \mathbb{E}[|\mathbb{E}[X_n - X | \mathcal{G}]|]$$

since $|\cdot|$ is convex, by Jensen's inequality:

$$\leq \lim_{n \rightarrow \infty} \mathbb{E}[\mathbb{E}[|X_n - X| | \mathcal{G}]]$$

by the Law of Total Expectation:

$$= \lim_{n \rightarrow \infty} \mathbb{E}[|X_n - X|]$$

by the def. of the $L_1(\Omega)$ -norm:

$$= \lim_{n \rightarrow \infty} \|X_n - X\|_{L_1(\Omega)}$$

by assumption:

$$= 0$$

Therefore $\lim_{n \rightarrow \infty} \|\mathbb{E}[X_n | \mathcal{G}] - \mathbb{E}[X | \mathcal{G}]\|_{L_1(\Omega)} = 0$ as required. $\square$

Exercise 1.5.9. Let $X \in L_2(\Omega)$ and let $\mathcal{G} \subset \mathcal{F}$ be a sub- σ -algebra of $\mathcal{F}$. Show that

$$\|X - \mathbb{E}[X | \mathcal{G}]\|_{L_2} = \min_{Z \in L_2^{\mathcal{G}}} \|X - Z\|_{L_2},$$

where we have denoted by $L_2^{\mathcal{G}}$ the collections of all $\mathcal{G}$ -measurable random variables that belong in $L_2(\Omega)$.

Solution: By definition of the conditional expectation we have that $\mathbb{E}[X | \mathcal{G}]$ is $\mathcal{G}$ -measurable. Moreover, by the conditional Jensen's inequality we see that

$$\mathbb{E}[|\mathbb{E}[X | \mathcal{G}]|^2] \leq \mathbb{E}[\mathbb{E}[|X|^2 | \mathcal{G}]] = \mathbb{E}[|X|^2] < \infty.$$

Consequently, $\mathbb{E}[X | \mathcal{G}] \in L_2^{\mathcal{G}}$, and by definition of the minimum we get

$$\|X - \mathbb{E}[X | \mathcal{G}]\|_{L_2} \geq \min_{Z \in L_2^{\mathcal{G}}} \|X - Z\|_{L_2}.$$

On the other hand, for any $Z \in L_2^{\mathcal{G}}$ we have

$$\begin{aligned} \|X - Z\|_{L_2}^2 &= \|X - \mathbb{E}[X | \mathcal{G}] + \mathbb{E}[X | \mathcal{G}] - Z\|_{L_2}^2 \\ &= \|X - \mathbb{E}[X | \mathcal{G}]\|_{L_2}^2 + 2\mathbb{E}[(X - \mathbb{E}[X | \mathcal{G}])(\mathbb{E}[X | \mathcal{G}] - Z)] + \|\mathbb{E}[X | \mathcal{G}] - Z\|_{L_2}^2 \\ &\geq \|X - \mathbb{E}[X | \mathcal{G}]\|_{L_2}^2 + 2\mathbb{E}[(X - \mathbb{E}[X | \mathcal{G}])(\mathbb{E}[X | \mathcal{G}] - Z)]. \end{aligned}$$

We claim the last term on the RHS is zero. Indeed, since $(\mathbb{E}[X | \mathcal{G}] - Z)$ and $\mathbb{E}[X | \mathcal{G}]$ are $\mathcal{G}$ -measurable, we have

$$\begin{aligned} \mathbb{E}[(X - \mathbb{E}[X | \mathcal{G}])(\mathbb{E}[X | \mathcal{G}] - Z)] &= \mathbb{E}\left[\mathbb{E}[(X - \mathbb{E}[X | \mathcal{G}])(\mathbb{E}[X | \mathcal{G}] - Z) | \mathcal{G}]\right] \\ &= \mathbb{E}\left[(\mathbb{E}[X | \mathcal{G}] - Z)\mathbb{E}[(X - \mathbb{E}[X | \mathcal{G}]) | \mathcal{G}]\right] \end{aligned}$$

$$\begin{aligned}
&= \mathbb{E}\left[(\mathbb{E}[X|\mathcal{G}] - Z)(\mathbb{E}[X|\mathcal{G}] - \mathbb{E}[X|\mathcal{G}])\right] \\
&= \mathbb{E}\left[(\mathbb{E}[X|\mathcal{G}] - Z) \cdot 0\right] \\
&= 0.
\end{aligned}$$

Therefore we have shown that for any $Z \in L_2^{\mathcal{G}}$, we have

$$\|X - \mathbb{E}[X|\mathcal{G}]\|_{L_2} \leq \|X - Z\|_{L_2}.$$

Hence,

$$\|X - \mathbb{E}[X|\mathcal{G}]\|_{L_2} \leq \min_{Z \in L_2^{\mathcal{G}}} \|X - Z\|_{L_2},$$

and since we have also shown the converse inequality, we get that

$$\min_{Z \in L_2^{\mathcal{G}}} \|X - Z\|_{L_2} = \|X - \mathbb{E}[X|\mathcal{G}]\|_{L_2},$$

□

2 Wiener Process or Brownian motion

2.1 Definition and the martingale property

In this section we will study the Wiener process or Brownian motion, one of the most fundamental stochastic processes. Its role among stochastic processes is similar to the standard normal among random variables.

The term Brownian motion comes from the name of the botanist R. Brown, who observed (1828) that pollen particles suspended in water perform a chaotic motion. It is now known that this motion is due to the cumulative effect of water molecules hitting the particle at various angles. L. Bachelier (1900) derived the law governing the position $W(t)$ at time t of a single particle performing a one-dimensional Brownian motion. In 1905, Einstein also derived the law from statistical mechanics considerations and applied it to the determination of molecular diameters.

The rigorous definition and the first mathematical proof of the existence of Brownian motion are due to N. Wiener, who studied Brownian motion in the 1920s, almost a century after it was observed by R. Brown. Wiener process is another term for Brownian motion, both terms being used equally often. Brownian motion and more general diffusion processes are extremely important in physics, economics, finance, and many branches of mathematics beyond probability theory.

Let us start with some basics about stochastic processes. For the rest of notes, all the random objects will be defined on a fixed probability space $(\Omega, \mathcal{F}, \mathbb{P})$ which we assume to be complete. For simplicity, we fix a time horizon $T > 0$, and we will only consider stochastic processes on the interval $[0, T]$. In the next definition, we denote by $\mathcal{F} \otimes \mathcal{B}([0, T])$ the sigma algebra on $\Omega \times [0, T]$ which is generated by sets of the form $A \times (a, b)$, where $A \in \mathcal{F}$ and $a, b \in [0, T]$.

Definition 2.1.1. A stochastic process is a map $X : \Omega \times [0, T] \rightarrow \mathbb{R}$ which is $\mathcal{F} \otimes \mathcal{B}([0, T])$ -measurable.

Notice that a stochastic process is a function of two variables

$$(\omega, t) \mapsto X(\omega, t).$$

If we fix $t \in [0, T]$, then the function $X(\cdot, t) : \Omega \rightarrow \mathbb{R}$ is $\mathcal{F}$ -measurable, hence it is a random variable. That is, we can see a stochastic process as a collection of random variables $X(t)$ indexed by $t \in [0, T]$. Sometimes we will denote the stochastic process X by $(X(t))_{t \in [0, T]}$. On the other hand, if we fix $\omega \in \Omega$, then we have a function of time $t \mapsto X(\omega, t)$, which will be called the trajectory or path corresponding to the fixed ω . We will say that a stochastic process X is continuous if for all $\omega \in \Omega$ the trajectory $t \mapsto X(\omega, t)$ is a continuous function of time. From now on, unless otherwise indicated, we drop the ω dependence as usual.

Remark 2.1.2. Functions $Y : \Omega \times [0, T] \rightarrow \mathbb{R}$ which satisfy the following

- for each $\omega \in \Omega$, the map $t \mapsto Y(\omega, t)$ is either left or right continuous
- for each $t \in [0, T]$, the map $\omega \mapsto Y(\omega, t)$ is $\mathcal{F}$ -measurable,

are called Caratheodory functions. It can be checked that Caratheodory functions are $\mathcal{F} \otimes \mathcal{B}([0, T])$ -measurable.

Definition 2.1.3. A continuous stochastic process $(W(t))_{t \in [0, T]}$ is called Wiener process if

1. $W(0) = 0$

2. For all $s < t$, $W(t) - W(s) \sim \mathcal{N}(0, |t - s|)$
3. For all $0 \leq t_1 < t_2 < \dots < t_n \leq T$, the random variables

$$W(t_2) - W(t_1), \dots, W(t_n) - W(t_{n-1})$$

are independent.

Exercise 2.1.4. Let $(W(t))_{t \in [0, T]}$ be a Wiener process, let $c > 0$ and set $\widetilde{W}(t) = cW(t/c^2)$ for $t \in [0, c^2T]$. Show that $\widetilde{W}$ is a Wiener processes on $[0, c^2T]$.

Solution: Since $W(t)$ is a Wiener process, it is continuous in t . Therefore $\widetilde{W}(t) = cW(t/c^2)$ is also continuous in t . Note furthermore the following:

- (i) $\widetilde{W}_0 = cW(0/c^2) = cW(0) = 0$ where the last equality holds almost surely because W is a Wiener process.

- (ii) Let $0 \leq s < t$. Then

$$\begin{aligned} \widetilde{W}(t) - \widetilde{W}_s &= cW(t/c^2) - cW(s/c^2) \\ &= c(W(t/c^2) - W(s/c^2)). \end{aligned}$$

Since $W(t/c^2) - W(s/c^2) \sim \mathcal{N}(0, t/c^2 - s/c^2)$:

$$\sim \mathcal{N}(0, t - s).$$

- (iii) Let $0 \leq t_1 < t_2 < \dots < t_n$. Then

$$\begin{aligned} (\widetilde{W}(t_2) - \widetilde{W}(t_1), \dots, \widetilde{W}(t_n) - \widetilde{W}(t_{n-1})) &= (cW(t_2/c^2) - cW(t_1/c^2), \dots, cW(t_n/c^2) - cW(t_{n-1}/c^2)) \\ &= (c(W(t_2/c^2) - W(t_1/c^2)), \dots, c(W(t_n/c^2) - W(t_{n-1}/c^2))). \end{aligned}$$

We can see that the components of this vector are independent, as W is a Wiener-process and $0 \leq t_1/c^2 < t_2/c^2 < \dots < t_n/c^2$, hence $W(t_2/c^2) - W(t_1/c^2), \dots, W(t_n/c^2) - W(t_{n-1}/c^2)$ are independent.

Therefore $\widetilde{W}$ has independent increments.

By (i),(ii),(iii): $\widetilde{W}$ is indeed a Wiener process. □

Exercise 2.1.5. Let $(W(t))_{t \in [0, T]}$ be a Wiener process and let $c \in (0, T)$. Set $\widetilde{W}(t) = W(c + t) - W(c)$ for $t \in [0, T - c]$. Show that $\widetilde{W}$ is a Wiener process on $[0, T - c]$.

Solution: Very similar to the exercise above. □

Definition 2.1.6. A family $\mathbb{F} := (\mathcal{F}_t)_{t \in [0, T]}$ of σ -algebras $\mathcal{F}_t \subset \mathcal{F}$ is called *filtration* if $\mathcal{F}_s \subset \mathcal{F}_t$ for $s < t$. The filtration $\mathbb{F}$ is called *complete* if $\mathcal{F}_0$ contains all the null sets of $\mathcal{F}$. A filtration is called *right continuous* if $\mathcal{F}_t := \bigcap_{s > t} \mathcal{F}_s$.

From now on, we fix a filtration $\mathbb{F} := (\mathcal{F}_t)_{t \in [0, T]}$.

Definition 2.1.7. A stochastic process $(X(t))_{t \in [0, T]}$ is called adapted to the filtration $\mathbb{F}$ if $X(t)$ is $\mathcal{F}_t$ -measurable for each $t \in [0, T]$,

Notice that every stochastic process X is adapted to its own filtration, that is, the family of σ -algebras given by $\mathcal{F}^X(t) = \sigma(X(s), s \leq t)$.

Definition 2.1.8. We will say that $(W(t))_{t \in [0, T]}$ is a $\mathbb{F}$ -Wiener process if

1. $(W(t))_{t \in [0, T]}$ is a Wiener process
2. $(W(t))_{t \in [0, T]}$ is adapted to the filtration $\mathbb{F}$
3. for all $0 \leq s \leq t \leq T$, the random variable $W(t) - W(s)$ is independent from $\mathcal{F}_s$.

Definition 2.1.9. A stochastic process $(X(t))_{t \in [0, T]}$ is an $\mathbb{F}$ -martingale if

1. $(X(t))_{t \in [0, T]}$ is $\mathbb{F}$ -adapted,
2. $\mathbb{E}[|X(t)|] < \infty$ for all $t \in [0, T]$,
3. $\mathbb{E}[X(t)|\mathcal{F}_s] = X(s)$ for all $s \leq t$.

Exercise 2.1.10. Let $(W(t))_{t \in [0, T]}$ be a $\mathbb{F}$ -Wiener process. Show that it is a martingale with respect to $\mathbb{F}$.

Solution: Let $(W(t))_{t \in [0, T]}$ be an $\mathbb{F}$ -Wiener process. Then

- (i) $(W(t))_{t \in [0, T]}$ is adapted to $\mathbb{F}$ by assumption.
- (ii) Let $t \in [0, T]$. Then using Hölder's inequality, and that since W is a Wiener process, $W(t) \sim \mathcal{N}(0, t)$, we have

$$\begin{aligned} \mathbb{E}[|W(t)|] &\leq (\mathbb{E}[|W(t)|^2])^{1/2} \\ &= (t)^{1/2} \\ &< \infty. \end{aligned}$$

- (iii) For $0 \leq s < t$:

$$\begin{aligned} \mathbb{E}[W(t)|\mathcal{F}_s] &= \mathbb{E}[W(t) - W(s) + W(s)|\mathcal{F}_s] \\ &= \mathbb{E}[W(t) - W(s)|\mathcal{F}_s] + \mathbb{E}[W(s)|\mathcal{F}_s] \\ &=: (\star). \end{aligned}$$

Note

- $W(t) - W(s)$ is independent of $\mathcal{F}_s$, as W is an $\mathbb{F}$ -Wiener process
- $W(s)$ is $\mathcal{F}_s$ -measurable, as W is adapted (again, because W is an $\mathbb{F}$ -Wiener process).

Thus

$$\begin{aligned}
 (\star) &= \mathbb{E}(W(t) - W(s)) + W(s) \\
 &= W(s).
 \end{aligned}
 \quad \text{as } W(t) - W(s) \sim \mathcal{N}(0, t - s):$$

By (i),(ii),(iii): W is indeed an $\mathbb{F}$ -martingale. □

Exercise 2.1.11. Let $(W(t))_{t \in [0, T]}$ be an $\mathbb{F}$ -Wiener process. Show that $((W(t))^2 - t)_{t \in [0, T]}$ is a martingale with respect to $\mathbb{F}$.

Solution: Let $(W(t))_{t \in [0, T]}$ be a an $\mathbb{F}$ -Wiener process. Then

- (i) Let $t \in [0, T]$. By assumption $W(t)$ is $\mathcal{F}_t$ -measurable. So since the map $x \mapsto x^2 - t$ is continuous and thus Borel, the composition $(W(t))^2 - t$ is also $\mathcal{F}_t$ -measurable.
- (ii) Let $t \in [0, T]$. Then (using the triangle inequality and that $W(t) \sim \mathcal{N}(0, t)$), we have

$$\begin{aligned}
 \mathbb{E}[|(W(t))^2 - t|] &\leq \mathbb{E}[(W(t))^2] + t \\
 &= t + t \\
 &< \infty.
 \end{aligned}$$

- (iii) For $0 \leq s < t$:

$$\begin{aligned}
 \mathbb{E}[(W(t))^2 - t | \mathcal{F}_s] &= \mathbb{E}[(W(t))^2 - 2W(t)W(s) + (W(s))^2 + 2W(t)W(s) - (W(s))^2 - t | \mathcal{F}_s] \\
 &= \mathbb{E}[(W(t) - W(s))^2 | \mathcal{F}_s] + \mathbb{E}[2W(t)W(s) | \mathcal{F}_s] - \mathbb{E}[(W(s))^2 | \mathcal{F}_s] - t \\
 &=: (\star)
 \end{aligned}$$

Note the following:

- $W(t) - W(s)$ is independent of $\mathcal{F}_s$, therefore

$$\begin{aligned}
 \mathbb{E}[(W(t) - W(s))^2 | \mathcal{F}_s] &= \mathbb{E}[(W(t) - W(s))^2] \\
 &= t - s
 \end{aligned}
 \quad \text{as } W(t) - W(s) \sim \mathcal{N}(0, t - s):$$

- $W(s)$ is $\mathcal{F}_s$ -measurable, hence

$$\begin{aligned}
 \mathbb{E}[2W(t)W(s) | \mathcal{F}_s] &= 2W(s)\mathbb{E}[W(t) | \mathcal{F}_s] \\
 &= 2W(s)W(s) \\
 &= 2(W(s))^2
 \end{aligned}
 \quad \text{since } \mathbb{F}\text{-Wiener processes are } \mathbb{F}\text{-martingales:}$$

- $W(s)$ is $\mathcal{F}_s$ -measurable, hence $\mathbb{E}[(W(s))^2 | \mathcal{F}_s] = (W(s))^2$

Substituting these into $(\star)$, we get

$$\begin{aligned} (\star) &= t - s + 2(W(s))^2 - (W(s))^2 - t \\ &= (W(s))^2 - s. \end{aligned}$$

That is:

$$\mathbb{E}[(W(t))^2 - t | \mathcal{F}_s] = (W(s))^2 - s.$$

By (i),(ii),(iii): $((W(t))^2 - t)_{t \in [0, T]}$ is indeed a martingale. $\square$

Theorem 2.1.12 (Doob's inequality). *Let $(M(t))_{t \in [0, T]}$ be a continuous martingale. Then, for any $p \in (1, \infty)$ we have*

$$\mathbb{E} \left[\sup_{t \in [0, T]} |M(t)|^p \right] \leq \left(\frac{p}{p-1} \right)^p \mathbb{E} [|M(T)|^p]$$

2.2 Construction of the Wiener Process (Level 5 only)

In this section we construct the Wiener process. We will need the Haar basis: Let us define

$$h_0(t) = \mathbf{1}_{[0, 1]}, \quad h_1(t) = \mathbf{1}_{[0, 1/2]}(t) - \mathbf{1}_{(1/2, 1]}(t),$$

and for k such that $2^n \leq k < 2^{n+1}$ we set

$$h_k(t) = 2^{n/2} \mathbf{1}_{[\frac{k-2^n}{2^n}, \frac{k-2^n+2^{-1}}{2^n}]}(t) - 2^{n/2} \mathbf{1}_{(\frac{k-2^n+2^{-1}}{2^n}, \frac{k-2^n+1}{2^n}]}(t)$$

Lemma 2.2.1. *The family $(h_k)_{k=1}^\infty$ is an orthonormal basis of $L_2([0, 1])$.*

Next, let us set

$$\xi_k(t) = \int_0^t h_k(s) ds,$$

and notice that for $2^n \leq k < 2^{n+1}$, we have

$$\sup_{t \in [0, 1]} |\xi_k(t)| = 2^{-n/2-1}.$$

We want to define

$$W(t) = \sum_{k=0}^{\infty} A_k \xi_k(t), \tag{2.1}$$

where $(A_k)_{k=0}^\infty$ are independent $\mathcal{N}(0, 1)$ random variables, and to show that $(W(t))_{t \in [0, T]}$ is a Wiener process. First of all, we need to show that the series in (2.1) converges a.s. Let us start with a convergence result in the case where the coefficients are deterministic.

Lemma 2.2.2. *Let $(a_k)_{k=1}^\infty$ be a sequence of real numbers such that $a_k = O(k^\delta)$ as $k \rightarrow \infty$, for some $\delta \in [0, 1/2)$. Then the series $\sum_{k=0}^\infty a_k \xi_k(t)$ converges uniformly in $t \in [0, 1]$.*

Proof. Notice that for $k \neq j$, with $2^n \leq k, j < 2^{n+1}$, the functions ξ_k and ξ_j have disjoint supports. Moreover, notice that by assumption, there exists a constant C such that $\max_{2^n \leq k < 2^{n+1}} |a_k| \leq C(2^{n+1})^\delta$. Let $\varepsilon > 0$. For any $t \in [0, 1]$ we have

$$\begin{aligned} \sum_{k=2^m}^{\infty} |a_k| |\xi_k(t)| &= \sum_{n=m}^{\infty} \sum_{k=2^n}^{2^{n+1}-1} |a_k| |\xi_k(t)| \leq \sum_{n=m}^{\infty} \left(\max_{2^n \leq k < 2^{n+1}} |a_k| \right) \left(\max_{\substack{2^n \leq k < 2^{n+1} \\ t \in [0, 1]}} |\xi_k(t)| \right) \\ &\leq \sum_{n=m}^{\infty} C(2^{n+1})^\delta 2^{-n/2-1} \leq \varepsilon, \end{aligned}$$

for m large enough, since $\delta < 1/2$. $\square$

Lemma 2.2.3. *Let $(A_k)_{k=0}^{\infty}$ be a sequence of independent $\mathcal{N}(0, 1)$ random variables. Then, almost surely, $|A_k| = O(\sqrt{\log k})$, as $k \rightarrow \infty$.*

Proof. For $r > 0$ we have

$$\mathbb{P}(|A_k| > x) = \frac{1}{\sqrt{2\pi}} \int_r^{\infty} e^{-\frac{s^2}{2}} dr \leq \frac{1}{\sqrt{2\pi}} e^{-\frac{r^2}{4}} \int_r^{\infty} e^{-\frac{s^2}{4}} dr \leq C e^{-\frac{r^2}{4}}.$$

Applying this with $r = 4\sqrt{\log k}$, we get $\mathbb{P}(|A_k| > 4\sqrt{\log k}) \leq C \frac{1}{k^4}$, which in particular implies that

$$\sum_{k=1}^{\infty} \mathbb{P}(|A_k| > 4\sqrt{\log k}) < \infty.$$

Consequently, by the Borel-Cantelli lemma, we get $\mathbb{P}(|A_k| > 4\sqrt{\log k} \text{ i. o. }) = 0$, which means that for a.e. $\omega \in \Omega$, there exists $K(\omega)$ such that $|A_k(\omega)| \leq 4\sqrt{\log k}$, for all $k \geq K(\omega)$. $\square$

Lemma 2.2.4. *For $s, t \in [0, 1]$, we have $\sum_{k=0}^{\infty} \xi_k(s) \xi_k(t) = s \wedge t$.*

Proof. Notice that $\mathbf{1}_{[0,s]}, \mathbf{1}_{[0,t]} \in L_2([0, 1])$. By Lemma 2.2.1 we have that

$$\int_0^1 \mathbf{1}_{[0,s]}(r) \mathbf{1}_{[0,t]}(r) dr = \sum_{k=0}^{\infty} a_k b_k,$$

where

$$a_k = \int_0^1 \mathbf{1}_{[0,s]}(r) h_k(r) dr = \xi_k(s), \quad b_k = \int_0^1 \mathbf{1}_{[0,t]}(r) h_k(r) dr = \xi_k(t).$$

The result now follows, since $\int_0^1 \mathbf{1}_{[0,s]}(r) \mathbf{1}_{[0,t]}(r) dr = s \wedge t$. $\square$

Theorem 2.2.5. *Let $(A_k)_{k=0}^{\infty}$ be a sequence of independent $\mathcal{N}(0, 1)$ random variables. Then, the sum $W(t) = \sum_{k=0}^{\infty} A_k \xi_k(t)$ converges uniformly in $t \in [0, 1]$ almost surely and $(W(t))_{t \in [0, 1]}$ is a Wiener process.*

Proof. The fact that $W_0 = 0$ follows easily since $\xi_k(0) = 0$ for all $k \in \mathbb{N}$. The convergence of the series is a consequence of Lemma 2.2.3 and Lemma 2.2.2. The fact that $(W(t))_{t \in [0, 1]}$ is a continuous process, follows from the fact for each $m \in \mathbb{N}$, the process $W_n(t) = \sum_{k=0}^n A_k \xi_k(t)$ is

continuous in t and that, almost surely, $\sup_{t \in [0,1]} |W_n(t) - W(t)| \rightarrow 0$ as $n \rightarrow \infty$ (we can simply set $W(\omega, \cdot) \equiv 0$ for ω in the null set where the convergence fails). Finally, we show that for all $0 \leq t_1 < t_2 < \dots < t_n \leq 1$, the random variables $W(t_2) - W(t_1), \dots, W(t_{n+1}) - W(t_n)$ are independent and that $W(t_{j+1}) - W(t_j) \sim \mathcal{N}(0, |t_{j+1} - t_j|)$. For this, it suffices to show that for $\lambda_1, \dots, \lambda_n \in \mathbb{R}$

$$\mathbb{E} \left[\exp \left(i \sum_{j=1}^n \lambda_j (W(t_{j+1}) - W(t_j)) \right) \right] = \prod_{j=1}^n \exp \left(-\frac{\lambda_j^2}{2} (t_{j+1} - t_j) \right).$$

We have

$$\begin{aligned} & \mathbb{E} \left[\exp \left(i \sum_{j=1}^n \lambda_j (W(t_{j+1}) - W(t_j)) \right) \right] \\ &= \mathbb{E} \left[\exp \left(i \sum_{k=1}^{\infty} A_k \sum_{j=1}^n \lambda_j (\xi_k(t_{j+1}) - \xi_k(t_j)) \right) \right] \\ &= \prod_{k=0}^{\infty} \exp \left(-\frac{\left(\sum_{j=1}^n \lambda_j (\xi_k(t_{j+1}) - \xi_k(t_j)) \right)^2}{2} \right) \\ &= \prod_{k=0}^{\infty} \exp \left(-\frac{\left(\sum_{j=1}^n \lambda_j (\xi_k(t_{j+1}) - \xi_k(t_j)) \right)^2}{2} \right) \\ &= \exp \left(-\frac{\left(\sum_{j,l=1}^n \lambda_j \lambda_l \sum_{k=0}^{\infty} (\xi_k(t_{j+1}) - \xi_k(t_j)) (\xi_k(t_{l+1}) - \xi_k(t_l)) \right)}{2} \right) \\ &= \exp \left(-\frac{\left(\sum_{j,l=1}^n \lambda_j \lambda_l (t_{j+1} \wedge t_{l+1} - t_{j+1} \wedge t_l - t_j \wedge t_{l+1} + t_j \wedge t_l) \right)}{2} \right). \end{aligned}$$

Notice that if $j \neq l$, then $t_{j+1} \wedge t_{l+1} - t_{j+1} \wedge t_l - t_j \wedge t_{l+1} + t_j \wedge t_l = 0$, which now gives,

$$\mathbb{E} \left[\exp \left(i \sum_{j=1}^n \lambda_j (W(t_{j+1}) - W(t_j)) \right) \right] = \exp \left(-\frac{\sum_{j=1}^n \lambda_j^2 (t_{j+1} - t_j)}{2} \right),$$

which is exactly what we wanted to prove. $\square$

We have now constructed a Wiener process on $[0, 1]$. We can now construct one on the half line $[0, \infty)$ (hence also on $[0, T]$), by "stitching" together independent Wiener processes defined on $[0, 1]$. For this, for $n \in \mathbb{N}$, let $(W_n(t))_{t \in [0,1]}$ be independent Wiener processes. Then, we set $W(t) = W_1(t)$ for $t \in [0, 1]$ and inductively for $n \geq 1$,

$$W(t) = W(n) + W_{n+1}(t - n), \quad \text{for } t \in (n, n + 1].$$

The process W defined above is a Wiener Process on $[0, \infty)$.

2.3 Properties of the trajectories

In this section we study how the Wiener process behaves as a function of time.

Definition 2.3.1. Let $X, \tilde{X} : \Omega \times [0, T] \rightarrow \mathbb{R}$ be stochastic processes. We call $\tilde{X}$ a modification of X if for all $t \in [0, T]$ we have $\mathbb{P}(\tilde{X}(t) = X(t)) = 1$.

Definition 2.3.2. Let $\alpha \in (0, 1]$. A function $f : [0, T] \rightarrow \mathbb{R}$ is called α -Hölder continuous if there exists a constant C_f such that $|f(t) - f(s)| \leq C_f |t - s|^\alpha$ for all $t, s \in [0, T]$. The collection of all α -Hölder continuous functions on $[0, T]$ will be denoted by $\mathcal{C}^\alpha$.

Theorem 2.3.3 (Kolmogorov). *Let $X : \Omega \times [0, T] \rightarrow \mathbb{R}$ be a process such that there exists positive constants α, K and β , such that*

$$\mathbb{E}[|X(t) - X(s)|^\alpha] \leq K |t - s|^{1+\beta}$$

for all $s, t \in [0, T]$. Then, there exists a continuous modification $\tilde{X} : \Omega \times [0, T] \rightarrow \mathbb{R}$ of X such that for all $\gamma < \beta/\alpha$, we have $\mathbb{P}(\tilde{X} \in \mathcal{C}^\gamma) = 1$. If X is continuous then it holds that $\mathbb{P}(X(t) = \tilde{X}(t), \forall t \in [0, T]) = 1$ and for all $\gamma < \beta/\alpha$

$$\mathbb{P}(X \in \mathcal{C}^\gamma) = 1.$$

We can apply this to see that the Wiener process is γ -Hölder continuous for all $\gamma < 1/2$. Indeed, since $W(t) - W(s) \sim \mathcal{N}(0, |t - s|)$ for $s < t$, we have that

$$\begin{aligned} \mathbb{E}[|W(t) - W(s)|^{2m}] &= \frac{1}{\sqrt{2\pi|t-s|}} \int_0^\infty |x|^{2m} e^{-\frac{x^2}{|t-s|}} dx \\ &= \frac{1}{\sqrt{2\pi|t-s|}} \int_0^\infty |x|^{2m} e^{-\frac{x^2}{|t-s|}} dx \\ &= \frac{|t-s|^m}{\sqrt{2\pi|t-s|}} \int_0^\infty \left(\frac{|x|^2}{|t-s|}\right)^m e^{-\frac{x^2}{|t-s|}} dx \\ &= C_m |t-s|^m. \end{aligned}$$

From Kolmogorov's theorem we have that W is Hölder continuous with exponent $\gamma < (m-1)/2m$. But m can be arbitrarily large, hence γ can be arbitrarily close to $1/2$.

Theorem 2.3.4 (Quadratic Variation). *Let $s < t$ and let $(t_i^n)_{i=1}^{m(n)}$, for $n \in \mathbb{N}$ be a sequence of partitions of $[s, t]$ such that $\max_{i \leq m(n)} |t_{i+1}^n - t_i^n| \rightarrow 0$ as $n \rightarrow \infty$. Then,*

$$\sum_{i=1}^{m(n)-1} |W(t_{i+1}^n) - W(t_i^n)|^2 \rightarrow (t - s),$$

in $L_2(\Omega)$, as $n \rightarrow \infty$.

Proof. We have

$$\mathbb{E} \left[\left(\sum_{i=1}^{m(n)-1} |W(t_{i+1}^n) - W(t_i^n)|^2 - (t - s) \right)^2 \right]$$

$$\begin{aligned}
&= \mathbb{E} \left[\left(\sum_{i=1}^{m(n)-1} \left(|W(t_{i+1}^n) - W(t_i^n)|^2 - (t_{i+1}^n - t_i^n) \right) \right)^2 \right] \\
&= \sum_{i=1}^{m(n)-1} \mathbb{E} \left[\left(|W(t_{i+1}^n) - W(t_i^n)|^2 - (t_{i+1}^n - t_i^n) \right)^2 \right] \\
&\leq 4 \sum_{i=1}^{m(n)-1} \mathbb{E} \left[|W(t_{i+1}^n) - W(t_i^n)|^4 + (t_{i+1}^n - t_i^n)^2 \right] \\
&\leq N \sum_{i=1}^{m(n)-1} (t_{i+1}^n - t_i^n)^2 \\
&\leq N \max_i |t_{i+1}^n - t_i^n| \sum_{i=1}^{m(n)-1} (t_{i+1}^n - t_i^n) \\
&= N \max_i |t_{i+1}^n - t_i^n| |t - s| \rightarrow 0,
\end{aligned}$$

as $n \rightarrow \infty$. For the second equality we have used that if $i \neq j$, then

$$\mathbb{E} \left[\left(|W(t_{i+1}^n) - W(t_i^n)|^2 - (t_{i+1}^n - t_i^n) \right) \left(|W(t_{j+1}^n) - W(t_j^n)|^2 - (t_{j+1}^n - t_j^n) \right) \right] = 0.$$

This shows that

$$\sum_{i=1}^{m(n)-1} |W(t_{i+1}^n) - W(t_i^n)|^2 \rightarrow (t - s),$$

in $L_2(\Omega)$ as $n \rightarrow \infty$. □

Corollary 2.3.5. *Let $(W(t))_{t \in [0, T]}$ be a Wiener process. Then $\mathbb{P}(\text{Var}_{[0, T]} W = \infty) = 1$.*

Proof. Assume the contrary, that is, $\mathbb{P}(\text{Var}_{[0, 1]} W < \infty) > 0$. From Theorem 2.3.4, it follows that there exists a sequence of partitions $[0, 1] (t_i^n)_{i=1}^{m(n)}$, for $n \in \mathbb{N}$, such that $\max_i |t_{i+1}^n - t_i^n| \rightarrow 0$ as $n \rightarrow \infty$ and

$$\sum_{i=1}^{m(n)-1} |W(t_{i+1}^n) - W(t_i^n)|^2 \rightarrow 1, \tag{2.2}$$

almost surely. On the other hand, on the event $\{\text{Var}_{[0, 1]} W < \infty\}$ we have

$$\sum_{i=1}^{m(n)-1} |W(t_{i+1}^n) - W(t_i^n)|^2 \leq \max_i |W(t_{i+1}^n) - W(t_i^n)| \text{Var}_{[0, 1]} W \rightarrow 0,$$

which contradicts (2.2). □

A consequence of Corollary 2.3.5 is that with probability one $W \notin C^1([0, 1])$. Below we will prove something stronger, namely, we will show that with probability one, W is *nowhere* differentiable on $[0, 1]$.

Theorem 2.3.6 (Nowhere differentiability). *Let $(W(t))_{t \in [0, 1]}$ be a Wiener process. Then with probability one, W is nowhere differentiable on $[0, 1]$.*

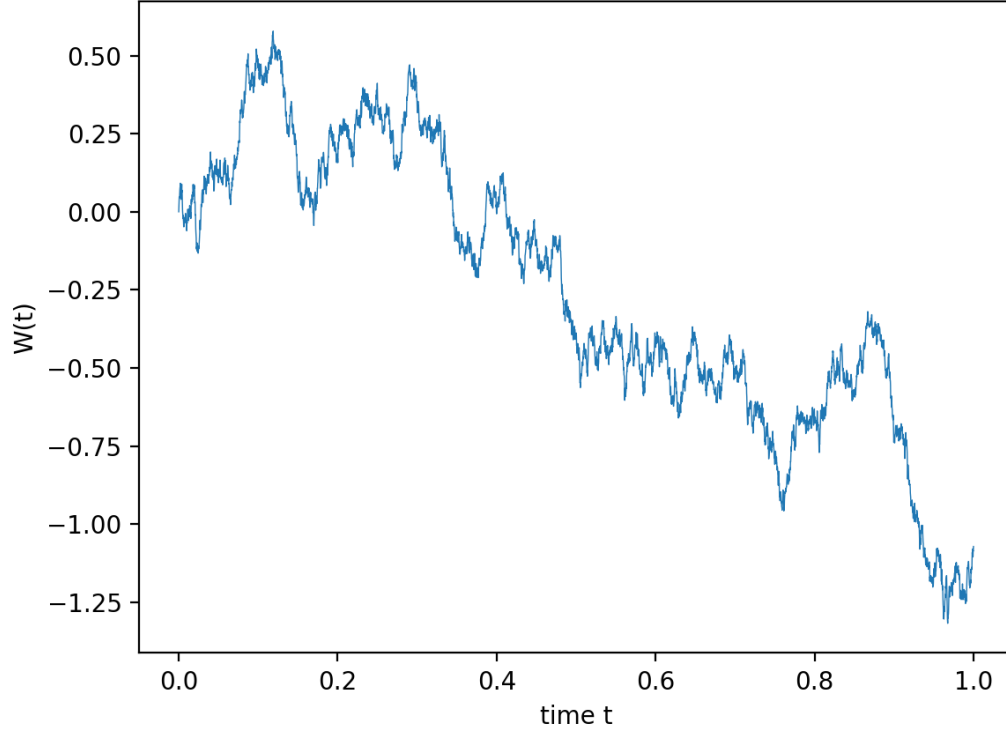


Figure 1: A typical Brownian path

Proof. For $h > 0$ and $M, n \in \mathbb{N}$ let us set

$$A_{M,h} := \{ \exists s \in [0, 1] \text{ s.t. } |W(t) - W(s)| \leq M|t - s| \text{ for all } t \in [0, 1] \text{ with } |t - s| \leq h \}.$$

and

$$B_n := \{ \exists k \in \{0, \dots, n\} \text{ s.t. } |W(\frac{k+1+i}{n}) - W(\frac{k+i}{n})| \leq 6Mn^{-1} \text{ for all } i = 0, 1, 2 \}.$$

We claim that $A_{M,h} \subset B_n$ for $n \geq 3/h$. Indeed, if $\omega \in A_{M,h}$ and $n \geq 3/h$, then there exists $s \in [0, 1]$ such that $|W(\omega, t) - W(\omega, s)| \leq M|t - s|$ for all $|t - s| \leq 3n^{-1}$. Let us set $k = [ns]$, and notice that $|(k+j)n^{-1} - s| \leq 3n^{-1}$ for $j \in \{0, 1, 2, 3\}$. Consequently, for $i = 0, 1, 2$ we have

$$\begin{aligned} & |W(\omega, \frac{k+1+i}{n}) - W(\omega, \frac{k+i}{n})| \\ & \leq |W(\omega, \frac{k+1+i}{n}) - W(\omega, s)| + |W(\omega, s) - W(\omega, \frac{k+i}{n})| \\ & \leq M|\frac{k+1+i}{n} - s| + M|s - \frac{k+i}{n}| \leq 6Mn^{-1}, \end{aligned}$$

which shows that $\omega \in B_n$, and consequently, $A_{M,h} \subset B_n$. Next, we see that

$$\mathbb{P}(B_n) \leq (n+1) \sup_{k \in \{0, \dots, n\}} \mathbb{P}(|W(\frac{k+i}{n}) - W(\frac{k+i-1}{n})| \leq 6Mn^{-1}, \text{ for } i = 0, 1, 2)$$

$$= (n+1) \left(\mathbb{P}(|W(\frac{1}{n})| \leq 6Mn^{-1}) \right)^3,$$

where we have used the independence and stationarity of the increments. Notice that by Exercise 2.1.4, we have

$$\mathbb{P}(|W(\frac{1}{n})| \leq 6Mn^{-1}) = \mathbb{P}(|W(1)| \leq 6Mn^{-1/2}) = 2 \int_0^{6Mn^{-1/2}} \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx \lesssim NMn^{-1/2}.$$

Consequently,

$$\mathbb{P}(B_n) \lesssim M(n+1)n^{-3/2} \lesssim Mn^{-1/2} \rightarrow 0,$$

as $n \rightarrow \infty$. This implies that for all $M \in \mathbb{N}$, $h > 0$, we have $\mathbb{P}(A_{M,h}) = 0$. Finally, notice that

$$\{\exists s \in [0, 1] \text{ s.t. } W \text{ is differentiable at } s\} \subset \bigcup_{M \in \mathbb{N}} \bigcup_{m \in \mathbb{N}} A_{M, \frac{1}{m}},$$

and the latter has obviously probability zero. □

3 Itô Calculus

3.1 Motivation

In this section we will define the stochastic integral, the class of Itô processes, and the calculus developed for the study of these processes.

Itô calculus is the main tool for studying stochastic differential equations. Let us start with some motivation. In Figure 2 below is the graph of a trajectory of the dynamical system

$$X(t_{n+1}) = X(t_n) + (t_{n+1} - t_n)f(X(t_n)), \quad X(0) = x,$$

with $f(x) = 6x$, starting from $x = 0.1$, and time-step given by $t_{n+1} - t_n = 1/100$ (interpolated between the discrete time points). This type of dynamics is a good approximation of the evolution

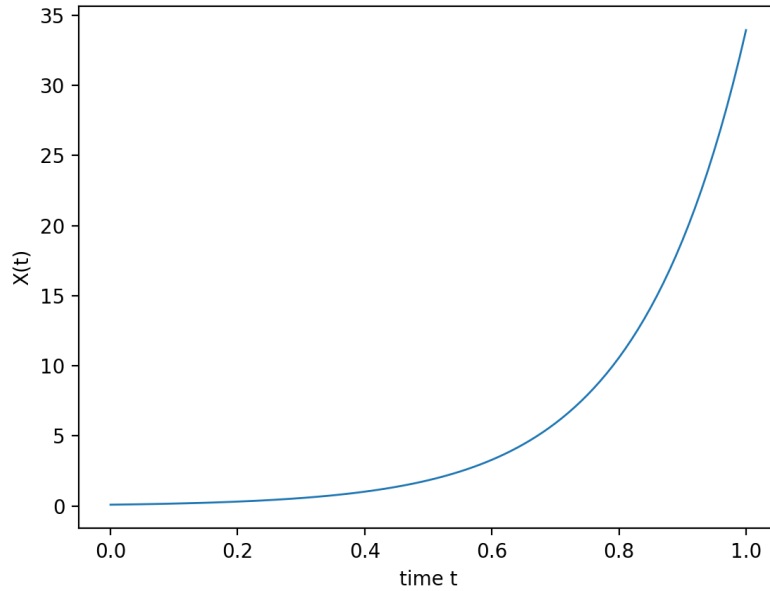


Figure 2: "Smooth" trajectory

of many quantities found in nature. However, in practice, the majority of systems in physics, finance, and biology, among others, are *noisy*. Let us look for example a stock price, and more specifically, let's say the stock price of Apple the last 10 years. While there is obviously an exponential trend, the price does not follow a "smooth" trajectory as in Figure 2 but rather fluctuates around it as shown in Figure 3. The dynamics that we used in order to produce the path shown in Figure 3 are

$$X(t_{n+1}) = X(t_n) + (t_{n+1} - t_n)f(X(t_n)) + \sigma(X(t_n))\xi_n, \quad X(0) = x,$$

where ξ_n are independent random variables with distribution $\mathcal{N}(0, 1/n)$ and $\sigma(x) = x$. Notice that ξ_n can be modelled by Brownian increments $W(t_{n+1}) - W(t_n)$. Hence, by replacing ξ_n with $W(t_{n+1}) - W(t_n)$, upon rearrangement, we get

$$\frac{X(t_{n+1}) - X(t_n)}{t_{n+1} - t_n} = f(X(t_n)) + \sigma(X(t_n)) \frac{W(t_{n+1}) - W(t_n)}{t_{n+1} - t_n}$$

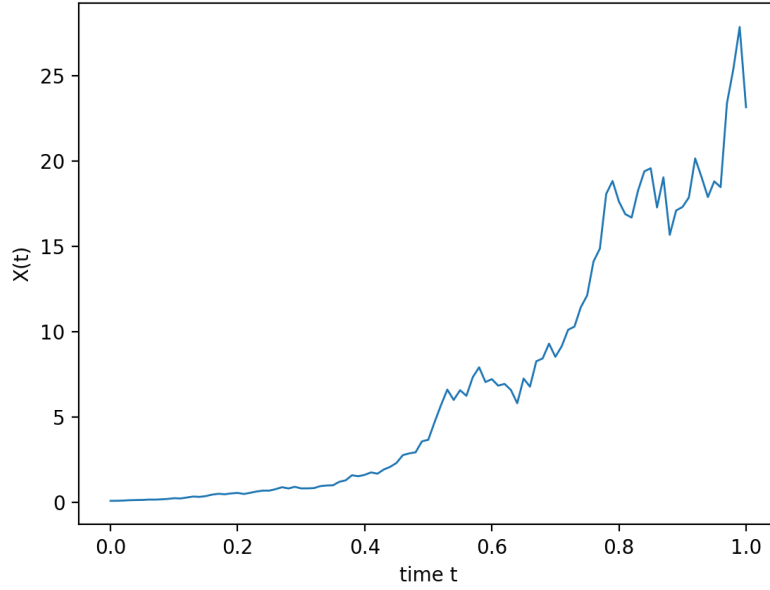


Figure 3: Noisy trajectory

The natural question now is what happens when $|t_{n+1} - t_n| \rightarrow 0$? Of course, when we are in the situation of Figure 2, that is, when $\sigma \equiv 0$ the above system leads to

$$\frac{dX(t)}{dt} = f(X(t)), \quad X(0) = x \quad (3.1)$$

What happens however if $\sigma \neq 0$? Formally, we have

$$\frac{dX(t)}{dt} = f(X(t)) + \sigma(X(t)) \frac{dW(t)}{dt}, \quad X(0) = x.$$

But now there is a problem. We have proved in Theorem 2.3.6 that with probability one, the derivative of $W(t)$ does not exist at any $t \geq 0$! Is there a way to overcome this problem? Lets see first a simple case where $\sigma(X(t))$ is a constant, lets say, $\sigma(x) \equiv 1$. Then we can integrate the above equation to get

$$X(t) = x + \int_0^t f(X(s)) ds + W(t). \quad (3.2)$$

Notice that this makes perfect sense as an equation. If we do the same for non-constant σ , then we arrive at

$$X(t) = x + \int_0^t f(X(s)) ds + \int_0^t \sigma(X(s)) dW(s). \quad (3.3)$$

The question now is: what is the meaning of $\int_0^t \sigma(X(s)) dW(s)$? Let us recall, more generally, when we can define integrals of the form $\int_0^1 g(s) dv(s)$, where g is a continuous function. Obviously,

if $v \in C^1$, then the above integral is defined and is given by

$$\int_0^1 g(s) dv(s) = \int_0^1 g(s) \frac{dv(s)}{ds} ds.$$

However, as it was already discussed, $W \notin C^1$ with probability one. One can relax the condition for v . Actually, for partitions $(t_i^n)_{i=1}^n$ of $[0, 1]$, $n \in \mathbb{N}$, with $\max_i |t_{i+1}^n - t_i^n| \rightarrow 0$ as $n \rightarrow \infty$, the partial sums

$$\sum_{i=1}^n g(t_i^n)(v(t_{i+1}^n) - v(t_i^n))$$

converge for all $g \in C$ if $v \in BV$. In this case the limit (which does not depend on the sequence of partitions) is known as Stieljes integral. Unfortunately, we have shown in Corollary 2.3.5 that with probability one $W \notin BV$. Actually, there is one case that we define the integral of g with respect to v even if v is not of bounded variation. Namely, one needs that $g \in C^\alpha$, $\alpha \in (0, 1)$ and $v \in C^\beta$ with $\beta + \alpha > 1$. This is known as Young integral. However, there is still a problem. We have shown that the paths of the Wiener process are $(1/2 - \varepsilon)$ -Hölder continuous, for any $\varepsilon > 0$, and this is actually quite sharp (one can show similarly to Theorem 2.3.6 that for $\delta > 0$, with probability one, the paths of W are *nowhere* $(1/2 + \delta)$ -Hölder continuous). The solution $X(t)$ of (3.3) (whatever this means) can not have, in general, regularity better than the regularity of W itself (look at the trivial case $f = 0$, $g = 1$) which means that in $\sigma(X.)$ can not be better than $(1/2 - \varepsilon)$ -Hölder continuous. Notice that the regularities of $\sigma(X.)$ and W sum up to $1 - 2\varepsilon < 1$ for any $\varepsilon > 0$. Therefore, we can not define the integral of $\sigma(X.)$ with respect to W as an Young integral. Long story short: If we fix $\omega \in \Omega$ and apply classical analysis techniques with the path $t \mapsto W(\omega, t)$, then with probability one we can not have an integral powerful enough to deal with (3.3)). This dictates that we should try to take advantage of the probabilistic structure of W . This is done in the next section.

3.2 Itô's stochastic integral

In this section, we will construct the integral

$$\int_0^t Y(s) dW(s).$$

as a stochastic process. We fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ with a complete and right continuous filtration $\mathbb{F} := (\mathcal{F}_t)_{t \in [0, T]}$. We assume that on Ω we are given an $\mathbb{F}$ -Wiener process W .

As in any integration theory, we will start with the stochastic integral for a rather simple class of integrands. It is very important that these integrands "can not see" the future of the Wiener process.

Definition 3.2.1. We say that a stochastic process $Y : \Omega \times [0, T] \rightarrow \mathbb{R}$ is in the class $\mathcal{H}_T^{step}$ if there exists $n \in \mathbb{N}$ and $0 \leq t_1 < \dots < t_n < t_{n+1} \leq T$ such that

1. $Y(t) = \sum_{i=0}^n Y_{t_i} \mathbf{1}_{[t_i, t_{i+1})}(t)$,
2. Y_{t_i} is $\mathcal{F}_{t_i}$ -measurable for all $i = 1, \dots, n$,
3. $\|Y_{t_i}\|_{L_\infty} < \infty$ for all $i = 1, \dots, n$.

The elements of the class $\mathcal{H}_T^{step}$ will be called *simple processes*.

For $Y \in \mathcal{H}_T^{step}$ we set

$$I(Y) = \int_0^T Y(s) dW(s) := \sum_{i=0}^n Y_{t_i} (W(t_{i+1}) - W(t_i)) \quad (3.4)$$

Remark 3.2.2. Notice that the stochastic integral $I(Y)$ is well defined (does not depend on the partition $t_1, \dots, t_n$) and is a linear operator on $\mathcal{H}_T^{step}$. Also notice that for $Y \in \mathcal{H}_T^{step}$ we have that $I(Y) \in L_2(\Omega)$. Indeed, it suffices to check that this is the case for $Y(t) = Y_{t_1} \mathbf{1}_{[t_1, t_{1+1})}$. From Proposition 1.5.7 we have

$$\begin{aligned} \mathbb{E} \left[|Y_{t_1}|^2 (W(t_2) - W(t_1))^2 \right] &= \mathbb{E} \left[\mathbb{E} \left[|Y_{t_1}|^2 (W(t_2) - W(t_1))^2 \middle| \mathcal{F}_{t_1} \right] \right] \\ &= \mathbb{E} \left[|Y_{t_1}|^2 \mathbb{E} \left[(W(t_2) - W(t_1))^2 \middle| \mathcal{F}_{t_1} \right] \right] \\ &= (t_2 - t_1) \mathbb{E} [|Y_{t_1}|^2] \leq (t_2 - t_1) \|Y_{t_1}\|_{L_\infty}^2 < \infty, \end{aligned}$$

which shows that $I(Y) \in L_2(\Omega)$.

Exercise 3.2.3. Show that the stochastic integral $I : \mathcal{H}_T^{step} \rightarrow L_2(\Omega)$ is a linear operator, that is, if $Y_1, Y_2 \in \mathcal{H}_T$ and $a_1, a_2 \in \mathbb{R}$, then we have that $I(a_1 Y_1 + a_2 Y_2) = a_1 I(Y_1) + a_2 I(Y_2)$.

Lemma 3.2.4. For $Y \in \mathcal{H}_T^{step}$ we have

$$\mathbb{E} [|I(Y)|^2] = \mathbb{E} \left[\int_0^T |Y(s)|^2 ds \right], \quad (3.5)$$

$$\mathbb{E} [I(Y)] = 0. \quad (3.6)$$

Proof. We start with (3.5). Suppose that Y has the representation $Y(t) = \sum_{i=0}^n Y_{t_i} \mathbf{1}_{[t_i, t_{i+1})}(t)$. First of all, notice that

$$\begin{aligned} \mathbb{E} [Y_{t_i}^2 (W(t_i) - W(t_{i+1}))^2] &= \mathbb{E} [Y_{t_i}^2 \mathbb{E} ((W(t_i) - W(t_{i+1}))^2 | \mathcal{F}_{t_i})] \\ &= (t_{i+1} - t_i) \mathbb{E} [|Y_{t_i}|^2], \end{aligned} \quad (3.7)$$

where we have used that $W(t_i) - W(t_{i+1})$ is independent from $\mathcal{F}_{t_i}$, which in particular implies that $\mathbb{E} [(W(t_i) - W(t_{i+1}))^2 | \mathcal{F}_{t_i}] = \mathbb{E} [(W(t_i) - W(t_{i+1}))^2] = (t_{i+1} - t_i)$.

If $i < j$, then we have

$$\begin{aligned} &\mathbb{E} [Y_{t_i} (W(t_i) - W(t_{i+1})) Y_{t_j} (W(t_j) - W(t_{j+1}))] \\ &= \mathbb{E} [Y_{t_i} (W(t_i) - W(t_{i+1})) Y_{t_j} \mathbb{E} ((W(t_j) - W(t_{j+1})) | \mathcal{F}_{t_j})] = 0 \end{aligned} \quad (3.8)$$

Consequently, we have

$$\begin{aligned} &\mathbb{E} [|I(Y)|^2] \\ &= \mathbb{E} \left[\left(\sum_{i=0}^n Y_{t_i} (W(t_{i+1}) - W(t_i)) \right)^2 \right] \\ &= \sum_{i=0}^n \mathbb{E} [Y_{t_i}^2 (W(t_{i+1}) - W(t_i))^2] + \sum_{i \neq j} \mathbb{E} [Y_{t_i} (W(t_{i+1}) - W(t_i)) Y_{t_j} (W(t_{j+1}) - W(t_j))] \end{aligned}$$

$$= \sum_{i=0}^n (t_{i_1} - t_i) \mathbb{E}[|Y_{t_i}|^2] = \mathbb{E} \left[\sum_{i=0}^n (t_{i_1} - t_i) |Y_{t_i}|^2 \right] = \mathbb{E} \left[\int_0^T |Y(s)|^2 ds \right]$$

The equality in (3.6) follows from

$$\mathbb{E}[Y_{t_i}(W(t_{i+1}) - W(t_i))] = \mathbb{E}[Y_{t_i} \mathbb{E}[(W(t_{i+1}) - W(t_i)) | \mathcal{F}_{t_i}]] = 0.$$

□

The equality in (3.5) is known as *Itô's Isometry* because it shows that if we see $\mathcal{H}_T^{step}$ as a subspace of $L_2(\Omega \times \mathbb{R}_+)$, then the stochastic integral $I : \mathcal{H}_T^{step} \rightarrow L_2(\Omega)$ is an isometry.

Definition 3.2.5. For $Y \in \mathcal{H}_T^{step}$, let us define for $t \in [0, T]$

$$I(Y, t) = \int_0^t Y(s) dW(s) := I(Y \mathbf{1}_{[0, t)}).$$

Lemma 3.2.6. *The process $(I(Y, t))_{t \in [0, T]}$ is a continuous martingale with respect to $\mathbb{F}$.*

Proof. Since sum of continuous martingales is a continuous martingale, it suffices to prove the claim for Y of the form

$$Y(t) = Y_{t_1} \mathbf{1}_{[t_1, t_2)}(t).$$

First, we have for each $t \in [0, T]$ that by Remark 3.2.2 we have $I(Y, t) \in L_2(\Omega) \subset L_1(\Omega)$. Let us check that $I(Y, t)$ is $\mathcal{F}_t$ -measurable. We clearly have

$$I(Y, t) = Y_{t_1}(W(t \wedge t_2) - W(t \wedge t_1)).$$

Notice that if $t \leq t_1$ the above expression is 0, hence $\mathcal{F}_t$ -measurable. If $t > t_1$, then clearly the above expression is $\mathcal{F}_t$ -measurable.

Finally, let $s < t \leq T$ and let us show that $\mathbb{E}[I(Y, t) | \mathcal{F}_s] = I(Y, s)$. There are three cases, namely, $s < t \leq t_1$, $s \leq t_1 < t$, or $t_1 < s < t$. In the first case, we have that $I(Y, t) = I(Y, s) = 0$, which in particular implies that

$$\mathbb{E}[I(Y, t) | \mathcal{F}_s] = 0 = I(Y, s).$$

In the second case, notice that still $I_s(Y) = 0$. Moreover, we have

$$\begin{aligned} \mathbb{E}[I(Y, t) | \mathcal{F}_s] &= \mathbb{E}[Y_{t_1}(W(t \wedge t_2) - W(t_1)) | \mathcal{F}_s] = \mathbb{E}[\mathbb{E}[Y_{t_1}(W(t \wedge t_2) - W(t_1)) | \mathcal{F}_{t_1}] | \mathcal{F}_s] \\ &= \mathbb{E}[Y_{t_1} \mathbb{E}[W(t \wedge t_2) - W(t \wedge t_1)] | \mathcal{F}_s] = 0 = I_s(Y). \end{aligned}$$

Finally, for the third case, we have

$$\begin{aligned} \mathbb{E}[I(Y, t) | \mathcal{F}_s] &= \mathbb{E}[Y_{t_1}(W(t \wedge t_2) - W(t_1)) | \mathcal{F}_s] \\ &= Y_{t_1} \mathbb{E}[(W(t \wedge t_2) - W(t_1)) | \mathcal{F}_s] = Y_{t_1}(W_{s \wedge t_2} - W(t_1)) = I_s(Y). \end{aligned}$$

This finishes the proof. □

Lemma 3.2.7. *For $Y \in \mathcal{H}_T^{step}$ we have*

$$\mathbb{E} \left[\sup_{t \in [0, T]} |I(Y, t)|^2 \right] \leq 4 \mathbb{E} \left[\int_0^T |Y(s)|^2 ds \right].$$

Proof. This is a direct consequence of Doob's inequality. Indeed, since the stochastic integral is a continuous martingale, we have by Theorem 2.1.12 and Lemma 3.2.4

$$\mathbb{E} \left[\sup_{t \in [0, T]} |I(Y, t)|^2 \right] \leq 4\mathbb{E} [|I(Y, T)|^2] = 4\mathbb{E} \left[\int_0^T |Y(s)|^2 ds \right].$$

□

Definition 3.2.8. We say that a stochastic process $Y : \Omega \times [0, T] \rightarrow \mathbb{R}$ is in the class $\mathcal{H}_T$, if it is adapted to the filtration $\mathbb{F}$ and it satisfies

$$\mathbb{E} \left[\int_0^T |Y(s)|^2 ds \right] < \infty.$$

Lemma 3.2.9. If $Y \in \mathcal{H}_T$ then there exists $Y_n \in \mathcal{H}_T^{step}$, for $n \in \mathbb{N}$ such that

$$\lim_{n \rightarrow \infty} \mathbb{E} \left[\int_0^T |Y(s) - Y_n(s)|^2 ds \right] = 0 \quad (3.9)$$

Proposition 3.2.10. Let $Y \in \mathcal{H}$ and let $(Y_n)_{n=1}^\infty \subset \mathcal{H}_T^{step}$ be the sequence from Lemma 3.2.9. Then there exists a random variable $Z \in L_2(\Omega)$, such that $\lim_{n \rightarrow \infty} \|I(Y_n) - Z\| = 0$.

Exercise 3.2.11. Show that Z does not depend on the approximating sequence $(Y_n)_{n=1}^\infty$. That is, show that if $(\tilde{Y}_n)_{n=1}^\infty \subset \mathcal{H}_T^{step}$ is another sequence satisfying (3.9), then

$$\lim_{n \rightarrow \infty} \|I(\tilde{Y}_n) - Z\|_{L_2} = 0.$$

Solution: Let $(\tilde{Y}_n)_{n=1}^\infty \subset \mathcal{H}_T^{step}$ be as above and let $(Y_n)_{n=1}^\infty \subset \mathcal{H}_T^{step}$ be the sequence from Lemma 3.2.9. Then, we have

$$\begin{aligned} \|I(\tilde{Y}_n) - Z\|_{L_2} &\leq \|I(\tilde{Y}_n) - I(Y_n)\|_{L_2} + \|I(Y_n) - Z\|_{L_2} \\ &= \|I(\tilde{Y}_n - Y_n)\|_{L_2} + \|I(Y_n) - Z\|_{L_2} \\ &= \left(\mathbb{E} \left[\int_0^T (\tilde{Y}_n(s) - Y_n(s))^2 ds \right] \right)^{1/2} + \|I(Y_n) - Z\|_{L_2}, \end{aligned}$$

where we have used the triangle inequality, the additivity of the stochastic integral, and Itô's isometry. Notice that by definition of Z , we have that $\|I(Y_n) - Z\|_{L_2} \rightarrow 0$ as $n \rightarrow \infty$. For the first term on the right hand side of the above inequality, we have by the triangle inequality that

$$\begin{aligned} \left(\mathbb{E} \left[\int_0^T (\tilde{Y}_n(s) - Y_n(s))^2 ds \right] \right)^{1/2} &\leq \left(\mathbb{E} \left[\int_0^T (\tilde{Y}_n(s) - Y(s))^2 ds \right] \right)^{1/2} \\ &\quad + \left(\mathbb{E} \left[\int_0^T (Y(s) - Y_n(s))^2 ds \right] \right)^{1/2}, \end{aligned}$$

and then by our assumption on Y_n and $\tilde{Y}_n$ we have that both terms on the right hand side, and therefore the left hand side too, tend to zero as $n \rightarrow \infty$. Hence, we have shown that $\|I(\tilde{Y}_n) - Z\|_{L_2} \rightarrow 0$ as $n \rightarrow \infty$, which finishes the proof. □

Definition 3.2.12. Let $Y \in \mathcal{H}_T$. We define the stochastic integral of Y to be

$$I(Y) = \int_0^T Y(s) dW(s) := Z,$$

where Z is the random variable from Proposition 3.2.10. Moreover, for $t \in [0, T]$, we set

$$I(Y, t) = I(Y \mathbf{1}_{[0, t]})$$

Lemma 3.2.13. Let $Y \in \mathcal{H}_T$. Then the process $(I(Y, t))_{t \geq 0}$ admits a continuous modification.

Exercise 3.2.14. Suppose that Y is as in Definition 3.2.1 with the only difference that condition (iii) is replaced by $\|Y_{t_i}\|_{L_2} < \infty$. Show that $Y \in \mathcal{H}_T$ and that $I(Y)$ is given (3.4).

From now on, whenever we talk about the stochastic integral of Y , we will always mean its continuous modification. That is, from now on, we assume that $(I(Y, t))_{t \geq 0}$ is a continuous processes.

The next theorem can be easily proved by approximating $Y \in \mathcal{H}_T$ by $Y_n \in \mathcal{H}_T^{step}$, since for the latter, the corresponding conclusions of the theorem have already been proved.

Theorem 3.2.15. Let $Y, Z \in \mathcal{H}_T$ and $a, b \in \mathbb{R}$. Then,

(i) Almost surely,

$$\int_0^t (aY(s) + bZ(s)) dW(s) = a \int_0^t Y(s) dW(s) + b \int_0^t Z(s) dW(s),$$

for all $t \in [0, T]$.

(ii) For all $t \in [0, T]$

$$\mathbb{E} \left[\int_0^T Y(s) dW_s \right] = 0.$$

(iii) (Itô's isometry) For all $t \in [0, T]$

$$\mathbb{E} \left[\left| \int_0^t Y(s) dW(s) \right|^2 \right] = \mathbb{E} \left[\int_0^t |Y(s)|^2 ds \right]$$

(iv) $(I(Y, t))_{t \geq 0}$ is a martingale with respect to $\mathbb{F}$.

(v) Doob's inequality holds,

$$\mathbb{E} \left[\sup_{t \in [0, T]} \left(\int_0^t Y(s) dW_s \right)^2 \right] \leq 4 \mathbb{E} \int_0^T |Y(s)|^2 ds.$$

Exercise 3.2.16. (PG) Let $Y \in \mathcal{H}_T$ and W be a real $\mathbb{F}$ -Wiener process. Show that

$$Z(t) := \left| \int_0^t Y(s) dW(s) \right|^2 - \int_0^t |Y(s)|^2 ds$$

is a martingale.

Hint: Show it first for $Y \in \mathcal{H}_T^{step}$.

Theorem 3.2.17 (Martingale Representation Theorem). *Let $\mathbb{G}$ be the augmentation of the natural filtration of W . Let $(M(t))_{t \leq T}$ be a continuous, martingale with respect to $\mathbb{G}$ such that $M(T) \in L_2(\Omega)$. Then, there exists a unique $(Y(t))_{t \leq T} \in \mathcal{H}_T$, adapted to $\mathbb{G}$, such that for each $t \in [0, T]$, with probability one*

$$M(t) = M_0 + \int_0^t Y(s) dW(s)$$

Exercise 3.2.18. Let Z be a random variable with $\mathbb{E}[|Z|] < \infty$. Show that $M(t) := \mathbb{E}[Z|\mathcal{F}_t]$ is a martingale with respect to $(\mathcal{F}_t)_{t \leq T}$.

Solution: First, notice that $M(t)$ is well-defined because $Z \in L_1(\Omega)$

- By definition of the conditional expectation, $\mathbb{E}[Z|\mathcal{F}_t]$ is $\mathcal{F}_t$ -measurable.
- By the conditional Jensen's inequality we have

$$\mathbb{E}[|\mathbb{E}[Z|\mathcal{F}_t]|] \leq \mathbb{E}[\mathbb{E}[|Z||\mathcal{F}_t]] = \mathbb{E}[|Z|] < \infty,$$

by assumption.

- For $s < t$, we have by the tower property of conditional expectation

$$\mathbb{E}[M(t)|\mathcal{F}_s] = \mathbb{E}[\mathbb{E}[Z|\mathcal{F}_t]|\mathcal{F}_s] = \mathbb{E}[Z|\mathcal{F}_s] = M(s).$$

We have shown that M is a martingale. □

Lemma 3.2.19. *Let $(Y(t))_{t \in [0, T]}$ be a non-negative $\mathbb{F}$ -adapted process. Then, for any $t \geq 0$, $\int_0^t Y(s) ds$ is $\mathcal{F}_t$ -measurable.*

Proof. If the claim holds for $Y \wedge n$, for $n \in \mathbb{N}$, then it also holds for Y from the monotone convergence theorem, Exercise 1.1.13, and the completeness of $\mathcal{F}_t$. Consequently, we may assume that Y is bounded. In this case, we have that $Y \in \mathcal{H}_T$ and it follows that there exists $Y_n \in \mathcal{H}_T^{\text{step}}$ such that

$$\begin{aligned} \mathbb{E} \left[\left| \int_0^t Y(s) ds - \int_0^t Y_n(s) ds \right| \right] &\leq \mathbb{E} \left[\int_0^t |Y(s) - Y_n(s)| ds \right] \\ &\leq \sqrt{t} \left(\mathbb{E} \left[\int_0^T |Y(s) - Y_n(s)|^2 ds \right] \right)^{1/2} \rightarrow 0, \end{aligned}$$

as $n \rightarrow \infty$. In addition, one can easily see that $\int_0^t Y_n(s) ds$ is the sum of $\mathcal{F}_t$ -measurable random variables, hence it is $\mathcal{F}_t$ -measurable. As usual, one can find a subsequence such that almost surely

$$\int_0^t Y_n(s) ds \rightarrow \int_0^t Y(s) ds,$$

and the claim follows from the completeness of $\mathcal{F}_t$ and Exercise 1.1.13. □

3.3 Proofs of Lemmata 3.2.9 and 3.2.13 (Level 5 only)

Proof of Lemma 3.2.9. Let $Y \in \mathcal{H}_T$. We want to show that for every $\varepsilon > 0$, there exists $Z \in \mathcal{H}_T^{step}$ such that

$$\mathbb{E} \int_0^T |Y(s) - Z(s)|^2 ds < \varepsilon.$$

First of all, notice that by the dominated convergence theorem we have that

$$\lim_{C \rightarrow \infty} \mathbb{E} \int_0^T |Y(s) - Y_C(s)|^2 ds = 0,$$

where $Y_C(s) = (Y(s) \wedge C) \vee (-C)$. Consequently we can and we will assume that $|Y(s)| \leq C$ for some C , for all $(\omega, s) \in \Omega \times [0, T]$. For convenience, let us also define $Y(s)$ for s outside of $[0, T]$ simply by extending it to zero. Let $\rho \in C_c((0, 1))$ be a non-negative function with unit integral and set $\rho_n(t) := n\rho(nt)$. It is well known that $\rho_n * Y(\omega) \in C_c^\infty(\mathbb{R})$, and that

$$\lim_{n \rightarrow \infty} \int_0^T |Y(s)(\omega) - (\rho_n * Y(\omega))(s)|^2 ds = 0,$$

for each $\omega \in \Omega$, where

$$(\rho_n * Y(\omega))(t) := \int_{\mathbb{R}} \rho_n(t - s)Y(s)(\omega) ds.$$

Notice that $(\rho_n * Y(t))_{t \geq 0}$ is $\mathbb{F}$ -adapted since ρ_n is supported on $(0, \infty)$. Moreover, since for all $\omega \in \Omega$

$$\int_0^T |Y(s)(\omega) - (\rho_n * Y(\omega))(s)|^2 ds \leq T4C^2,$$

we get by the DC theorem,

$$\lim_{n \rightarrow \infty} \mathbb{E} \int_0^T |Y(s)(\omega) - (\rho_n * Y(\omega))(s)|^2 ds = 0.$$

This further shows that we can assume that for each $\omega \in \Omega$, Y is smooth. To summarise, we can and we will assume that $Y \in \mathcal{H}_T$ is bounded by a constant C and it is smooth. In this case let us set

$$Y_m(t) := \sum_{i=1}^m Y(t_i^m) I_{[t_i^m, t_{i+1}^m)}(t),$$

where $(t_i^m)_{i=1}^m$ are partitions of $[0, T]$ such that $\max_i |t_i^m - t_{i+1}^m| \rightarrow 0$ as $m \rightarrow \infty$. Since, Y is $\mathbb{F}$ -adapted, it follows that $Y_m \in \mathcal{H}_T^{step}$. From the uniform continuity of Y on $[0, T]$ it follows easily that for each $\omega \in \Omega$ we have

$$\lim_{m \rightarrow \infty} \int_0^T |Y(\omega, s) - Y_m(\omega, s)|^2 ds.$$

Moreover, since Y, Y^m are bounded by C , we have by the DC theorem that

$$\lim_{m \rightarrow \infty} \mathbb{E} \int_0^T |Y(s) - Y_m(s)|^2 ds.$$

This proves the claim. □

Proof of Lemma 3.2.13. Since $Y \in \mathcal{H}_T$, by Lemma 3.2.9 there exists a sequence $Y_n \in \mathcal{H}_T^{step}$ which approximates Y in $L_2(\Omega \times [0, T])$. It is straightforward that for a process $Z \in \mathcal{H}_T^{step}$ with a representation

$$Z(t) = \sum_{i=0}^n Z_{t_i} \mathbf{1}_{[t_i, t_{i+1})}(t),$$

we have

$$I(Z, t) = \sum_{i=0}^n Z_{t_i} (W(t \wedge t_{i+1}) - W(t \wedge t_i)),$$

which is obviously continuous in t (since W is). Hence, $I(Y_n, \cdot)$ are continuous functions of time. Then, by Doob's inequality and Itô's isometry we have

$$\begin{aligned} \mathbb{E} \left[\sup_{t \in [0, T]} |I(Y_n, t) - I(Y_m, t)|^2 \right] &\leq C \mathbb{E} [|I(Y_n, T) - I(Y_m, T)|^2] \\ &= \mathbb{E} \left[\int_0^T |Y_n(s) - Y_m(s)|^2 ds \right] \rightarrow 0, \end{aligned}$$

as $n, m \rightarrow \infty$. This shows that $(I(Y_n, \cdot))_{n=1}^\infty$ is a Cauchy sequence in $L_2(\Omega; C([0, T]))$ and since the latter is complete, there exists an element $J \in L_2(\Omega; C([0, T]))$ such that $I(Y_n, \cdot) \rightarrow J$ in $L_2(\Omega; C([0, T]))$. In particular for all $t \in [0, T]$ have that $I(Y_n, t) \rightarrow J(t)$ in $L_2(\Omega)$. By definition the map $t \mapsto J(\omega, t)$ is continuous for each $\omega \in \Omega$. On the other hand, for all $t \in [0, T]$ we have that $I(Y_n, t) \rightarrow J(t)$ in $L_2(\Omega)$. This shows that for all $t \in [0, T]$ we have that $J(t) = I(Y, t)$ as elements of $L_2(\Omega)$, therefore, also almost surely. Consequently, J is the continuous modification that we were looking for. $\square$

3.4 Stopping times

Definition 3.4.1. A map $\tau : \Omega \rightarrow [0, +\infty]$ is called *stopping time* with respect to a filtration $\mathbb{F}$ if for all $t \in [0, T]$ we have $\{\tau > t\} \in \mathcal{F}_t$.

Exercise 3.4.2. Let $X : \Omega \times [0, T] \rightarrow \mathbb{R}$ be a continuous process adapted to the filtration $\mathbb{F}$. Show that

$$\tau_b := \inf\{t \in [0, T] : X(t) \geq b\},$$

where recall that $\inf \emptyset = +\infty$. Show that τ_b is a stopping time.

Solution: Let $t \in [0, T]$. Then, by the continuity of X we see that

$$\begin{aligned} \{\tau_b > t\} &= \{\inf\{s \in [0, T] : X(s) \geq b\} > t\} \\ &= \left\{ \sup_{s \in [0, t]} X(s) < b \right\} \\ &= \left\{ \sup_{s \in [0, t] \cap \mathbb{Q}} X(s) < b \right\} \end{aligned}$$

Since X is adapted, $X(s)$ is $\mathcal{F}_s$ -measurable. Since $\mathcal{F}_s \subset \mathcal{F}_t$ for $s \leq t$, we get that for each $s \in [0, t] \cap \mathbb{Q}$, $X(s)$ is $\mathcal{F}_t$ -measurable. Since $[0, t] \cap \mathbb{Q}$ is a countable set, by Exercise 1.1.13 we get that $\sup_{s \in [0, t] \cap \mathbb{Q}} X(s)$ is $\mathcal{F}_t$ -measurable, which implies that $\{\sup_{s \in [0, t] \cap \mathbb{Q}} X(s) < b\} \in \mathcal{F}_t$. Consequently, $\{\tau_b > t\} \in \mathcal{F}_t$. Hence τ_b is indeed an $\mathbb{F}$ -stopping time. $\square$

Exercise 3.4.3. Let τ_1, τ_2 be stopping times with respect to the filtration $\mathbb{F}$. Show that $\tau_1 \wedge \tau_2$ is also a stopping time with respect to the filtration $\mathbb{F}$.

Solution: Let τ_1, τ_2 be stopping times with respect to the filtration $\mathbb{F}$. Then

$$\{\tau_1 > t\}, \{\tau_2 > t\} \in \mathcal{F}_t \quad \forall t \in [0, T].$$

Now,

$$\begin{aligned} \{\tau_1 \wedge \tau_2 > t\} &= \{\tau_1 > t \text{ and } \tau_2 > t\} \\ &= \{\tau_1 > t\} \cap \{\tau_2 > t\} \end{aligned}$$

which is in $\mathcal{F}_t$, because σ -algebras are closed under (countable and thus also finite) intersection.

We have shown that $\{\tau_1 \wedge \tau_2 > t\} \in \mathcal{F}_t$ for all $t \in [0, T]$. Therefore $\tau_1 \wedge \tau_2$ is indeed an $\mathbb{F}$ -stopping time. $\square$

Exercise 3.4.4. Let τ be a stopping time. Show that for each $t \geq 0$ we have $\{\tau \geq t\} \in \mathcal{F}_t$. Show that for all $t \in [0, T]$, the random variable $\mathbf{1}_{[0, \tau)}(t)$ and $\mathbf{1}_{[0, \tau]}(t)$ are $\mathcal{F}_t$ -measurable. Conclude that they are both $\mathbb{F}$ -adapted stochastic processes (to show that they are $\mathcal{F} \otimes \mathcal{B}([0, T])$ -measurable, recall Remark 2.1.2).

Solution: Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space with filtration $\mathbb{F} = (\mathcal{F}_t)_{t \in [0, T]}$, and let τ be an $\mathbb{F}$ -stopping time. Then for each $t \in [0, T]$,

$$\begin{aligned} \{\tau \geq t\} &= \{\tau > t - 1/n \mid \forall n \in \mathbb{N}\} \\ &= \bigcap_{n \in \mathbb{N}} \{\tau > t - 1/n\}. \end{aligned}$$

Since τ is a stopping time: $\{\tau > t - 1/n\}$ is $\mathcal{F}_{t-1/n}$ -measurable. So since for any $n \in \mathbb{N}$ we have $\mathcal{F}_{t-1/n} \subset \mathcal{F}_t$, it follows that $\{\tau > t - 1/n\}$ is $\mathcal{F}_t$ -measurable for all $n \in \mathbb{N}$.

We have shown that $\{\tau \geq t\}$ is a countable intersection of $\mathcal{F}_t$ -measurable sets. Therefore $\{\tau \geq t\}$ is $\mathcal{F}_t$ -measurable.

We will now show that the process $(\mathbf{1}_{[0, \tau)}(t))_{t \in [0, T]}$ is adapted, i.e. that for all $t \in [0, T]$, $\mathbf{1}_{[0, \tau)}(t)$ is $\mathcal{F}_t$ -measurable. To this end, let $t \in [0, T]$ and $B \in \mathcal{B}([0, \infty])$. Then

$$\begin{aligned} (\mathbf{1}_{[0, \tau)}(t))^{-1}(B) &= \{\omega \in \Omega : \mathbf{1}_{[0, \tau(\omega))}(t) \in B\} \\ &= \begin{cases} \Omega & \text{if } 0, 1 \in B \\ \emptyset & \text{if } 0, 1 \notin B \\ \{t < \tau\} & \text{if } 1 \in B, 0 \notin B \\ \{t < \tau\}^c & \text{if } 0 \in B, 1 \notin B \end{cases} \end{aligned}$$

Since $\mathcal{F}_t$ is a σ -algebra: $\Omega, \emptyset \in \mathcal{F}_t$. Furthermore since τ is a stopping time: $\{\tau > t\} \in \mathcal{F}_t$. So since σ -algebras are closed under complementation, we also have that $\{\tau > t\}^c \in \mathcal{F}_t$. Therefore

$$(\mathbf{1}_{[0, \tau)}(t))^{-1}(B) \in \mathcal{F}_t \quad \forall B \in \mathcal{B}([0, \infty]).$$

Hence $\mathbf{1}_{[0,\tau)}(t)$ is $\mathcal{F}_t$ -measurable.

Similarly we can show that the process $(\mathbf{1}_{[0,\tau]}(t))_{t \in [0,T]}$ is adapted: For $t \in [0, T]$ and $B \in \mathcal{B}([0, \infty])$ we have

$$\begin{aligned} (\mathbf{1}_{[0,\tau]}(t))^{-1}(B) &= \{\omega \in \Omega : \mathbf{1}_{[0,\tau(\omega)]}(t) \in B\} \\ &= \begin{cases} \Omega & \text{if } 0, 1 \in B \\ \emptyset & \text{if } 0, 1 \notin B \\ \{t \leq \tau\} & \text{if } 1 \in B, 0 \notin B \\ \{t \leq \tau\}^c & \text{if } 0 \in B, 1 \notin B. \end{cases} \end{aligned}$$

As $\mathcal{F}_t$ is a σ -algebra: $\Omega, \emptyset \in \mathcal{F}_t$. As we have proven earlier: $\{t \leq \tau\} \in \mathcal{F}_t$. So since σ -algebras are closed under complementation we also have that $\{t \leq \tau\}^c \in \mathcal{F}_t$. Therefore

$$(\mathbf{1}_{[0,\tau]}(t))^{-1}(B) \in \mathcal{F}_t \quad \forall B \in \mathcal{B}([0, \infty]).$$

Hence $\mathbf{1}_{[0,\tau]}(t)$ is $\mathcal{F}_t$ -measurable.

Finally, since $\mathbf{1}_{[0,\tau)}(t)$ is right-continuous and $\mathbf{1}_{[0,\tau]}(t)$ is left-continuous, by Remark 2.1.2 both processes are $\mathcal{F} \otimes \mathcal{B}([0, T])$ -measurable. $\square$

Exercise 3.4.5. Let $X : \Omega \times [0, T] \rightarrow \mathbb{R}$ be a continuous and $\mathbb{F}$ -adapted process and take real numbers $a < b$. Let $\tau := \inf\{t \in [0, T] : X(t) \notin (a, b)\}$ ($\inf \emptyset := \infty$). Prove that τ is an $\mathbb{F}$ -stopping time.

The stopping time from the exercise above it is call the *first exit time of $X(t)$ from (a, b)* .

Solution: See Exercise 3.4.2 and Exercise 3.4.3. $\square$

3.5 Extending the stochastic integral

Definition 3.5.1. We denote by $\mathcal{S}_T$ the class of all $\mathbb{F}$ -adapted processes $Y : \Omega \times [0, T] \rightarrow \mathbb{R}$ such that $T \geq 0$, almost surely

$$\int_0^T |Y(s)|^2 ds < \infty.$$

Notice that

$$\mathcal{H}_T^{step} \subset \mathcal{H}_T \subset \mathcal{S}_T.$$

We want to extend the stochastic integral to integrands from the class $\mathcal{S}_T$. We will need the following two results.

Theorem 3.5.2. Let $Y \in \mathcal{H}_T$ and let τ be a stopping time bounded by T . Then, almost surely

$$I(Y, \tau) = I(\mathbf{1}_{[0,\tau]}Y) = I(\mathbf{1}_{[0,\tau)}Y)$$

Theorem 3.5.3. Let $Y \in \mathcal{H}_T$. Then, for all $\delta, \varepsilon > 0$ we have

$$\mathbb{P}\left(\sup_{t \in [0, T]} \left| \int_0^t Y(s) dW(s) \right| \geq \varepsilon\right) \leq \mathbb{P}\left(\int_0^T |Y(s)|^2 ds \geq \delta\right) + \frac{1}{\varepsilon^2} \mathbb{E}\left[\delta \wedge \int_0^T |Y(s)|^2 ds\right]$$

From this, one can prove *Davis's inequality*

Lemma 3.5.4. For $Y \in \mathcal{H}_T$ we have

$$\mathbb{E}\left[\sup_{t \in [0, T]} \left| \int_0^t Y(s) dW(s) \right|\right] \leq 3\mathbb{E}\left[\left(\int_0^T |Y(s)|^2 ds\right)^{1/2}\right].$$

Exercise 3.5.5. Use Theorem 3.5.3 in order to prove Lemma 3.5.4.

Hint: Use that $\mathbb{E}[X] = \int_0^\infty \mathbb{P}(X > \lambda) d\lambda$, for $X \geq 0$.

Solution: Let $Y \in \mathcal{H}_T$ and $T \in [0, \infty)$. Let us set

$$X := \sup_{t \leq T} \left| \int_0^t Y(s) dW(s) \right| \quad Z := \int_0^T (Y(s))^2 ds.$$

We want to prove that

$$\mathbb{E}[X] \leq 3\mathbb{E}[Z^{1/2}].$$

By using the hint we get

$$\begin{aligned} \mathbb{E}[X] &= \\ &= \int_0^\infty \mathbb{P}(X \geq \lambda) d\lambda \\ &\leq \int_0^\infty \mathbb{P}(Z \geq \lambda^2) d\lambda + \int_0^\infty \frac{1}{\lambda^2} \mathbb{E}(\lambda^2 \wedge Z) d\lambda \\ &=: I_1 + I_2. \end{aligned} \quad \begin{array}{l} \text{by Theorem 3.5.3 applied with } \delta = \lambda^2: \\ \\ (\diamond) \end{array}$$

The first integral can be computed as follows:

$$\begin{aligned} I_1 &= \int_0^\infty \mathbb{P}(Z^{1/2} \geq \lambda) d\lambda \\ &= \mathbb{E}[Z^{1/2}]. \end{aligned} \quad \begin{array}{l} \text{by the hint:} \\ (\diamond \diamond) \end{array}$$

We will now aim to compute the second integral. We have

$$\begin{aligned} I_2 &= \int_0^\infty \frac{1}{\lambda^2} \mathbb{E}[\lambda^2 \wedge Z] d\lambda \\ &= \mathbb{E}\left[\int_0^\infty \frac{1}{\lambda^2} (\lambda^2 \wedge Z) d\lambda\right] \\ &= \mathbb{E}\left[\int_0^{Z^{1/2}} \frac{1}{\lambda^2} (\lambda^2 \wedge Z) d\lambda\right] + \mathbb{E}\left[\int_{Z^{1/2}}^\infty \frac{1}{\lambda^2} (\lambda^2 \wedge Z) d\lambda\right] \end{aligned}$$

$$\begin{aligned}
&= \mathbb{E} \left[\int_0^{Z^{1/2}} \frac{1}{\lambda^2} (\lambda^2) d\lambda \right] + \mathbb{E} \left[\int_{Z^{1/2}}^{\infty} \frac{1}{\lambda^2} Z d\lambda \right] \\
&=: I_{2,1} + I_{2,2}
\end{aligned}$$

Now,

$$I_{2,1} = \mathbb{E} \left[\int_0^{Z^{1/2}} 1 d\lambda \right] = \mathbb{E}[Z^{1/2}],$$

furthermore

$$\begin{aligned}
\mathbf{1}_{2,2} &= \mathbb{E} \left[Z \int_{Z^{1/2}}^{\infty} \frac{1}{\lambda^2} d\lambda \right] \\
&= \mathbb{E} \left[Z \left[-\lambda^{-1} \right]_{Z^{1/2}}^{\infty} \right] \\
&= \mathbb{E} \left[Z \cdot Z^{-1/2} \right] \\
&= \mathbb{E}[Z^{1/2}].
\end{aligned}$$

Therefore

$$I_2 = I_{2,1} + I_{2,2} = 2\mathbb{E}[Z^{1/2}]. \quad (\diamond \diamond \diamond)$$

The statement is thus proven, because

$$\begin{aligned}
&\mathbb{E}[X] && \text{by } (\diamond): \\
&\leq I_1 + I_2 \\
&&& \text{by } (\diamond \diamond) \text{ and } (\diamond \diamond \diamond): \\
&= \mathbb{E}[Z^{1/2}] + 2\mathbb{E}[Z^{1/2}] \\
&= 3\mathbb{E}[Z^{1/2}],
\end{aligned}$$

as required. $\square$

Let $Y \in \mathcal{S}_T$ and for $n \in \mathbb{N}$ let us set

$$\tau_n = \inf \{ t \in [0, T] : \int_0^t |Y(s)|^2 ds \geq n \} \wedge T. \quad (3.10)$$

By Exercise 3.4.5 we know that τ_n is a stopping time. Next, notice that by Exercise 3.4.4, for each $n \in \mathbb{N}$, $(Y_n(t) := I_{[0, \tau_n]}(t)Y(t))_{t \in [0, T]}$ is an $\mathbb{F}$ -adapted process which moreover satisfies

$$\mathbb{E} \int_0^T |I_{[0, \tau_n]}(s)Y(s)|^2 ds \leq \mathbb{E} \int_0^{\tau_n} |Y(s)|^2 ds \leq nT < \infty.$$

In particular, we have that $(I_{[0, \tau_n]}(t)Y(t))_{t \in [0, T]} \in \mathcal{H}_T$. Therefore, we can define the stochastic integral process $(I(Y_n, t))_{t \in [0, T]}$. Moreover, from the definition of τ_n we have that if $m > n$, then $\tau_m > \tau_n$ and that for each $\omega \in \Omega$ $\tau_n(\omega) = T$ for all n large enough. If $m > n$, then we have

$$\mathbb{E} \left[\sup_{t \in [0, \tau_n]} |I(Y_n, t) - I(Y_m, t)|^2 \right]$$

$$\begin{aligned}
&= \mathbb{E} \left[\sup_{t \in [0, \tau_n]} \left| \int_0^t (I_{[0, \tau_n]}(s)Y(s) - I_{[0, \tau_m]}(s)Y(s)) dW(s) \right|^2 \right] \\
&= \mathbb{E} \left[\sup_{t \in [0, T]} \left| \int_0^{t \wedge \tau_n} (I_{[0, \tau_n]}(s)Y(s) - I_{[0, \tau_m]}(s)Y(s)) dW(s) \right|^2 \right] \\
&= \mathbb{E} \left[\sup_{t \in [0, T]} \left| \int_0^t (I_{[0, \tau_n]}(s)Y(s) - I_{[0, \tau_n]}(s)Y(s)) dW(s) \right|^2 \right] = 0,
\end{aligned}$$

where we have used Theorem 3.5.2. This means that $I(Y_n, t) = I(Y_m, t)$ with probability one, for all $t \leq \tau_n$. In particular this means that we can then define $I(Y, t) = I(Y_n, t)$ for $t \leq \tau_n$. It is easy to see that the extension of I on $\mathcal{S}_T$ is linear and for each $Y \in \mathcal{S}_T$ we have that $(I(Y, t))_{t \geq 0}$ is a continuous process and satisfies the conclusions of Theorem 3.5.3 and Lemma 3.5.4.

Remark 3.5.6. Notice that for $Y \in \mathcal{S}_T$, $(I(Y, t))_{t \geq 0}$ is not a martingale in general (might not even be in $L_1(\Omega)$). However, it is a *local martingale*.

Definition 3.5.7. A process $X : \Omega \times [0, T] \rightarrow \mathbb{R}$ is a local martingale if there exists a sequence of stopping times $(\tau_n)_{n=1}^\infty$ satisfying

- (i) for any $n \in \mathbb{N}$, $\mathbb{P}(\tau_n \leq \tau_{n+1}) = 1$
- (ii) for all $\omega \in \Omega$, $\tau_n(\omega) = T$ for all n large enough.
- (iii) for each $n \in \mathbb{N}$, the process $(X(t \wedge \tau_n))_{t \in [0, T]}$ is a martingale.

Such a sequence of stopping times is called *localising sequence* for $(X(t))_{t \in [0, T]}$.

For $Y \in \mathcal{S}_T$ it is easy to see that the sequence of stopping times defined in (3.10) is a localising sequence for $(I(Y, t))_{t \in [0, T]}$.

Exercise 3.5.8. Let $Y_n, Y \in \mathcal{S}_T$ such that

$$\int_0^T |Y_n(s) - Y(s)|^2 ds \rightarrow 0,$$

as $n \rightarrow \infty$, in probability. Show that the stochastic integrals of Y_n converge to the stochastic integral of Y , uniformly in time, in probability. That is, show that

$$\sup_{t \in [0, T]} \left| \int_0^t Y_n(s) dW(s) - \int_0^t Y(s) dW(s) \right| \rightarrow 0,$$

as $n \rightarrow \infty$, in probability.

Solution: We want to show that for any $\varepsilon > 0$, we have

$$\lim_{n \rightarrow \infty} \mathbb{P} \left(\sup_{t \leq T} \left| \int_0^t Y_n(s) dW(s) - \int_0^t Y(s) dW(s) \right| \geq \varepsilon \right) = 0.$$

For this it suffices to show that for any $\varepsilon > 0$ and for any $\delta > 0$,

$$\limsup_{n \rightarrow \infty} \mathbb{P} \left(\sup_{t \leq T} \left| \int_0^t Y_n(s) dW(s) - \int_0^t Y(s) dW(s) \right| \geq \varepsilon \right) \leq \delta.$$

To this end, let $\varepsilon, \delta, C > 0$. Since $Y, Y^n \in \mathcal{S}$, we have that $Y^n - Y \in \mathcal{S}$. So since Theorem 4.3.2. remains valid for processes of class $\mathcal{S}$, we can apply it here as follows:

$$\begin{aligned} & \mathbb{P} \left(\sup_{t \leq T} \left| \int_0^t Y_n(s) dW(s) - \int_0^t Y(s) dW(s) \right| \geq \varepsilon \right) = \\ & = \mathbb{P} \left(\sup_{t \leq T} \left| \int_0^t (Y_n(s) - Y(s)) dW(s) \right| \geq \varepsilon \right) \end{aligned}$$

by Theorem 3.5.3:

$$\begin{aligned} & \leq \mathbb{P} \left(\int_0^T |Y_n(s) - Y(s)|^2 ds \geq C \right) + \frac{1}{\varepsilon^2} \mathbb{E} \left[C \wedge \int_0^T |Y_n(s) - Y(s)|^2 ds \right] \\ & \leq \mathbb{P} \left(\int_0^T |Y_n(s) - Y(s)|^2 ds \geq C \right) + \frac{C}{\varepsilon^2} = \end{aligned}$$

Choosing $C = \varepsilon^2 \delta$:

$$= \mathbb{P} \left(\int_0^T |Y_n(s) - Y(s)|^2 ds \geq \varepsilon^2 \delta \right) + \delta.$$

Now taking the lim sup, and recalling the assumption that $\int_0^T |Y_n(s) - Y(s)|^2 ds \rightarrow 0$ as $n \rightarrow \infty$ in probability, we get

$$\begin{aligned} & \limsup_{n \rightarrow \infty} \mathbb{P} \left(\sup_{t \leq T} \left| \int_0^t Y_n(s) dW(s) - \int_0^t Y(s) dW(s) \right| \geq \varepsilon \right) \\ & \leq \limsup_{n \rightarrow \infty} \mathbb{P} \left(\int_0^T |Y_n(s) - Y(s)|^2 ds \geq \varepsilon^2 \delta \right) + \delta \\ & = \delta \end{aligned}$$

as required. □

3.6 Itô's formula

Let us recall that if we have two differentiable functions f and g , then the following chain rule holds

$$\frac{d}{dt} g(f(t)) = g'(f(t)) f'(t),$$

or, in other words, if f has differential

$$f(t) = f(0) + \int_0^t a(s) ds, \tag{3.11}$$

then $g(f(t))$ satisfies

$$g(f(t)) = g(f(0)) + \int_0^t g'(f(s)) a(s) ds.$$

This chain rule allows us to study equations of the type (3.1), exactly because the solutions have differential of the form (3.11). However, we want to deal with equations of the form (3.3).

Let us start with some terminology. We are working on the probability space from the previous section. We will say that an adapted process $(A(t))_{t \in [0, T]}$ is of class $\mathcal{D}_T$, if it satisfies

$$\mathbb{P} \left(\int_0^T |A(s)| ds < \infty \right) = 1$$

An adapted stochastic process $(X(t))_{t \geq 0}$ is an Itô process if there exist processes $A \in \mathcal{D}_T$ and $B \in \mathcal{S}_T$ such that with probability one,

$$X(t) = X(0) + \int_0^t A(s) ds + \int_0^t B(s) dW(s), \quad \forall t \in [0, T]. \quad (3.12)$$

In this case we will say that X has differential

$$dX(t) = A(t) dt + B(t) dW(t). \quad (3.13)$$

For Itô processes the following analogue of the chain rule holds.

Theorem 3.6.1 (Itô's formula in $d = 1$). *Let X be an Itô process with differential as in (3.13) and let $u : \mathbb{R} \rightarrow \mathbb{R}$ be a function of class C^2 . Then, $u(X)$ is an Itô process and with probability one,*

$$\begin{aligned} u(X(t)) = u(X(0)) + \int_0^t u'(X(s))A(s) ds + \int_0^t u'(X(s))B(s) dW(s) \\ + \frac{1}{2} \int_0^t u''(X(s))|B(s)|^2 ds, \quad \forall t \in [0, T]. \end{aligned}$$

In notation of differentials, the above theorem says that

$$du(X(t)) = u'(X(t))A(t) dt + u'(X(t))B(t) dW(t) + \frac{1}{2}u''(X(t))|B(t)|^2 dt. \quad (3.14)$$

A nice way to remember the above formula is to keep in mind (3.13) and to remember the following multiplication rules

$$(dt)^2 = 0, \quad dW(t)dt = 0, \quad (dW(t))^2 = dt.$$

The relation in (3.14) then becomes

$$du(X(t)) = u'(X(t)) dX(t) + \frac{1}{2}u''(X(t))(dX(t))^2,$$

which probably looks more familiar (something like a Taylor expansion).

We will not provide a proof of the above theorem but its proof heavily relies on the following product formula.

Lemma 3.6.2. *Let X, Y be Itô processes. Then XY is also an Itô process and*

$$d(X(t)Y(t)) = X(t) dY(t) + Y(t) dX(t) + dX(t) dY(t).$$

The above, in turn, can be proved on the basis of the following lemma.

Lemma 3.6.3. *The following relations hold*

$$(i) \quad d|W(t)|^2 = 2W(t) dW(t) + dt$$

$$(ii) \quad d(tW(t)) = t dW(t) + W(t) dt.$$

Proof. What we want to prove is

$$|W(t)|^2 = 2 \int_0^t W(s) dW(s) + t.$$

Let $t_i^n = t \frac{i}{n}$ for $i = 0, \dots, n$, and let us define the process $W_n(t) = \sum_{i=0}^n W(t_i) \mathbf{1}_{[t_i, t_{i+1})}(t) \in \mathcal{H}_T$. From the elementary identity $2ab = -(a-b)^2 + a^2 + b^2$, we get

$$\begin{aligned} 2I(W_n, t) &= 2 \sum_{i=0}^n W(t_i^n)(W(t_{i+1}^n) - W(t_i^n)) \\ &= - \sum_{i=1}^n |W(t_{i+1}^n) - W(t_i^n)|^2 + \sum_{i=1}^n (|W(t_{i+1}^n)|^2 - |W(t_i^n)|^2) \\ &= - \sum_{i=1}^n |W(t_{i+1}^n) - W(t_i^n)|^2 + |W(t)|^2. \end{aligned} \quad (3.15)$$

It is easy to see that

$$\lim_{n \rightarrow \infty} \mathbb{E} \left[\int_0^t |W_n(s) - W(s)|^2 ds \right] = 0,$$

which by Ito's isometry shows that $I(W_n, t) \rightarrow I(W, t)$ in $L_2(\Omega)$, as $n \rightarrow \infty$. On the other hand, by Theorem 2.3.4, we have that the first term at the right-hand side of (3.15) converges to t in $L_2(\Omega)$ as $n \rightarrow \infty$. Consequently, passing to the limit on (3.15), we get

$$2 \int_0^t W(s) dW(s) = -t + |W(t)|^2,$$

which shows (i). In a similar way one proves (ii). □

Exercise 3.6.4. Prove (ii) of Lemma 3.6.3.

Solution: We want to show that $d(tW(t)) = t dW(t) + W(t) dt$, i.e. that $tW(t) = \int_0^t s dW(s) + \int_0^t W(s) ds$. To this end, let $t_i^n := t \cdot \frac{i}{n}$ for $i = 0, \dots, n$, and define the simple (and deterministic) process

$$\tilde{t}^n(s) := \sum_{i=0}^{n-1} t_i^n \mathbf{1}_{[t_i^n, t_{i+1}^n)}(s).$$

Then, it is easy to see that

$$\lim_{n \rightarrow \infty} \int_0^t (\tilde{t}^n(s) - s)^2 ds = 0,$$

which implies that

$$\int_0^t \tilde{t}_i^n(s) dW(s) \rightarrow \int_0^t s dW(s)$$

as $n \rightarrow \infty$, in $L_2(\Omega)$, and therefore also in probability $(\dagger)$.

We also have that

$$\begin{aligned}
 \int_0^t \tilde{t}_i^n(s) dW(s) &= \int_0^t \sum_{i=0}^{n-1} t_i^n \mathbf{1}_{[t_i^n, t_{i+1}^n)}(s) dW(s) \\
 &= \sum_{i=0}^{n-1} t_i^n (W(t_{i+1}^n) - W(t_i^n)) \\
 &= \sum_{i=0}^{n-1} t_i^n W(t_{i+1}^n) - \sum_{i=0}^{n-1} t_i^n W(t_i^n) \\
 &= \sum_{i=0}^{n-1} t_i^n W(t_{i+1}^n) - \sum_{i=0}^{n-1} t_{i+1}^n W(t_{i+1}^n) + \sum_{i=0}^{n-1} t_{i+1}^n W(t_{i+1}^n) - \sum_{i=0}^{n-1} t_i^n W(t_i^n) \\
 &= - \sum_{i=0}^{n-1} W(t_{i+1}^n) (t_{i+1}^n - t_i^n) + \sum_{i=0}^{n-1} (t_{i+1}^n W(t_{i+1}^n) - t_i^n W(t_i^n)) \\
 &=: -S_1^n + S_2^n
 \end{aligned}$$

Note:

- S_1^n is a Riemann sum for $\int_0^t W(s) ds$, therefore $S_1^n \rightarrow \int_0^t W(s) ds$ as $n \rightarrow \infty$, almost surely, hence also in probability.
- S_2^n is a telescopic sum, so it can be evaluated as follows: $S_2^n = t_n W(t_n) - t_0 W(t_0) = tW(t) - 0W(0) = tW(t)$.

Therefore

$$\int_0^t \tilde{t}_i^n(s) dW(s) = -S_1^n + S_2^n \rightarrow - \int_0^t W(s) ds + tW(t)$$

as $n \rightarrow \infty$, in probability. $(\ddagger)$

By $(\dagger)$ and $(\ddagger)$, by uniqueness of limits, it follows that

$$\int_0^t s dW(s) = - \int_0^t W(s) ds + tW(t),$$

i.e.

$$tW(t) = \int_0^t s dW(s) + \int_0^t W(s) ds$$

as required. $\square$

3.7 Itô processes in higher dimensions

We will now present a multi-dimensional analog of Theorem 3.6.1. Let us first introduce some concepts. A d_0 -dimensional $\mathbb{F}$ -Wiener process W is a stochastic process taking values in $\mathbb{R}^{d_0}$ given by $W(t) = (W_1(t), \dots, W_{d_0}(t))^\top$, where W_i , for $i = 1, \dots, d_0$ are independent real valued $\mathbb{F}$ -Wiener

processes. From now on, we assume that we are given such a Wiener process on our probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

On the basis of independence one can easily see that for $F, G \in \mathcal{H}_T^{step}$, we have

$$\mathbb{E} \left[\int_0^T F(s) dW_i(s) \int_0^T G(s) dW_j(s) \right] = \delta_{ij} \mathbb{E} \left[\int_0^T F(s) G(s) ds \right],$$

where $\delta_{ij} = 1$ if $i = j$ and $\delta_{ij} = 0$ otherwise. The above by continuity extends to $F, G \in \mathcal{H}_T$.

Suppose now that we are given a process $B(t) = (B_{ij}(t))_{i,j}$ which takes values in $\mathbb{R}^{d \times d_0}$ (the space of all $d \times d_0$ matrices). We will write that $B \in \mathcal{S}_T^{d \times d_0}$ if $B_{ij} \in \mathcal{S}_T$ for all $i = 1, \dots, d$, $j = 1, \dots, d_0$. In this case, we can define the Itô integral as a d -dimensional process $I(B, t)$, given by

$$\begin{pmatrix} dI_1(B, t) \\ \vdots \\ dI_d(B, t) \end{pmatrix} = \begin{pmatrix} B_{11}(t) & \dots & B_{1d_0}(t) \\ \vdots & \ddots & \vdots \\ B_{d1}(t) & \dots & B_{dd_0}(t) \end{pmatrix} \begin{pmatrix} dW_1(t) \\ \vdots \\ dW_{d_0}(t) \end{pmatrix},$$

in other words, the i -th component is given by

$$\left(\int_0^t B(s) dW(s) \right)^i = I_i(B, t) = \sum_{j=1}^{d_0} \int_0^t B_{ij}(s) dW_j(s).$$

Form now on, for $x = (x_1, \dots, x_d) \in \mathbb{R}^d$ and $y = (y_{ij})_{i,j} \in \mathbb{R}^{d \times d_0}$ we will use the notation

$$|x|_d = \left(\sum_{i=1}^d |x_i|^2 \right)^{1/2}, \quad |y|_{d \times d_0} = \left(\sum_{i=1}^d \sum_{j=1}^{d_0} |y_{ij}|^2 \right)^{1/2}$$

Exercise 3.7.1. Show that for $B \in \mathcal{H}_T^{d \times d_0}$ (which is defined in the obvious way) we have

$$\mathbb{E} \left[\left| \int_0^T B(s) dW(s) \right|_d^2 \right] = \mathbb{E} \left[\int_0^T |B(s)|_{d \times d_0}^2 ds \right] = \mathbb{E} \int_0^T \left[\text{tr}(B(s) B_s^\top) \right] ds,$$

where for a square matrix G , $\text{tr}(G) := \sum_{ii} G_{ii}$ is the trace. Similarly, show the Davis' inequality: For $B \in \mathcal{S}_T^{d \times d_0}$, we have

$$\mathbb{E} \left[\sup_{t \in [0, T]} \left| \int_0^t B(s) dW(s) \right|_d \right] \leq C \mathbb{E} \left[\left(\int_0^T |B(s)|_{d \times d_0}^2 ds \right)^{1/2} \right]$$

Definition 3.7.2. An Itô process with values in $\mathbb{R}^d$ is a process $X(t) = (X_1(t), \dots, X_d(t))$ such that for each i , X_i is an adapted process and there exist $A \in \mathcal{D}_T^d$ and $B \in \mathcal{S}_T^{d \times d_0}$ such that

$$X_i(t) = X_i(0) + \int_0^t A_i(s) ds + \sum_{j=1}^{d_0} \int_0^t B_{ij}(s) dW_j(s).$$

In matrix and differential notation, we have

$$\begin{pmatrix} dX_1(t) \\ \vdots \\ dX_d(t) \end{pmatrix} = \begin{pmatrix} A_1(t) \\ \vdots \\ A_d(t) \end{pmatrix} dt + \begin{pmatrix} B_{11}(t) & \dots & B^{1d_0}(t) \\ \vdots & \ddots & \vdots \\ B_{d1}(t) & \dots & B_{dd_0}(t) \end{pmatrix} \begin{pmatrix} dW_1(t) \\ \vdots \\ dW_{d_0}(t) \end{pmatrix} \quad (3.16)$$

Remark 3.7.3. Notice that not only we allowed our process to take values in higher dimensional spaces but we also allowed to have input from multiple noises.

The Itô formula in this setting takes the following form.

Theorem 3.7.4. Let $u : \mathbb{R}^d \rightarrow \mathbb{R}$ be of class C^2 and let X be an Itô process with differential (3.16). Then,

$$\begin{aligned} du(X(t)) &= \sum_{i=1}^d u_{x_i}(X(t)) A_i(t) dt + \sum_{i=1}^d \sum_{j=1}^{d_0} u_{x_i}(X(t)) B_{ij}(t) dW_j(t) \\ &\quad + \frac{1}{2} \sum_{i,j} u_{x_i x_j}(X(t)) \sum_k B_{ik}(t) B_{jk}(t) dt. \end{aligned} \quad (3.17)$$

Using the following multiplication table

$$dW_i(t) dW_j(t) = \delta_{ij} dt, \quad dW_i(t) dt = (dt)^2 = 0,$$

we can write (3.17) in the form

$$du(X(t)) = \sum_{i=1}^d u_{x_i}(X(t)) dX_i(t) + \frac{1}{2} \sum_{i,j=1}^d u_{x_i x_j}(X(t)) dX_i(t) dX_j(t)$$

Exercise 3.7.5. Let X be an Itô process with differential (3.16) and let $u : [0, \infty) \times \mathbb{R}^d \rightarrow \mathbb{R}$ be of class C^2 . Show that the following holds

$$\begin{aligned} du(t, X(t)) &= u_t(t, X(t)) dt + \sum_{i=1}^d u_{x_i}(t, X(t)) A_i(t) dt + \sum_{i=1}^d \sum_{j=1}^{d_0} u_{x_i}(t, X(t)) B_{ij}(t) dW_j(t) \\ &\quad + \frac{1}{2} \sum_{i,j} u_{x_i x_j}(t, X(t)) \sum_k B_{ik}(t) B_{jk}(t) dt. \end{aligned}$$

Show that the above remains valid for $u \in C^{1,2}$, that is, for u such that $u_t, u_{x_i}, u_{x_i x_j}$ are continuous.

Solution: Let $Y(t) := (t, X(t)) = (t, X_1, \dots, X_d)$. Then Y is a $(d+1)$ -dimensional Itô process. Now,

$$du(t, X(t)) = du(Y(t))$$

by Itô's Lemma:

$$\begin{aligned} &= \sum_{i=1}^{d+1} \partial_{y_i} u(Y(t)) dY_i(t) + \frac{1}{2} \sum_{i,j=1}^{d+1} \partial_{y_i y_j} u(Y(t)) dY_i(t) dY_j(t) \\ &=: (\star) \end{aligned}$$

Note: for $i = 1$ we have that

- if $j = 1$, then $dY_i(t)dY_j(t) = dt dt = 0$
- if $j \geq 2$, then $dY_i(t)dY_j(t) = dt dX_{j-1}(t) = dt(A_{j-1}(t)dt + \sum_{k=1}^{d_0} B_{j-1,k}(t)dW_k(t)) = 0$ because $(dt)^2 = dt dW_k(t) = 0$.

This shows that in the second sum in $(\star)$, if either $i = 1$ or $j = 1$, then the summand is zero. Using this, and also separating the first term of the first sum, we have

$$\begin{aligned}
 (\star) &:= \partial_t u(Y(t))dt + \sum_{i=2}^{d+1} \partial_{y_i} u(Y(t))dY_i(t) + \frac{1}{2} \sum_{i,j=2}^{d+1} \partial_{y_i y_j} u(Y(t))dY_i(t)dY_j(t) \\
 &= \partial_t u(t, X(t))dt + \sum_{i=1}^d \partial_{x_i} u(t, X(t))dX(t)^i + \frac{1}{2} \sum_{i,j=1}^d \partial_{x_i x_j} u(t, X(t))dX(t)^i dX(t)^j \\
 &= \partial_t u(t, X(t))dt + \sum_{i=1}^d \partial_{x_i} u(t, X(t)) \left(A_i(t)dt + \sum_{j=1}^{d_0} B_{ij}(t)dW_t^j \right) \\
 &\quad + \frac{1}{2} \sum_{i,j=1}^d \partial_{x_i x_j} u(t, X(t)) \left(A_i(t)dt + \sum_{k=1}^{d_0} B_{ik}(t)dW_k(t) \right) \left(A_j(t)dt + \sum_{l=1}^{d_0} B_{jl}(t)dW_l(t) \right) \\
 &=: (\star\star).
 \end{aligned}$$

Note that

$$\begin{aligned}
 \left(A_i(t)dt + \sum_{k=1}^{d_0} B_{ik}(t)dW_k(t) \right) \left(A_j(t)dt + \sum_{l=1}^{d_0} B_{jl}(t)dW_l(t) \right) &= 0 + 0 + 0 + \sum_{k=1}^{d_0} \sum_{l=1}^{d_0} B_{ik}(t)B_{jl}(t)dW_k(t)dW_l(t) \\
 &= \sum_{k=1}^{d_0} \sum_{l=1}^{d_0} B_{ik}(t)B_{jl}(t)\delta_{kl}dt \\
 &= \sum_{k=1}^{d_0} B_{ik}(t)B_{jk}(t)dt,
 \end{aligned}$$

therefore

$$\begin{aligned}
 (\star\star) &= \partial_t u(t, X(t))dt + \sum_{i=1}^d \partial_{x_i} u(t, X(t)) \left(A_i(t)dt + \sum_{j=1}^{d_0} B_{ij}(t)dW_t^j \right) \\
 &\quad + \frac{1}{2} \sum_{i,j=1}^d \partial_{x_i x_j} u(t, X(t)) \sum_{k=1}^{d_0} B_{ik}(t)B_{jk}(t)dt \\
 &= \partial_t u(t, X(t))dt + \sum_{i=1}^d \partial_{x_i} u(t, X(t))A_i(t)dt + \sum_{i=1}^d \sum_{j=1}^{d_0} \partial_{x_i} u(t, X(t))B_{ij}(t)dW_t^j \\
 &\quad + \frac{1}{2} \sum_{i,j=1}^d \partial_{x_i x_j} u(t, X(t)) \sum_{k=1}^{d_0} B_{ik}(t)B_{jk}(t)dt
 \end{aligned}$$

as required. If $u \in C^{1,2}$ then there exists u_n which are twice continuously differentiable in (t, x) such that $u_n, \partial_t u_n, \partial_{x_i} u_n, \partial_{x_i x_j} u_n$ converge uniformly to $u, \partial_t u, \partial_{x_i} u, \partial_{x_i x_j} u$. Applying the result to u_n and letting $n \rightarrow \infty$ gives the desired result (check the details on your own!). $\square$

Exercise 3.7.6. Let W be a one dimensional $\mathbb{F}$ -Wiener processes and let $\sigma > 0$. Set

$$Z(t) := \exp\left(\sigma W(t) - \frac{1}{2}\sigma^2 t\right)$$

1. Show that Z_t satisfies the equation

$$dZ(t) = \sigma Z_t dW(t), \quad Z_0 = 1.$$

2. Conclude that $(Z(t))_{t \in [0, T]}$ is a martingale.

Hint: Recall that if $X \sim \mathcal{N}(0, 1)$ then $\mathbb{E}[\exp(aX)] = \exp(\frac{1}{2}a^2)$.

Solution:

1. Let $(W(t))_{t \in [0, T]}$ be a real $\mathbb{F}$ -Wiener process, and define Z as above. Define

$$X(t) = \sigma W(t) - \frac{1}{2}\sigma^2 t.$$

Then X is an Itô process with differential

$$dX(t) = -\frac{1}{2}\sigma^2 dt + \sigma dW(t).$$

(so in the notation of the notes: $A(t) = -\frac{1}{2}\sigma^2$ and $B(t) = \sigma$)

Let $u : \mathbb{R} \rightarrow \mathbb{R}$ be given by $u(x) = \exp(x)$. Then $u'' = u' = u$. By Itô's lemma:

$$\begin{aligned} dZ(t) &= du(X(t)) \\ &= u'(X(t))A(t)dt + u'(X(t))B(t)dW(t) + \frac{1}{2}u''(X(t))|B(t)|^2 dt \\ &= -\exp(X(t))\left(-\frac{1}{2}\sigma^2\right)dt + \exp(X(t))\sigma dW(t) + \frac{1}{2}\exp(X(t))|\sigma|^2 dt \\ &= \exp(X(t))\sigma dW(t) \\ &= \sigma Z(t)dW(t) \end{aligned}$$

as required. Furthermore

$$Z_0 = \exp(\sigma W(0) - \frac{1}{2}\sigma^2 0) = \exp(0) = 1,$$

also as required.

2. By part 1. we have

$$dZ(t) = \sigma Z(t)dW(t), \quad Z_0 = 1,$$

thus

$$Z(t) = 1 + \sigma \int_0^t Z(s)dW(s).$$

So to show that Z is a martingale, it suffices to show that $\int_0^t Z(s)dW(s)$ is a martingale. For this, it suffices to show that the integrand $Z \in \mathcal{H}_T$.

(i) Let $t \in [0, T]$. Since W is an $\mathbb{F}$ -Wiener process, $W(t)$ is $\mathcal{F}_t$ -measurable. So since the map $x \mapsto \exp(\sigma x - \frac{1}{2}\sigma^2 t)$ is (continuous and thus) Borel-measurable, it follows that the composition $Z(t) = \exp(\sigma W(t) - \frac{1}{2}\sigma^2 t)$ is also $\mathcal{F}_t$ -measurable. Hence Z is adapted.

(ii) We have

$$\begin{aligned} & \mathbb{E} \int_0^T |Z(t)|^2 dt \\ &= \mathbb{E} \int_0^T \left| \exp(\sigma W(t) - \frac{1}{2}\sigma^2 t) \right|^2 dt \end{aligned}$$

using Fubini's theorem:

$$\begin{aligned} &= \int_0^T \mathbb{E} \exp(2\sigma W(t) - \sigma^2 t) dt \\ &= \int_0^T \exp(-\sigma^2 t) \mathbb{E} \exp(2\sigma W(t)) dt \\ &= \int_0^T \exp(-\sigma^2 t) \mathbb{E} \left(2\sigma \sqrt{t} \frac{W(t)}{\sqrt{t}} \right) dt \end{aligned}$$

as $\frac{W(t)}{\sqrt{t}} \sim \mathcal{N}(0, 1)$, by the hint:

$$\begin{aligned} &= \int_0^T \exp(-\sigma^2 t) \exp(\frac{1}{2}4\sigma^2 t) dt \\ &= \int_0^T \exp(\sigma^2 t) dt \\ &\leq T \exp(\sigma^2 T) \\ &< \infty. \end{aligned}$$

By (i) and (ii): the process Z is indeed in $\mathcal{H}_T$. This concludes the proof.

□

Exercise 3.7.7. Show that $Z(t) = e^{t/2} \cos(W(t))$ is a martingale.

Solution: By Itô's product rule:

$$dZ(t) = e^{t/2} d(\cos(W(t))) + d(e^{t/2}) \cos(W(t)) + d(e^{t/2}) d(\cos(W(t))). \quad (\dagger)$$

Now,

- $\frac{de^{t/2}}{dt} = e^{t/2} \cdot \frac{1}{2}$, therefore $d(e^{t/2}) = \frac{1}{2}e^{t/2}dt$.
- Let $u(x) = \cos(x)$. Then $u'(x) = -\sin(x)$, and $u''(x) = -\cos(x)$.

Note that W is an Itô process with differential $dW(t) = 0dt + 1dW(t)$ (so in the notation of the notes, we have $A(t) = 0$ and $B(t) = 1$).

By Itô's Lemma:

$$\begin{aligned}
 d \cos(W(t)) &= du(W(t)) \\
 &= u'(W(t))A(t)dt + u'(W(t))B(t)dW(t) + \frac{1}{2}u''(W(t))|B(t)|^2dt \\
 &= -\sin(W(t))0dt + (-\sin(W(t)))1dW(t) + \frac{1}{2}(-\cos(W(t)))|1|^2dt \\
 &= -\sin(W(t))dW(t) - \frac{1}{2}\cos(W(t))dt
 \end{aligned}$$

Substituting these into (†), we get

$$\begin{aligned}
 dZ(t) &= e^{t/2} \left(-\sin(W(t))dW(t) - \frac{1}{2}\cos(W(t))dt \right) + \left(\frac{1}{2}e^{t/2}dt \right) \cos(W(t)) \\
 &\quad + \left(\frac{1}{2}e^{t/2}dt \right) \left(-\sin(W(t))dW(t) - \frac{1}{2}\cos(W(t))dt \right) \\
 &= -e^{t/2}\sin(W(t))dW(t) - \frac{1}{2}e^{t/2}\cos(W(t))dt + \frac{1}{2}e^{t/2}\cos(W(t))dt + 0 + 0 \\
 &= -e^{t/2}\sin(W(t))dW(t)
 \end{aligned}$$

Therefore

$$dZ(t) = -e^{t/2}\sin(W(t))dW(t).$$

By this, and since $Z(0) = e^{0/2}\cos(W(0)) = e^0\cos(0) = 1$, we have

$$Z(t) = 1 - \int_0^t e^{t/2}\sin(W(t))dW(t).$$

So to show that Z is a martingale, it suffices to show that $\int_0^t e^{t/2}\sin(W(t))dW(t)$ is a martingale. For this, it suffices to show that the process $e^{t/2}\sin(W(t))$ is in $\mathcal{H}_T$.

- (i) Since W is adapted, $W(t)$ is $\mathcal{F}_t$ -measurable. So since the map $x \mapsto e^{t/2}\cos(x)$ is (continuous and thus) Borel, it follows that the composition $e^{t/2}\sin(W(t))$ is also $\mathcal{F}_t$ -measurable.
- (ii) We also have

$$\begin{aligned}
 &\mathbb{E} \int_0^T |e^{t/2}\sin(W(t))|^2 dt \\
 &\leq \mathbb{E} \int_0^T e^{T/2} \cdot 1 dt \\
 &= e^{T/2} \cdot T \\
 &< \infty.
 \end{aligned}$$

By (i) and (ii): the process $e^{t/2}\sin(W(t))$ is indeed of class $\mathcal{H}_T$. Thus we are done. $\square$

Exercise 3.7.8. Let W be a d -dimensional $\mathbb{F}$ -Wiener process and let $a_1, \dots, a_d \in \mathbb{R}$ be fixed. Find $c \in \mathbb{R}$ such that the process

$$Z(t) = \exp \left(ct + \sum_{i=1}^d a_i W_i(t) \right)$$

is a martingale.

Solution: Let

$$X(t) := ct + \sum_{i=1}^d a_i W_i(t)$$

and

$$u(x) := \exp(x).$$

Then by Itô's lemma:

$$\begin{aligned} dZ(t) &= du(X(t)) \\ &= u'(X(t))dX(t) + \frac{1}{2}u''(X(t))(dX(t))^2 \\ &= \exp(X(t)) \left(cdt + \sum_{i=1}^d a_i dW_i(t) \right) + \frac{1}{2} \exp(X(t)) \left(cdt + \sum_{i=1}^d a_i dW_i(t) \right)^2 \\ &= Z(t) \left(cdt + \sum_{i=1}^d a_i dW_i(t) \right) + \frac{1}{2} Z(t) \left(c^2(dt)^2 + 2cdt \sum_{i=1}^d a_i dW_i(t) + \sum_{i=1}^d a_i dW_i(t) \sum_{j=1}^d a_j dW_j(t) \right) \\ &=: (\star) \end{aligned}$$

Recall that $(dt)^2 = dt dW_i(t) = 0$, and $dW_i(t)dW_j(t) = \delta_{ij}dt$. Hence

$$\begin{aligned} (\star) &= Z(t) \left(cdt + \sum_{i=1}^d a_i dW_i(t) \right) + \frac{1}{2} Z(t) \left(0 + 0 + \sum_{i=1}^d \sum_{j=1}^d a_i a_j \delta_{ij} dt \right) \\ &= Z(t) \left(cdt + \sum_{i=1}^d a_i dW_i(t) \right) + \frac{1}{2} Z(t) \sum_{i=1}^d a_i^2 dt \\ &= Z(t) \left(c + \frac{1}{2} \sum_{i=1}^d a_i^2 \right) dt + Z(t) \sum_{i=1}^d a_i dW_i(t) \end{aligned}$$

If the drift term is zero, then Z is a martingale. So we need to choose c so that the drift term is zero. Hence we choose

$$c = -\frac{1}{2} \sum_{i=1}^d a_i^2.$$

To show that the sum of the stochastic integrals is indeed a martingale, you can argue as in the previous exercise. $\square$

Exercise 3.7.9. Let $u : [0, \infty) \times \mathbb{R} \rightarrow \mathbb{R}$ be a smooth bounded function satisfying the backward heat equation

$$u_t + \frac{1}{2}u_{xx} = 0.$$

Assume that there exists a constant $C > 0$ such that $|u_x(t, y)| \leq C$ for all $(t, y) \in [0, \infty) \times \mathbb{R}$. Show that if W is a one dimensional Wiener process, then for all $t \geq 0$, we have $\mathbb{E}[u(t, W(t))] = u(0, 0)$.

For the PG students: Solve the above exercises without assuming that u_x is bounded. *Hint:* Use the stopping times $\tau_n = \inf\{t \in [0, T] : \int_0^t |u_x(s, W(s))|^2 ds > n\} \wedge T$.

Solution: By Itô's lemma:

$$\begin{aligned}
du(t, W(t)) &= \partial_t u(t, W(t))dt + \sum_{i=1}^d \partial_{w_i} u(t, W(t))dW_i(t) + \frac{1}{2} \sum_{i,j=1}^d \partial_{w_i w_j} u(t, W(t))dW_i(t)dW_j(t) \\
&= \partial_t u(t, W(t))dt + \sum_{i=1}^d \partial_{w_i} u(t, W(t))dW_i(t) + \frac{1}{2} \sum_{i=1}^d \partial_{w_i w_i} u(t, W(t))dt \\
&= \left(\partial_t u(t, W(t)) + \frac{1}{2} \Delta u(t, W(t)) \right) dt + \sum_{i=1}^d \partial_{w_i} u(t, W(t))dW_i(t) \\
&= \sum_{i=1}^d \partial_{w_i} u(t, W(t))dW_i(t)
\end{aligned}$$

as $dW_i(t)dW_j(t)$

where the last equality holds because u satisfies the backward heat equation by assumption. Therefore

$$u(t, W(t)) = u(0, 0) + \sum_{i=1}^d \int_0^t \partial_{w_i} u(s, W(s))dW_i(s)$$

and thus

$$\mathbb{E}u(t, W(t)) = \mathbb{E}u(0, 0) + \sum_{i=1}^d \mathbb{E} \int_0^t \partial_{w_i} u(s, W(s))dW_i(s).$$

So to show that $\mathbb{E}u(t, W(t)) = \mathbb{E}u(0, 0)$, it suffices to show that $\mathbb{E} \int_0^t \partial_{w_i} u(s, W(s))dW_i(s) = 0$ for all $i \in \{1, \dots, d\}$. To this end, it suffices to show that the process $(\partial_{w_i} u(t, W(t)))_{t \in [0, T]}$ is of class $\mathcal{H}_T$ for all $i \in \{1, \dots, d\}$. This can be seen as follows:

- Note that $W(t)$ is $\mathcal{F}_t$ measurable. Furthermore, since u is smooth, ∂_{w_i} is continuous in x , so it is also Borel in x . Hence the composition $\partial_{w_i} u(t, W(t))$ is $\mathcal{F}_t$ -measurable.
- We also have

$$\mathbb{E} \int_0^T |\partial_{w_i} u(t, W(t))|^2 dt$$

by assumption:

$$\begin{aligned}
&\leq \mathbb{E} \int_0^T C^2 dt \\
&= C^2 T \\
&< \infty.
\end{aligned}$$

Hence we are done. □

3.8 Girsanov's theorem

In this section we will discuss Girsanov's theorem. To be more precise, an implication of Girsanov's theorem, which nevertheless will be sufficient for our purposes.

We fix a terminal time $T \in (0, \infty)$. Let us consider a process $B \in \mathcal{S}_T^{d \times 1}$ and let us denote by Z the process given by

$$Z(t) = \exp \left(- \sum_{i=1}^d \int_0^t B_i(s) dW_i(s) - \frac{1}{2} \sum_{i=1}^d \int_0^t |B_i(s)|^2 ds \right), \quad t \in [0, T]. \quad (3.18)$$

On $(\Omega, \mathcal{F})$ let us define a measure by

$$\mathbb{P}^*(A) = \mathbb{E}[\mathbf{1}_A Z(T)], \quad A \in \mathcal{F}. \quad (3.19)$$

Let us also define a new R^d valued Itô process W^* defined by

$$W^*(t) = \int_0^t B(s) ds + W(t), \quad t \in [0, T]. \quad (3.20)$$

Theorem 3.8.1 (Girsanov). *Let $B \in \mathcal{S}_T^{d \times 1}$ and let Z be defined by (3.18). Assume that $(Z(t))_{t \in [0, T]}$ is a martingale with respect to $\mathbb{F}$. Then the following hold*

- (i) *The map P^* defined in (3.19) is a probability measure on $(\Omega, \mathcal{F})$ and it is equivalent to $\mathbb{P}$.*
- (ii) *The process $(W^*(t))_{t \in [0, T]}$ defined in (3.20) is an $\mathbb{F}$ -Wiener process on $(\Omega, \mathcal{F}, P^*)$.*

Theorem 3.8.2 (Novikov). *Assume that $B \in \mathcal{S}_T^{d \times 1}$ satisfies*

$$\mathbb{E} \left[\exp \left(\frac{1}{2} \sum_{i=1}^d \int_0^T |B_i(s)|^2 ds \right) \right] < \infty.$$

Then $(Z(t))_{t \in [0, T]}$ is a martingale with respect to $\mathbb{F}$.

Corollary 3.8.3. *Assume that $B \in \mathcal{S}_T^{d \times 1}$ is bounded, that is, there exists a constant C such that for all $i = 1, \dots, d$ and all $(\omega, t) \in \Omega \times [0, T]$ we have $|B_i(\omega, t)| \leq C$. Then, the conclusions of Theorem 3.8.1 hold.*

4 Stochastic Differential Equations

4.1 A theorem on existence and uniqueness of solutions

We are now in a position to study stochastic differential equations. Let $T > 0$ be the terminal time. We consider a fixed probability space $(\Omega, \mathcal{F}, \mathbb{P})$ with a right continuous and complete filtration $\mathbb{F} := (\mathcal{F}_t)_{t \in [0, T]}$, and a d_0 -dimensional $\mathbb{F}$ -Wiener process W .

Let us consider measurable functions $f : [0, T] \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ and $\sigma : [0, T] \times \mathbb{R}^d \rightarrow \mathbb{R}^{d \times d_0}$ and a point $x_0 \in \mathbb{R}^d$.

Assumption 4.1.1. There exists a constant $K \geq 0$ such that for all $t \leq 0$, $x, y \in \mathbb{R}^d$, we have

$$|f(t, x) - f(t, y)|_d \leq K|x - y|_d, \quad |\sigma(t, x) - \sigma(t, y)|_{d \times d_0} \leq K|x - y|_d, \quad (4.1)$$

and

$$|f(t, x)|_d \leq K(1 + |x|_d), \quad |\sigma(t, x)|_{d \times d_0} \leq K(1 + |x|_d). \quad (4.2)$$

Remark 4.1.2. For convenience we will slightly abuse the notation. For $x \in \mathbb{R}^d$ and $y \in \mathbb{R}^{d \times d_0}$ we used to write $|x|_d$ and $|y|_{d \times d_0}$ for their norms. From now on, we will drop the subscripts d and $d \times d_0$. Hence, if $x \in \mathbb{R}^d$, $|x|$ stands for $|x|_d$, and if $y \in \mathbb{R}^{d \times d_0}$, $|y|$ stands for $|y|_{d \times d_0}$. Of course it will be clear from the context which one is used.

On $[0, T]$, we consider the equation

$$dX(t) = f(t, X(t)) dt + \sigma(t, X(t)) dW(t), \quad X(0) = x_0. \quad (4.3)$$

Definition 4.1.3. A solution of (4.3) is an adapted, continuous stochastic process $(X(t))_{t \in [0, T]}$ such that with probability one,

$$X(t) = x_0 + \int_0^t f(s, X(s)) ds + \int_0^t \sigma(s, X(s)) dW(s), \quad \forall t \in [0, T]. \quad (4.4)$$

Exercise 4.1.4. Show that under Assumption 4.1.1 the integrals in (4.4) are well defined.

Lemma 4.1.5 (A priori estimates). *Let X be a solution of (4.3). Then, for any $p > 0$, there exists a constant N_0 depending only on p, T, K such that*

$$\mathbb{E} \left[\sup_{t \in [0, T]} |X(t)|^p \right] \leq N_0(1 + |x_0|^p).$$

Theorem 4.1.6. *Let Assumption 4.1.1 hold. Then, (4.3) admits a unique solution.*

Exercise 4.1.7. Let $G : [0, \infty) \rightarrow \mathbb{R}$ be function of class C^1 such that $G(0) = 0$. Let $a, b \in \mathbb{R}$. Find a solution to the equation

$$dX(t) = aX(t) dt + bX(t) dG_t, \quad X(0) = x, \quad t \in [0, T].$$

In the above, of course $dG_t = G'_t dt$, since G is continuously differentiable.

Hint: This is a separable equation. Solution: $X(t) = xe^{at+bG_t}$

Solution: Let us find a candidate solution. We want to solve the equation

$$\frac{dX(t)}{dt} = aX(t) + bX(t)G'(t).$$

Let us assume for now that such X exists and is positive. By dividing the equation by $X(t)$ we see that

$$\begin{aligned} \implies \frac{1}{X(t)} \frac{dX(t)}{dt} &= a + bG'(t) \\ \implies \frac{d}{dt}(\ln(X(t))) &= a + bG'(t) \\ \implies \int_0^t \frac{d}{ds}(\ln(X_s))ds &= \int_0^t (a + bG'_s)ds \\ \implies \ln(X(t)) - \ln(X_0) &= at + bG(t) - a \cdot 0 + bG(0) \\ &\qquad\qquad\qquad \text{as } X_0 = x \text{ and } G(0) = 0: \\ \implies \ln(X(t)) &= \ln(x) + at + bG(t) \\ \implies X(t) &= xe^{at+bG(t)}. \end{aligned}$$

We now have a candidate solution given by $X(t) = xe^{at+bG(t)}$. It is now easy to see by differentiation that X is indeed a solution. $\square$

Exercise 4.1.8. Let W be a one-dimensional Wiener processes. In the solution that you found in the exercise above, replace $G(t)$ by $W(t)$, that is, set $X(t) = xe^{at+bW(t)}$. Write an equation for X .

Solution: Let $(W(t))_{t \in [0, T]}$ be a one-dimensional Wiener process. In the solution that you found in the exercise above, replace $G(t)$ by $W(t)$, that is, set $X(t) = xe^{at+bW(t)}$. Write an equation for X .

Solution. Set

$$X(t) := x \exp(at + bW(t)).$$

Define

$$Y(t) := at + bW(t)$$

and let

$$u(y) = x \exp(y).$$

Then Y is an Itô process with differential

$$dY(t) = adt + bW(t),$$

and $u \in C^2$. So by Itô's lemma:

$$\begin{aligned} dX(t) &= du(Y(t)) = u'(Y(t))adt + u'(Y(t))bW(t) + \frac{1}{2}u''(Y(t))|b|^2dt \\ &= x \exp(Y(t))adt + x \exp(Y(t))bW(t) + \frac{1}{2}x \exp(Y(t))b^2dt \\ &= X(t)adt + X(t)bW(t) + \frac{1}{2}X(t)b^2dt \end{aligned}$$

$$= \left(a + \frac{1}{2}b^2 \right) X(t)dt + bX(t)dW(t).$$

Also note that the initial condition is: $X(0) = xe^{a \cdot 0 + bW(0)} = xe^0 = x$.

So the equation we were looking for is

$$dX(t) = \left(a + \frac{1}{2}b^2 \right) X(t)dt + bX(t)dW(t), \quad X(0) = x.$$

□

4.2 Proof of Theorem 4.1.6 (Level 5 only)

First, let us prove the a priori estimate of Lemma 4.1.5. For this, we will need Gronwall's inequality:

Lemma 4.2.1 (Gronwall's inequality/lemma). *Let $g : [0, T] \rightarrow [0, \infty)$ be a measurable function such that*

$$g(t) \leq \alpha + \beta \int_0^t g(s) ds < \infty, \quad \forall t \in [0, T].$$

Then, for all $t \in [0, T]$

$$g(t) \leq \alpha e^{\beta t}.$$

Proof of Lemma 4.1.5. By Itô's formula, we have in vector/matrix notation

$$\begin{aligned} |X(t)|^p &= |x_0|^p + \int_0^t p|X(s)|^{p-2} X(s) f(s, X(s)) ds + \int_0^t p|X(s)|^{p-2} X(s) \sigma(X(s)) dW(s) \\ &\quad + \frac{1}{2} \int_0^t (p(p-1)|X(s)|^{p-4} |\sigma^*(X(s))X(s)|^2 + p|\sigma(X(s))|^2 |X(s)|^{p-2}) ds. \end{aligned}$$

By (4.2), there exists a constant $N = N(p, K)$ such that

$$|X(t)|^p \leq |x_0|^p + N + N \int_0^t |X(s)|^p ds + M_t,$$

where

$$M(t) = \int_0^t p|X(s)|^{p-2} X(s) \sigma(X(s)) dW(s).$$

Let $\tau_n = \inf\{t \in [0, T]; |M(t)| + |X(t)| \geq n\} \wedge T$. Notice that $(M(t \wedge \tau_n))_{t \leq T}$ is a martingale starting from zero, and we have

$$|X(t \wedge \tau_n)|^p \leq |x_0|^p + N + N \int_0^{t \wedge \tau_n} |X(s)|^p ds + M(t \wedge \tau_n).$$

Upon taking suprema and expectations, we see that

$$\mathbb{E} \left[\sup_{t \in [0, r]} |X(t \wedge \tau_n)|^p \right] \leq |x_0|^p + N + N \int_0^r \mathbb{E} \left[\sup_{t \in [0, s]} |X(t \wedge \tau_n)|^p \right] ds + \mathbb{E} \left[\sup_{t \in [0, r]} |M(t \wedge \tau_n)| \right].$$

By Davis' inequality, we get

$$\begin{aligned}
\mathbb{E} \left[\sup_{t \in [0, r]} |M(t \wedge \tau_n)| \right] &\leq N \mathbb{E} \left[\left(\int_0^{r \wedge \tau_n} (|X(s)|^{p-1} |\sigma(X(s))|)^2 ds \right)^{1/2} \right] \\
&\leq N \mathbb{E} \left[\left(\int_0^r (1 + \sup_{t \in [0, s]} |X(t \wedge \tau_n)|^{2p}) ds \right)^{1/2} \right] \\
&\leq N + N \mathbb{E} \left[\left(\int_0^r \sup_{t \in [0, s]} |X(t \wedge \tau_n)|^{2p} ds \right)^{1/2} \right] \\
&\leq N + \mathbb{E} \left[\left(\sup_{t \in [0, r]} |X(t \wedge \tau_n)|^p \int_0^r \sup_{t \in [0, s]} |X(t \wedge \tau_n)|^p ds \right)^{1/2} \right] \\
&\leq N + \frac{1}{2} \mathbb{E} \left[\sup_{t \in [0, r]} |X(t \wedge \tau_n)|^p \right] + N \int_0^r \mathbb{E} \left[\sup_{t \in [0, s]} |X(t \wedge \tau_n)|^p \right] ds.
\end{aligned}$$

Consequently, we get

$$\mathbb{E} \left[\sup_{t \in [0, r]} |X(t \wedge \tau_n)|^p \right] \leq |x_0|^p + N + N \int_0^r \mathbb{E} \left[\sup_{t \in [0, s]} |X(t \wedge \tau_n)|^p \right] ds.$$

Gronwall's lemma gives

$$\mathbb{E} \left[\sup_{t \in [0, T]} |X(t \wedge \tau_n)|^p \right] \leq e^{NT} (N + |x_0|^p).$$

By Fatou's lemma we get

$$\mathbb{E} \left[\sup_{t \in [0, T]} |X(t)|^p \right] \leq e^{NT} (N + |x_0|^p),$$

which finishes the proof. $\square$

We will also need the following.

Theorem 4.2.2 (Banach's fixed point theorem). *Let $(B, \|\cdot\|_B)$ be a Banach space and let $J : B \rightarrow B$ be a contraction, that is, there exists $c < 1$ such that*

$$\|J(b_1) - J(b_2)\|_B \leq c \|b_1 - b_2\|_B, \quad \forall b_1, b_2 \in B.$$

Then, there exists a unique $b \in B$ such that $J(b) = b$.

We will choose an appropriate Banach space and a mapping J such that the fixed point of J is the solution of (4.3). Let us denote by $\mathcal{L}^2$ the space of all adapted process $(X(t))_{t \in [0, T]}$ such that

$$\|X\|_{\mathcal{L}^2}^2 := \sup_{t \in [0, T]} (e^{-Ct} \mathbb{E}[|X(t)|^2]) < \infty,$$

where the constant C is given by $C = 4K^2(1 + T)$, for reasons that will become apparent in the proof of Theorem 4.1.6 below. It is easy to see that $\mathcal{L}^2$ is a Banach space. Let us consider the operator J on $\mathcal{L}^2$, $X \mapsto J(X) = (J(X, t))_{t \geq 0}$, where

$$J_t(X) := x_0 + \int_0^t f(s, X(s)) ds + \int_0^t \sigma(s, X(s)) dW(s).$$

Exercise 4.2.3. Consider $(J(X, t))_{t \geq 0}$ for $X \in \mathcal{L}^2$. Under Assumption 4.1.1, show that

$$\sup_{t \in [0, T]} \mathbb{E} |J(X, t)|^2 < \infty,$$

which obviously implies that

$$\sup_{t \in [0, T]} (e^{-Ct} \mathbb{E} |J(X, t)|^2) < \infty.$$

Hence, notice that $J : \mathcal{L}^2 \mapsto \mathcal{L}^2$.

We are now ready to proceed with the proof of Theorem 4.1.6.

Proof. Notice that a fixed point of J is a solution of (4.3). Indeed, if X is a fixed point of J , then for all $t \geq 0$, we have

$$X(t) = J(X, t), \quad a.s.$$

In addition, since $J(X, t)$ is continuous in t , it follows that $X(t)$ is also continuous (or to be more precise, it has a continuous modification which is exactly given by $J(X, t)$). In this sense, we have that almost surely,

$$X(t) = J(X, t) = x_0 + \int_0^t f(s, X(s)) ds + \int_0^t \sigma(s, X(s)) dW(s), \quad \forall t \in [0, T].$$

In addition, by Lemma 4.1.5, it follows that any solution of (4.3) in $\mathcal{L}^2$. Consequently, in order to show that (4.3) has a unique solution, it suffices to show that J is a contraction. In other words, it suffices to show that there exists $\alpha \in (0, 1)$ such that for all $X, Y \in \mathcal{L}^2$, we have

$$\|J(X) - J(Y)\|_{\mathcal{L}^2} \leq \alpha \|X - Y\|_{\mathcal{L}^2},$$

To this end, let $X, Y \in \mathcal{L}^2$. Be the inequality $(a + b)^2 \leq 2(a^2 + b^2)$ we have

$$\begin{aligned} \mathbb{E} [|J(X, t) - J(Y, t)|^2] &\leq 2\mathbb{E} \left[\left| \int_0^t (f(s, X(s)) - f(s, Y(s))) ds \right|^2 \right] \\ &\quad + 2\mathbb{E} \left[\left| \int_0^t (\sigma(s, X(s)) - \sigma(s, Y(s))) dW(s) \right|^2 \right]. \end{aligned} \quad (4.5)$$

By Itô's isometry and Assumption 4.1.1, we get

$$\begin{aligned} &\mathbb{E} \left[\left| \int_0^t (\sigma(s, X(s)) - \sigma(s, Y(s))) dW(s) \right|^2 \right] \\ &= \mathbb{E} \left[\int_0^t |\sigma(s, X(s)) - \sigma(s, Y(s))|^2 ds \right] \\ &\leq K^2 \int_0^t \mathbb{E} [|X(s) - Y(s)|^2] ds \\ &\leq K^2 \left(\int_0^t e^{Cs} ds \right) \sup_{s \in [0, T]} (e^{-Cs} \mathbb{E} [|X(s) - Y(s)|^2]) \end{aligned}$$

$$= \frac{K^2}{C}(e^{Ct} - 1)\|X - Y\|_{\mathcal{L}^2}.$$

This implies that

$$\sup_{t \in [0, T]} \left(e^{-Ct} 2\mathbb{E} \left[\left| \int_0^t (\sigma(s, X(s)) - \sigma(s, Y(s))) dW(s) \right|^2 \right] \right) \leq 2 \frac{K^2}{C} \|X - Y\|_{\mathcal{L}^2}.$$

Similarly, by using Hölder's inequality in place of Itô's and proceeding as above, one gets

$$\sup_{t \in [0, T]} \left(e^{-Ct} 2\mathbb{E} \left[\left| \int_0^t (f(s, X(s)) - f(s, Y(s))) ds \right|^2 \right] \right) \leq 2T \frac{K^2}{C} \|X - Y\|_{\mathcal{L}^2}.$$

By the above inequalities and (4.5), we get

$$\|J(X) - J(Y)\|_{\mathcal{L}^2} = \sup_{t \in [0, T]} (e^{-Ct} \mathbb{E} [|J(X, t) - J(Y, t)|^2]) \leq 2 \frac{K^2}{C} (1 + T) \|X - Y\|_{\mathcal{L}^2}.$$

Keeping in mind that $C = 4K^2(1 + T)$, we get

$$\|J(X) - J(Y)\|_{\mathcal{L}^2} \leq \frac{1}{2} \|X - Y\|_{\mathcal{L}^2},$$

which shows that J is a contraction. This finishes the proof. $\square$

4.3 Feynman-Kac formula

Next, we consider a simple version of the Feynman-Kac formula (for the general version, see Section 3.8 in [?]). Let f and σ be functions as in Assumption 4.1.1. For simplicity, let us assume that they do not depend on the time parameter t . Let $g : \mathbb{R}^d \rightarrow \mathbb{R}$ be a continuous function with at most polynomial growth. We consider the parabolic PDE:

$$\begin{aligned} u_t(t, x) + Lu(t, x) &= 0, & (t, x) &\in [0, T) \times \mathbb{R}^d \\ u(T, x) &= g(x), & x &\in \mathbb{R}^d, \end{aligned} \tag{4.6}$$

where the differential operator is given by

$$L\phi(x) = \sum_{i,j=1}^d \frac{1}{2} (\sigma(x)\sigma^\top(x))_{ij} \phi_{x_i x_j}(x) + \sum_{i=1}^d f^i(x) \phi_{x_i}(x).$$

For each $x \in \mathbb{R}^d$, let us consider the SDE

$$dX^x(t) = f(X^x(t)) dt + \sigma(X^x(t)) dW(t), \quad X^x(0) = x. \tag{4.7}$$

Finally, by $C_{Pol}([0, T] \times \mathbb{R}^d)$ we denote the set of all real-valued, continuous functions u on $[0, T] \times \mathbb{R}^d$ for which there exist constants C and $p > 0$ such that

$$\sup_{t \in [0, T]} |u(t, x)| \leq C(1 + |x|^p).$$

By $C^{1,2}([0, T] \times \mathbb{R}^d)$ we denote the set of all real functions u on $[0, T] \times \mathbb{R}^d$ whose derivatives u_t , u_{x_i} , and $u_{x_i x_j}$ exist for all $i, j = 1, \dots, d$ and are continuous.

Theorem 4.3.1. *Let $u \in C_{Pol}([0, T] \times \mathbb{R}^d) \cap C^{1,2}([0, T] \times \mathbb{R}^d)$ be a solution of (4.6). Then, we have that*

$$u(t, x) = \mathbb{E}[g(X^x(T - t))],$$

where X is the unique solution of (4.7).

The proof of the above theorem is an easy consequence of Itô's formula.

4.4 Markov Property

Definition 4.4.1. A family of stochastic processes X^x , $x \in \mathbb{R}^d$, is called a time-homogeneous Markov family if the following are satisfied:

1. For each $t \geq 0$ and $A \in \mathcal{B}(\mathbb{R}^d)$, the map $x \mapsto \mathbb{P}(X^x(t) \in A)$ is Borel measurable
2. $\mathbb{P}(X^x(0) = x) = 1$
3. For all bounded measurable functions $g : \mathbb{R}^d \rightarrow \mathbb{R}$, and for all $t, s \geq 0$,

$$\mathbb{E}[g(X^x(t + s)) | \mathcal{F}_s] = \psi_g(t, X^x(s)),$$

where $\psi_g(t, x) := \mathbb{E}[g(X^x(t))]$

Exercise 4.4.2. Suppose that a family X^x , $x \in \mathbb{R}^d$, is a time homogeneous Markov family. Show that 3 above holds also for all measurable $g : \mathbb{R}^d \rightarrow \mathbb{R}$ such that $g(X_t^x) \in L_1(\Omega)$ for all $x \in \mathbb{R}^d$, $t \geq 0$.

Theorem 4.4.3. *For $x \in \mathbb{R}^d$, let X^x be the solution of equation (4.7). Then the family X^x , $x \in \mathbb{R}^d$, is a time-homogeneous Markov family.*

5 Applications in Finance

5.1 Basic concepts in the Black-Scholes model

In this section we consider a market with a riskless asset (i.e., a bank account) and a risky asset (i.e., a stock) and the goal is to price derivatives with maturity $T > 0$ written on the risky asset.

We consider a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ carrying a one dimensional Wiener process $(W(t))_{t \leq T}$ and let $\mathbb{F} = (\mathcal{F}_t)_{t \leq T}$ be the augmentation of the natural filtration of W . Notice that in this case, any $\mathcal{F}_0$ -measurable random variable is a constant. In this section $\mathcal{D}_T$ denotes the set of all $\mathbb{F}$ -adapted processes $(X(t))_{t \leq T}$ such that $\int_0^T |X(s)| ds < \infty$ a.s. and $\mathcal{S}_T$ denotes the set of all $\mathbb{F}$ -adapted processes $(X(t))_{t \leq T}$ such that $\int_0^T |X(s)|^2 ds < \infty$ a.s. The riskless asset is evolving according to the dynamics

$$dB(t) = rB(t) dt, \quad B_0 = 1, \quad (5.1)$$

that is, $B(t) = e^{rt}$. The coefficient $r > 0$ represents the risk-free interest rate. The risky asset is evolving according to

$$dS(t) = \mu S(t) dt + \sigma S(t) dW(t), \quad S(0) = s_0,$$

where $\mu > 0$ is known as the *trend* of the stock.

The model given above in (5.1)-(5.2) is known as the *Black-Scholes model*.

From now on, if $X(t)$ is a stochastic process, we will denote by $\tilde{X}_t$ its discounted value, that is $\tilde{X}_t := e^{-rt} X(t)$.

Theorem 5.1.1. *There exists a unique probability measure $\mathbb{P}^*$ on $\mathcal{F}_T$ such that*

- $\mathbb{P}$ and $\mathbb{P}^*$ are equivalent, that is, for all $A \in \mathcal{F}_T$, $\mathbb{P}(A) = 0$ if and only if $\mathbb{P}^*(A) = 0$
- The process $\tilde{S}$ is a martingale under $\mathbb{P}^*$

The above theorem is a consequence of Girsanov's theorem. Indeed, if we define

$$\mathbb{P}^*(A) := \mathbb{E} \left[\mathbf{1}_A \exp \left(-\theta W(T) - \frac{1}{2} \theta^2 T \right) \right], \quad \theta = \frac{\mu - r}{\sigma}$$

then under $\mathbb{P}^*$ the process $W^*(t) = \theta t + W(t)$ is an $\mathbb{F}$ -Wiener process. Then, we have that

$$\begin{aligned} dS(t) &= \mu S(t) dt + \sigma S(t) dW(t) \\ &= rS(t) dt + \sigma S(t) (\theta dt + dW(t)), \end{aligned}$$

which shows that

$$dS(t) = rS(t) dt + \sigma S(t) dW^*(t), \quad S(0) = s_0. \quad (5.2)$$

It is easy then to see that under $\mathbb{P}^*$ the process $\tilde{S}$ is a martingale (see the exercises below). One can also show that it is the unique equivalent measure to $\mathbb{P}$ with this property.

The measure $\mathbb{P}^*$ will be called the *risk-neutral* probability measure while $\mathbb{P}$ is called the *physical* probability measure. The expectation with respect to $\mathbb{P}^*$ will be denoted by $\mathbb{E}^*$.

From now on we are working on the probability space $(\Omega, \mathcal{F}_T, \mathbb{P}^*)$. To avoid confusion, martingales with respect to the filtration $\mathbb{F}$ under the measure $\mathbb{P}^*$ will be called $\mathbb{P}^*$ -martingales.

Exercise 5.1.2. Find the solution of equation (5.2).

Hint: See also Exercise 4.1.8

Solution: Recall that in Exercise 5.1.11 we found that the process

$$X(t) = x \exp(at + bW^*(t))$$

solves the equation

$$dX(t) = \left(a + \frac{1}{2}b^2\right) X(t)dt + bX(t)dW^*(t), \quad X_0 = x$$

We want to solve equation (6.2):

$$dS(t) = rS(t)dt + \sigma S(t)dW^*(t), \quad S(0) = s_0.$$

We can do this by choosing a, b, x as follows:

- Choose $b = \sigma$.
- We want to have $r = a + \frac{1}{2}b^2$, i.e. (as we chose $b = \sigma$.) $r = a + \frac{1}{2}\sigma^2$. For this we need to choose

$$a = r - \frac{1}{2}\sigma^2.$$

- For the initial condition, choose $x = s_0$.

Hence the solution is

$$S(t) = s_0 \exp\left(\left(r - \frac{1}{2}\sigma^2\right)t + \sigma W^*(t)\right).$$

We can check that this is indeed the solution to the Black-Scholes equation, using Ito's lemma: Let

$$Y(t) := \left(r - \frac{1}{2}\sigma^2\right)t + \sigma W^*(t)$$

(so we have $dY(t) = \left(r - \frac{1}{2}\sigma^2\right)dt + \sigma dW^*(t)$), and let

$$u(y) = s_0 \exp(y).$$

Then by Ito's lemma

$$\begin{aligned} dS(t) &= du(Y(t)) = u'(Y(t)) \left(r - \frac{1}{2}\sigma^2\right) dt + u'(Y(t))\sigma dW^*(t) + \frac{1}{2}u''(Y(t))|\sigma|^2 dt \\ &= s_0 \exp(Y(t)) \left(r - \frac{1}{2}\sigma^2\right) dt + s_0 \exp(Y(t))\sigma dW^*(t) + \frac{1}{2}s_0 \exp(Y(t))\sigma^2 dt \\ &= S(t)r dt + S(t) \left(-\frac{1}{2}\sigma^2\right) dt + S(t)\sigma dW^*(t) + \frac{1}{2}S(t)\sigma^2 dt \\ &= rS(t)dt + \sigma S(t)dW^*(t) \end{aligned}$$

So $S(t)$ solves the SDE. Also note, that

$$S(0) = s_0 \exp\left(\left(r - \frac{1}{2}\sigma^2\right) \cdot 0 + \sigma W^*(0)\right) = s_0 \exp(0) = s_0,$$

Therefore $S(t)$ also satisfies the initial condition. Therefore by uniqueness of solutions, $S(t)$ is indeed the solution. $\square$

Exercise 5.1.3. Show that $(\tilde{S}(t))_{t \leq T}$ satisfies

$$d\tilde{S}(t) = \sigma \tilde{S}(t) dW^*(t), \quad \tilde{S}(0) = s_0.$$

Conclude that it is a $\mathbb{P}^*$ -martingale.

Solution: Recall:

$$\tilde{S}(t) = e^{-rt} S(t).$$

So by Ito's product rule:

$$d\tilde{S}(t) = e^{-rt} dS(t) + S(t) d(e^{-rt}) + d(e^{-rt}) dS(t). \quad (\dagger)$$

Note:

- $\frac{de^{-rt}}{dt} = e^{-rt}(-r)$, thus $d(e^{-rt}) = -re^{-rt}dt$.
- Also: $dS(t) = rS(t)dt + \sigma S(t)dW^*(t)$,

Substituting these into $(\dagger)$, we get

$$\begin{aligned} d\tilde{S}(t) &= e^{-rt}(rS(t)dt + \sigma S(t)dW^*(t)) + (-re^{-rt}dt)S(t) + (-re^{-rt}dt)(rS(t)dt + \sigma S(t)dW^*(t)) \\ &\quad \text{as } (dt)^2 = dt dW^*(t) = 0: \\ &= e^{-rt}rS(t)dt + e^{-rt}\sigma S(t)dW^*(t) - re^{-rt}dtS(t) + 0 + 0 \\ &= e^{-rt}\sigma S(t)dW^*(t) \\ &= \sigma (e^{-rt}S(t)) dW^*(t) \\ &= \sigma \tilde{S}(t) dW^*(t) \end{aligned}$$

as required. Furthermore $\tilde{S}_0 = e^{-r \cdot 0} S(0) = s_0$. To show that it is a martingale it suffices to check that $\tilde{S} \in \mathcal{H}_T$. We have seen similar exercises. $\square$

We want to find the price at time zero of a derivative written on the risky asset which at maturity T gives to its owner the *pay-off* ξ). The pay-off ξ is assumed to be an $\mathcal{F}_T$ -measurable random variable which moreover satisfies $\mathbb{E}^*[|\xi|^2] < \infty$. Often in application, $\xi = g(S_T)$, for a Borel measurable function $g : \mathbb{R} \rightarrow [0, \infty)$, which we assume that it has polynomial growth.

The most classical derivatives in the market are the European options (call and put). In the European call option the owner has the right (but no obligation, in contrast to a forward contract) to buy the stock at time T at a predetermined price K , which is called the strike price. If the value of a stock is above K , that is, if $S_T > K$ then the owner of the contract will buy the stock for K , which leads to a profit $(S_T - K)$. If the price of the stock is below K , that is, if $S_T \leq K$ then the owner will not exercise his right to buy the option because that would lead to a loss. In this case the profit is just 0. Therefore, the pay-off is $(S_T - K)^+$, that is, $g(x) = (x - K)^+$. The European put option, is a derivative that gives to its owner the right to sell the stock at the maturity T for a fixed price K . Similarly to the above, it follows that the pay-off is $(K - S_T)^+$, that is, the pay-off function in this case is given by $g(x) = (K - x)^+$.

In order to find the fair price of a contract with pay-off $g(S_T)$, the idea is to "replicate" the derivative by trading the riskless and the risky asset. This means, we want to find a portfolio (a strategy for trading those two assets) which at the terminal time has value exactly $g(S_T)$. It is

natural (fair) then to say that the price of the derivative at time zero is equal to the value of the portfolio replicating the derivative (provided that this value is unique). Let us make this more precise mathematically.

A *portfolio* is a pair $H = (H_1, H_2)$ where $H_1 \in \mathcal{D}_T$ and $H_2 \in \mathcal{S}_T$. For a portfolio H we define its *value* at time $t \in [0, T]$ by $V_H(t) := H_1(t)B(t) + H_2(t)S(t)$. A portfolio H is said to be *self-financing* if the change of its value is only due to the changes in the assets prices weighted by the portfolio. In mathematical terms, this means

$$dV_H(t) = H_1(t) dB(t) + H_2(t) dS(t)$$

or in different notation

$$dV_H(t) = rV_H(t) dt + H_2(t)(dS(t) - rS(t)dt).$$

This last equation shows that if H is self-financing, then V_H is determined by H_2 and $V_H(0)$ (by solving the above linear SDE). Consequently, also H_1 is determined by H_2 and $V_H(0)$. A portfolio H is called *admissible* if for each t we have $V_H(t) \geq 0$ a.s. and the class of all admissible portfolios will be denoted by $\mathcal{P}_a$. A self-financing, admissible portfolio H *replicates* ξ if $V_H(T) = \xi$ a.s.

Exercise 5.1.4. Show that a portfolio $H = (H_1, H_2)$ is self financing if and only if

$$d\tilde{V}_H(t) = H_2(t) d\tilde{S}(t).$$

Solution: Let $H = (H_1, H_2)$ be a portfolio with value V^H (so we have $V_H(t) = H_1(t)B(t) + H_2(t)S(t)$).

(i) Suppose that H is self-financing. Recall that $\tilde{V}_H(t) = e^{-rt}V_H(t)$. Therefore

$$\begin{aligned} dV_H(t) &= d(e^{-rt}V_H(t)) \\ &\quad \text{by Ito's product rule:} \\ &= e^{-rt}dV_H(t) + d(e^{-rt})V_H(t) + d(e^{-rt})dV_H(t) \\ &=: (\star) \end{aligned}$$

Note:

- $\frac{de^{-rt}}{dt} = e^{-rt}(-r)$, thus $d(e^{-rt}) = -re^{-rt}dt$.
- Since by assumption H is self-financing, we have $dV_H(t) = rV_H(t)dt + H_2(t)(dS(t) - rS(t)dt)$.

Therefore

$$\begin{aligned} (\star) &= e^{-rt}(rV_H(t)dt + H_2(t)(dS(t) - rS(t)dt)) \\ &\quad + (-re^{-rt}dt)V_H(t) + (-re^{-rt}dt)(rV_H(t)dt + H_2(t)(dS(t) - rS(t)dt)) \\ &= (\star\star). \end{aligned}$$

We can see that the last term is zero, because (using that $(dt)^2 = dt dW^*(t) = 0$), we have that

$$(-re^{-rt}dt)(rV_H(t)dt + H_2(t)(dS(t) - rS(t)dt)) = 0 - re^{-rt}dtH_2(t)dS(t) + 0$$

as $S(t)$ solves the B-S SDE:

$$\begin{aligned} &= -re^{-rt}dtH_2(t)(rS(t)dt + \sigma S(t)dW^*(t)) \\ &= 0 + 0 \\ &= 0. \end{aligned}$$

Hence

$$\begin{aligned} (\star\star) &= e^{-rt}(rV_H(t)dt + H_2(t)(dS(t) - rS(t)dt)) + (-re^{-rt}dt)V_H(t) + 0 \\ &= e^{-rt}rV_H(t)dt + e^{-rt}H_2(t)(dS(t) - rS(t)dt) - re^{-rt}dtV_H(t) \\ &= e^{-rt}H_2(t)(dS(t) - rS(t)dt) \end{aligned}$$

as $S(t)$ solves the B-S SDE:

$$\begin{aligned} &= e^{-rt}H_2(t)(\sigma S(t)dW^*(t)) \\ &= H_2(t)\sigma e^{-rt}S(t)dW^*(t) \end{aligned}$$

as $\tilde{S}(t) = e^{-rt}S(t)$:

$$= H_2(t)\sigma\tilde{S}(t)dW^*(t)$$

by Exercise 5.1.3:

$$= H_2(t)d\tilde{S}(t).$$

Therefore

$$d\tilde{V}_t = H_2(t)d\tilde{S}(t).$$

(ii) Now suppose that $d\tilde{V}_t = H_2(t)d\tilde{S}(t)$. Recall that $V_H(t) = e^{rt}\tilde{V}_H(t)$. Therefore

$$dV_H(t) = d(e^{rt}\tilde{V}_H(t))$$

by Ito's product rule:

$$\begin{aligned} &= e^{rt}d\tilde{V}_H(t) + d(e^{rt})\tilde{V}_H(t) + d(e^{rt})d\tilde{V}_H(t) \\ &=: (\star\star\star) \end{aligned}$$

Note

- $\frac{de^{rt}}{dt} = e^{rt}r$, thus $d(e^{rt}) = re^{rt}dt$.
- By assumption: $d\tilde{V}_t = H_2(t)d\tilde{S}(t)$.

Therefore

$$\begin{aligned} (\star\star\star) &= e^{rt}(H_2(t)d\tilde{S}(t)) + (re^{rt}dt)\tilde{V}_H(t) + (re^{rt}dt)(H_2(t)d\tilde{S}(t)) \\ &= (\star\star\star\star). \end{aligned}$$

We can see that the last term is zero, because (using that $dt dW^*(t) = 0$.) we have that

$$(re^{rt}dt)(H_2(t)d\tilde{S}(t)) =$$

by Exercise 5.1.3:

$$\begin{aligned} &= re^{rt}dtH_2(t)\sigma\tilde{S}(t)dW^*(t) \\ &= 0. \end{aligned}$$

Thus

$$(\star \star \star \star) = e^{rt}(H_2(t)d\tilde{S}(t)) + (re^{rt}dt)\tilde{V}_H(t)$$

by Exercise 6.1.2, and by the def. of $\tilde{V}_H(t)$:

$$\begin{aligned} &= e^{rt}H_2(t)\sigma\tilde{S}(t)dW^*(t) + re^{-rt}dte^{-rt}V_H(t) \\ &= H_2(t)\sigma e^{rt}\tilde{S}(t)dW^*(t) + rV_H(t)dt \end{aligned}$$

as $S(t) = e^{rt}\tilde{S}(t)$:

$$= H_2\sigma S(t)dW^*(t) + rV_H(t)dt$$

as $S(t)$ solves the B-S SDE:

$$= H_2(dS(t) - rS(t)dt) + rV_H(t)dt$$

Therefore

$$dV_H(t) = rV_H(t)dt + H_2(t)(dS(t) - rS(t)dt).$$

Hence H is self financing.

In (i) we have proven that if H is self-financing, then $d\tilde{V}_H(t) = H_2(t)d\tilde{S}(t)$. In (ii) we have proven that if $d\tilde{V}_H(t) = H_2(t)d\tilde{S}(t)$, then H is self financing. Thus we are done. $\square$

5.2 Pricing

The following theorem provides the "fair" price of a derivative with pay-off ξ at the terminal time T .

Theorem 5.2.1. *There exists a unique self-financing admissible portfolio H which replicates ξ . Moreover, its value at time $t \in [0, T]$ satisfies*

$$V_H(t) = \mathbb{E}^* [e^{-r(T-t)}\xi | \mathcal{F}_t]$$

Proof of Theorem 5.2.1. Let $R(t) = \mathbb{E}^* [e^{-r(T-t)}\xi | \mathcal{F}_t]$. By Exercise 3.2.18, we have that $\tilde{R}(t)$ is a martingale such that $\mathbb{E}^* [|\tilde{R}(t)|^2] < \infty$. By the martingale representation theorem, we get that there exists a unique $Y \in \mathcal{H}_T$ such that

$$\tilde{R}(t) = \mathbb{E}^* [e^{rT}\xi] + \int_0^t Y(s) dW^*(s). \quad (5.3)$$

Set

$$H_2(t) := \frac{Y(t)}{\sigma\tilde{S}(t)}, \quad H_1(t) := \tilde{R}(t) - H_2(t)\tilde{S}(t). \quad (5.4)$$

Keeping in mind that $B(t) = e^{rt}$, we have for the value of the portfolio $H = (H_1, H_2)$

$$V_H(t) = H_1(t)B(t) + H_2(t)S(t) = (\tilde{R}(t) - \frac{Y(t)}{\sigma\tilde{S}(t)}\tilde{S}(t))B(t) + \frac{Y(t)}{\sigma\tilde{S}(t)}S(t) = R(t).$$

This shows that $V_H(t) \geq 0$, and it also shows (by choosing $t = T$) that H replicates $g(S_T)$. Also, by the above and (5.3) we see that

$$d\tilde{V}_H(t) = d\tilde{R}(t) = Y(t) dW^*(t) = H_2\sigma\tilde{S}(t) dW^*(t) = H_2(t) d\tilde{S}(t),$$

which according to Exercise 5.1.4, shows that H is self-financing.

Now we show that H is unique. Let $U = (U_1, U_2)$ be another self-financing admissible strategy that replicates $g(S_T)$. By Exercise 5.1.4, we have that

$$d\tilde{V}_t^U = U_2(t) d\tilde{S}(t) = U_2(t)\sigma\tilde{S}(t) dW^*(t)$$

and

$$d\tilde{V}_H(t) = H_2(t) d\tilde{S}(t) = H_2(t)\sigma\tilde{S}(t) dW^*(t)$$

Hence, since $\tilde{V}_H(T) = e^{-RT} V_H(T) = e^{-RT} g(S_T) = e^{-RT} V_U(T) = \tilde{V}_U(T)$, we have that

$$\tilde{V}_H(0) + \int_0^T H_2(t)\sigma\tilde{S}(t) dW^*(t) = \tilde{V}_U(0) + \int_0^T U_2(t)\sigma\tilde{S}(t) dW^*(t).$$

Taking expectations in the above relation we conclude that $\tilde{V}_H(0) = \tilde{V}_U(0)$, which in turn gives

$$\int_0^T (H_2(t) - U_2(t))\sigma\tilde{S}(t) dW^*(t) = 0.$$

Ito's isometry now gives that

$$\mathbb{E}^* \left[\int_0^T |(H_2(t) - U_2(t))\sigma\tilde{S}(t)|^2 dt \right] = 0,$$

which shows that $H_2 = U_2$ (since almost surely, $\sigma\tilde{S}(t) > 0$ for all $t > 0$). Since H and U are self financing, H_1 and U_1 are uniquely determined by H_2 and U_2 , respectively, which shows that $H = U$. This shows the uniqueness part and finishes the proof. $\square$

Exercise 5.2.2. Suppose that $r = 0.5$, $\sigma = 1$, $s_0 = 1$ and $T = 2$. Calculate the price at time zero of a derivative written on S that pays the owner the amount $(S(T))^2$ at the terminal time T .

Solution: Here the payoff is $g(x) = x^2$. So by Theorem 5.2.1, the price at time $t = 0$ is

$$\begin{aligned} P_0 &= e^{-rT} \mathbb{E}^* g(S(T)) \\ &= e^{-rT} \mathbb{E}^* |S(T)|^2 \end{aligned}$$

as S solves the B-S SDE:

$$\begin{aligned} &= e^{-rT} \mathbb{E}^* \left| \exp \left(\left(r - \frac{1}{2}\sigma^2 \right) T + \sigma W_T^* \right) \right|^2 \\ &= e^{-rT} \mathbb{E}^* \exp((2r - \sigma^2)T + 2\sigma W_T^*) \\ &= e^{-rT} \exp((2r - \sigma^2)T) \mathbb{E}^* \exp \left(2\sigma\sqrt{T} \frac{W_T^*}{\sqrt{T}} \right) \end{aligned}$$

as $\frac{W_T^*}{\sqrt{T}} \sim \mathcal{N}(0, 1)$:

$$\begin{aligned} &= e^{-rT} \exp((2r - \sigma^2)T) \exp \left(\frac{1}{2} \cdot 4\sigma^2 T \right) \\ &= \exp(-rT + 2rT - \sigma^2 T + 2\sigma^2 T) \end{aligned}$$

$$\begin{aligned}
&= \exp((r + \sigma^2)T) \\
&= \exp((0.5 + 1^2) \cdot 2) \\
&= \exp(3).
\end{aligned}$$

So the price at time zero is $P_0 = \exp(3)$. $\square$

Theorem 5.2.3. *Let H be the unique admissible, self-financing portfolio that replicates $g(S_T)$. Then, we have that*

$$V_H(t) = u(t, S(t)),$$

where $u \in C_{Pol}([0, T] \times \mathbb{R}^d) \cap C^{1,2}([0, T] \times \mathbb{R}^d)$ is the unique solution of

$$\begin{aligned}
u_t(t, x) + \frac{\sigma^2 x^2}{2} u_{xx}(t, x) + rxu_x(t, x) - ru(t, x) &= 0, & (t, x) &\in [0, T] \times \mathbb{R} \\
u(T, x) &= g(x), & x &\in \mathbb{R},
\end{aligned} \tag{5.5}$$

Proof. Let $v \in C_{Pol}([0, T] \times \mathbb{R}^d) \cap C^{1,2}([0, T] \times \mathbb{R}^d)$ be the unique solution of

$$\begin{aligned}
v_t(t, x) + \frac{\sigma^2 x^2}{2} v_{xx}(t, x) + rxv_x(t, x) &= 0, & (t, x) &\in [0, T] \times \mathbb{R} \\
v(T, x) &= g(x), & x &\in \mathbb{R},
\end{aligned}$$

It is the straightforward to see that

$$u(t, x) = e^{-r(T-t)} v(t, x) \tag{5.6}$$

From the Feynman-Kac formula, we have that

$$v(t, x) = \mathbb{E}^*[g(S^x(T-t))] = \psi_g(T-t, x), \tag{5.7}$$

where $S^x(t)$ is the solution of (5.2) with initial condition $S^x(0) = x$. Since the family $(S^x(t))_{t \leq T}$, $x \geq 0$ is a Markov family, we have

$$\mathbb{E}^*[g(S^{s_0}(T)) | \mathcal{F}_t] = \psi_g(T-t, S^{s_0}(t)) = v(t, S^{s_0}(t)).$$

Multiplying the above equality with $e^{-r(T-t)}$ gives

$$V_H(t) = e^{-r(T-t)} \mathbb{E}^*[g(S^{s_0}(T)) | \mathcal{F}_t] = e^{-r(T-t)} v(t, S^{s_0}(t)) = u(t, S^{s_0}(t)),$$

which finishes the proof. $\square$

When ξ is of the form $g(S_T)$, then we can have more explicit representations of the price.

Theorem 5.2.4. *Let H be the unique admissible, self-financing portfolio that replicates $g(S_T)$. Then, we have that*

$$V_H(t) = G(t, S(t)),$$

where

$$G(t, x) = e^{-r(T-t)} \int_{-\infty}^{\infty} g\left(xe^{(r-\sigma^2/2)(T-t)+\sigma y\sqrt{T-t}}\right) e^{-y^2/2} dy.$$

Proof. It suffices to show that $G(t, x) = u(t, x)$, where u is the solution of (5.5). This follows from (5.6)-(5.7) and the fact

$$\mathbb{E}^*[g(S^x(T-t))] = \int_{-\infty}^{\infty} g\left(xe^{(r-\sigma^2/2)(T-t)+\sigma y\sqrt{T-t}}\right) e^{-y^2/2} dy. \quad (5.8)$$

□

Exercise 5.2.5. Prove (5.8).

Theorem 5.2.6. *The prices at time t of the European call and put options with strike K is given by $F_c(t, S(t))$ and $F_p(t, S(t))$, respectively, where*

$$\begin{aligned} F_c(t, x) &= x\Phi(a) - Ke^{-r(T-t)}\Phi(b) \\ F_p(t, x) &= -x\Phi(-a) - Ke^{-r(T-t)}\Phi(-b) \end{aligned}$$

and

$$a = \frac{\log(x/K) + (r + \sigma^2/2)(T-t)}{\sigma\sqrt{T-t}}, \quad b = a - \sigma\sqrt{T-t}$$

5.3 Hedging

We have shown that there exists a unique admissible, self-financing portfolio H which replicates ξ . In particular, if $\xi = g(S_T)$, we know by the above theorem that the value of the derivative at time $t \in [0, T]$ is given by $\mathbb{E}^*[e^{-r(T-t)}g(S_T)|\mathcal{F}_t]$. Notice that at time zero, the price is given by $\mathbb{E}^*[e^{-rT}g(S_T)]$ (since $\mathcal{F}_0$ is the trivial σ -algebra $\{\emptyset, \Omega\}$).

The next question, is whether we can find "explicitly" this portfolio. Since, H is self financing, it suffices to determine H_2 . It turns out that in the case $\xi = g(S_T)$, the answer to this question is affirmative.

We start with the following representation of the value of the derivative.

We can now find "explicitly" the hedging portfolio.

Theorem 5.3.1. *Let $H = (H_1, H_2)$ be the unique admissible, self-financing portfolio that replicates $g(S_T)$. Then we have*

$$H_2(t) = u_x(t, S(t)),$$

where $u \in C_{Pol}([0, T] \times \mathbb{R}^d) \cap C^{1,2}([0, T] \times \mathbb{R}^d)$ is the unique solution of (5.5).

Proof. Recall that by (5.3) and (5.4) we have

$$\tilde{V}_H(t) = \tilde{V}_H(0) + \int_0^t \sigma \tilde{S}(s) H_2(s) dW^*(s). \quad (5.9)$$

By Itô's formula we have that

$$\begin{aligned} u(t, S(t)) &= u(0, S(0)) + \int_0^t (u_t(s, S(s)) + rS(s)u_x(s, S(s)) + \frac{(\sigma S(s))^2}{2}u_{xx}(s, S(s))) ds \\ &\quad + \int_0^t u_x(s, S(s))\sigma S(s) dW^*(s) \end{aligned}$$

$$= u(0, S(0)) + \int_0^t ru(s, S(s)) ds + \int_0^t \sigma u_x(s, S(s)) S(s) dW^*(s).$$

By Itô's product rule, we have

$$\tilde{u}(t, S(t)) = \tilde{u}(0, S(0)) + \int_0^t \sigma u_x(s, S(s)) \tilde{S}(s) dW^*(s) \quad (5.10)$$

By Theorem 5.2.3, we have that $\tilde{V}_H(t) = \tilde{u}(t, S(t))$ for all $t \in [0, T]$. Consequently, subtracting (5.10) from (5.10), we get

$$\int_0^T (\sigma \tilde{S}(s) H_2(s) - \sigma u_x(s, S(s)) \tilde{S}(s)) dW^*(s) = 0.$$

Itô's isometry now gives

$$\begin{aligned} & \mathbb{E}^* \left[\int_0^T (\sigma \tilde{S}(s) H_2(s) - \sigma u_x(s, S(s)) \tilde{S}(s))^2 ds \right] \\ &= \mathbb{E}^* \left[\left| \int_0^T (\sigma \tilde{S}(s) H_2(s) - \sigma u_x(s, S(s)) \tilde{S}(s)) dW^*(s) \right|^2 \right] = 0. \end{aligned}$$

This implies that

$$(\sigma \tilde{S}(s) H_2(s) - \sigma \tilde{S}(s) u_x(s, S(s))) = 0$$

and since for all $s \in [0, T]$ we have that $\sigma \tilde{S}(s) > 0$, we conclude that $H_2(s) - u_x(s, S(s)) = 0$ for almost all $(\omega, s) \in \Omega \times [0, T]$. This finishes the proof. $\square$